

## Introduction

### About This Report

This report is for the Mid-Atlantic Fishery Management Council (MAFMC). The purpose of this report is to synthesize ecosystem information to allow the MAFMC to better meet fishery management objectives, and to update the MAFMC's Ecosystem Approach to Fishery Management (EAFM) risk assessment. The major messages of the report are synthesized on pages 1 and 2, with highlights of 2025 ecosystem events on page 3.

The information in this report is organized into two main sections; **performance measured against ecosystem-level management objectives** (Table 1), and potential **risks to meeting fishery management objectives** (Table 2: **climate change** and **other ocean uses**). A final section highlights **notable 2025 ecosystem observations**.

### Report structure

A glossary of terms<sup>1</sup>, detailed technical methods documentation<sup>2</sup>, indicator data<sup>3</sup>, and detailed indicator descriptions<sup>4</sup> are available online. We recommend new readers first review the details of standard figure formatting (Fig. 66a), categorization of fish and invertebrate species into feeding guilds (Table 6), and definitions of ecological production units (EPUs, including the Mid-Atlantic Bight, MAB; Fig. 66b) provided at the end of the document.

The two main sections contain subsections for each management objective or potential risk. Within each subsection, we first review observed trends for indicators representing each objective or risk, including the status of the most recent data year relative to a threshold (if available) or relative to the long-term average. Second, we identify potential drivers of observed trends, and synthesize results of indicators related to those drivers to outline potential implications for management. For example, if there are multiple drivers related to an indicator trend, do indicators associated with the drivers have similar trends, and can any drivers be affected by management action(s)? We emphasize that these implications are intended to represent testable hypotheses at present, rather than “answers,” because the science behind these indicators and syntheses continues to develop.

Table 1: Ecosystem-scale fishery management objectives in the Mid-Atlantic Bight

| Objective categories                                         | Indicators reported                                                           |
|--------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Objectives: Provisioning and Cultural Services</b>        |                                                                               |
| Seafood Production                                           | Landings; commercial total and by feeding guild; recreational harvest         |
| Commercial Profits                                           | Revenue decomposed to price and volume                                        |
| Recreational Opportunities                                   | Angler trips; recreational fleet diversity                                    |
| Stability                                                    | Fishery and ecosystem volatility, adaptive capacity, and shifts from baseline |
| Social & Cultural                                            | Community fishing engagement and social vulnerability status                  |
| Protected Species                                            | Bycatch; population (adult and juvenile) numbers; mortalities                 |
| <b>Potential Drivers: Supporting and Regulating Services</b> |                                                                               |
| Management                                                   | Stock status; catch compared with catch limits                                |
| Biomass                                                      | Biomass or abundance by feeding guild from surveys                            |
| Environment                                                  | Climate and ecosystem risk indicators listed in Table 2                       |

<sup>1</sup><https://noaa-edab.github.io/tech-doc/glossary.html>

<sup>2</sup><https://noaa-edab.github.io/tech-doc/>

<sup>3</sup><https://noaa-edab.github.io/ecodata/>

<sup>4</sup><https://noaa-edab.github.io/catalog/index.html>

Table 2: Risks to meeting fishery management objectives in the Mid-Atlantic Bight

| Risk categories                    | Observation indicators reported                                        | Potential driver indicators reported                                                          |
|------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Climate and Ecosystem Risks</b> |                                                                        |                                                                                               |
| Risks to Managing Spatially        | Managed species (fish and cetacean) distribution shifts                | Benthic and pelagic forage distribution; ocean temperature, changes in currents and cold pool |
| Risks to Managing Seasonally       | Managed species spawning and migration timing changes                  | Habitat timing: Length of ocean summer, cold pool seasonal persistence                        |
| Risks to Setting Catch Limits      | Managed species body condition and recruitment changes                 | Benthic and pelagic forage quality & abundance: ocean temperature & acidification             |
| <b>Other Ocean Uses Risks</b>      |                                                                        |                                                                                               |
| Offshore Wind Risks                | Fishery revenue and landings from wind lease areas by species and port | Wind development speed; Protected species presence and hotspots                               |

## Performance Relative to Fishery Management Objectives

In this section, we examine indicators related to broad, ecosystem-level fishery management objectives. We also provide hypotheses on the implications of these trends—why we are seeing them, what’s driving them, and potential or observed regime shifts or changes in ecosystem structure. Identifying multiple drivers, regime shifts, and potential changes to ecosystem structure, as well as identifying the most vulnerable resources, can help managers determine whether anything needs to be done differently to meet objectives and how to prioritize upcoming issues/risks.

### Seafood Production

#### Indicators: Landings; commercial and recreational

In the Mid-Atlantic, total [commercial landings](#) (includes bait and industrial landings) have declined over the long-term, and both total U.S. seafood (excludes bait and industrial uses) and MAFMC-managed seafood landings are at their all time low in 2024 (Fig. [1](#)). Commercial landings by [guild](#) include all species and all uses caught within the MAB, and are reported in total and for MAFMC managed species only. Landings of benthos have been below the long-term average since 2010, primarily driven by surf clam, ocean quahog, and recently scallops. Planktivores show a long-term decline, primarily due to decreases in species not managed by the MAFMC (Atlantic herring and Atlantic menhaden; Fig. [2](#)).

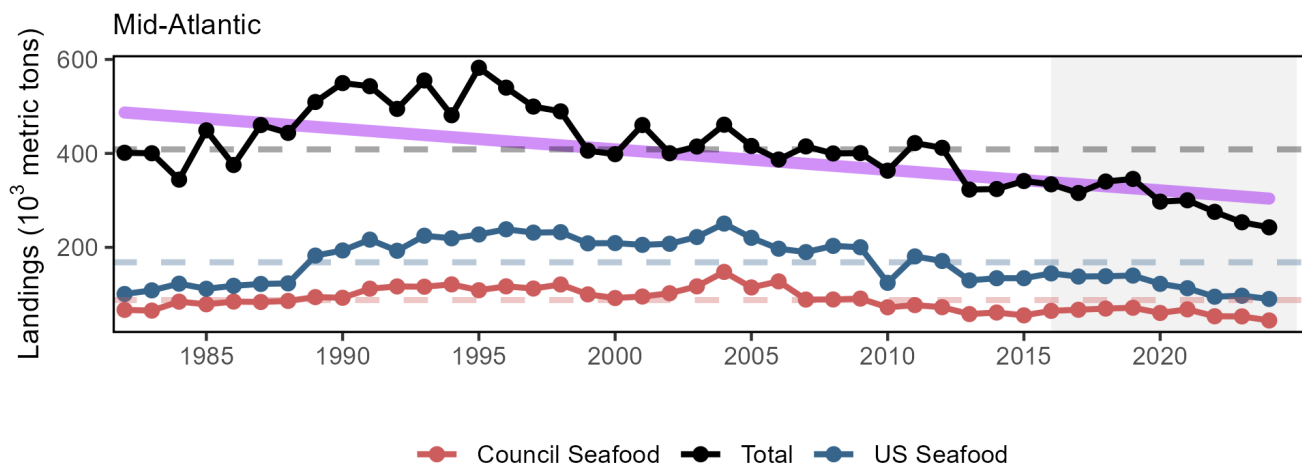


Figure 1: Total commercial landings (black), total U.S. seafood landings (blue), and Mid-Atlantic managed U.S. seafood landings (red), with significant decline (purple) in total landings.

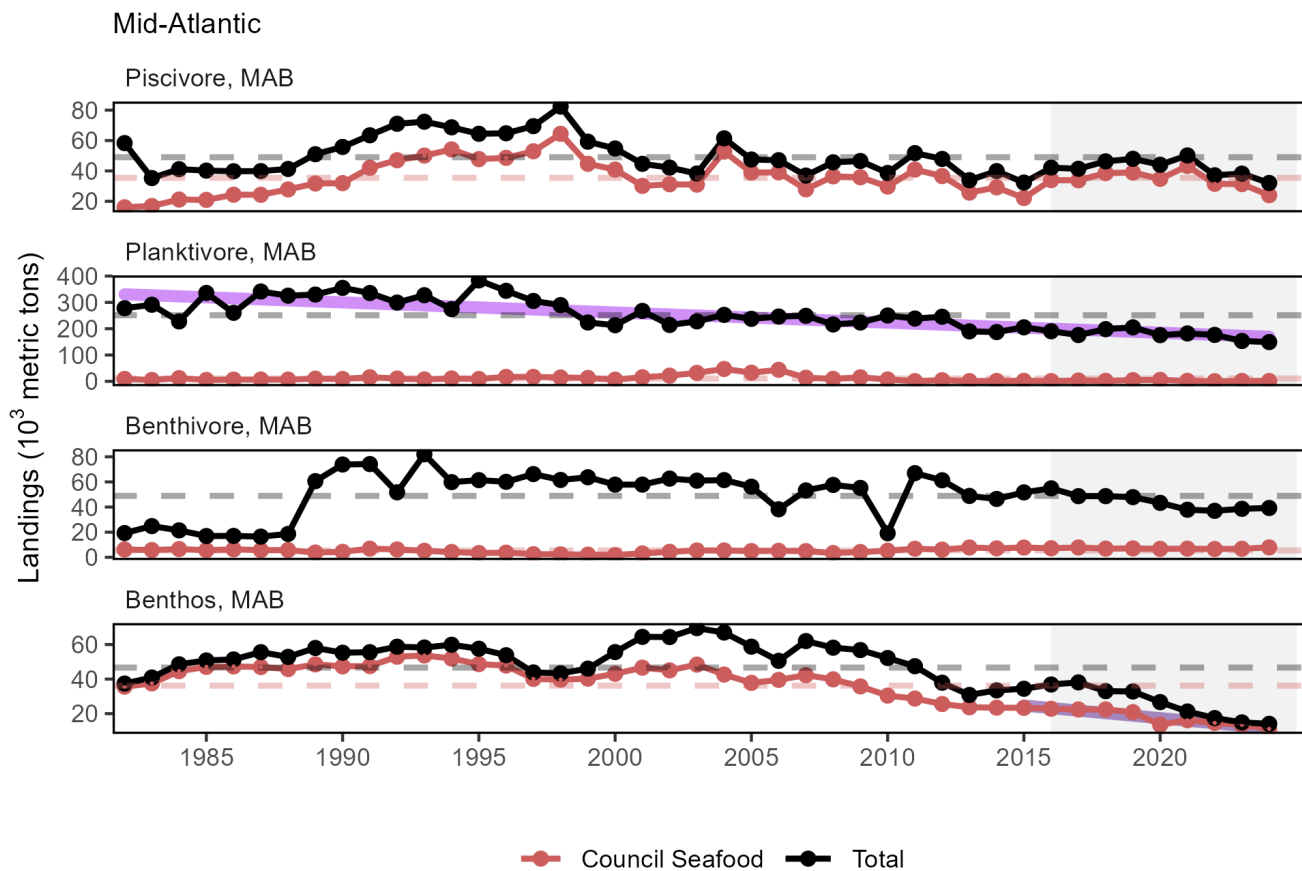


Figure 2: Total commercial landings in the Mid-Atlantic Bight (black) and MAFMC-managed U.S. seafood landings (red) by feeding guild, with significant declines (purple) in total planktivore landings.

Mid-Atlantic ports face a moderate to high risk from environmental variability, as evaluated by the 2025 [Community Environmental Variability Risk Indicators](#) (Fig. 3). These indicators assess port level risk to environmental variability based on dependence on species and their respective bioenvironmental vulnerabilities as assessed by regional experts. Total Vulnerability measures how much a region's landings (or revenue) is dependent on species that are sensitive to different climate and environmental change factors including temperature and acidification.

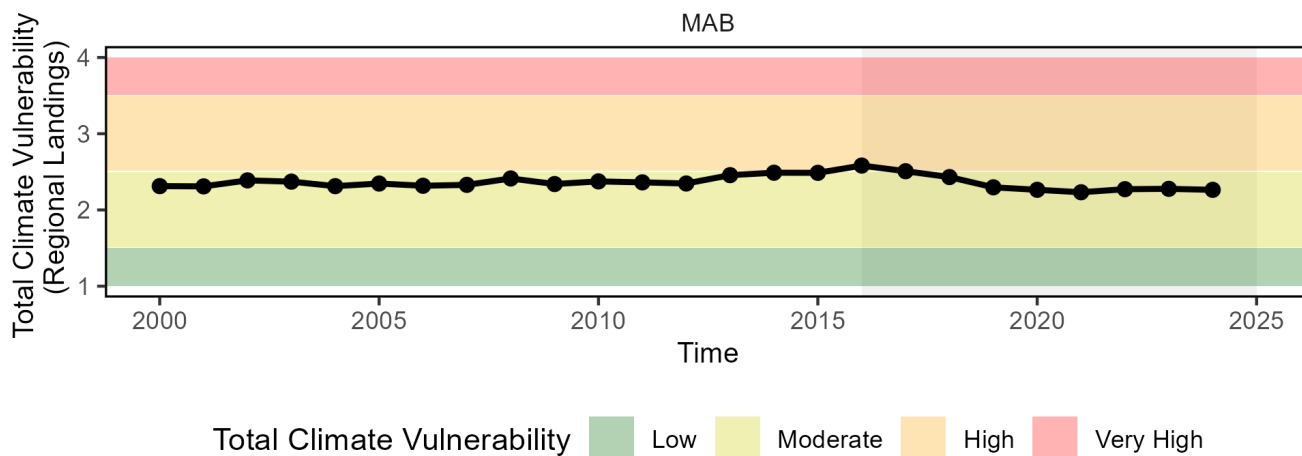


Figure 3: Mid-Atlantic region Total Vulnerability of commercial landings (sum of Mid-Atlantic port landings weighted by species climate vulnerability from Hare et al. 2016 and Loughran et al. 2025). Horizontal colored bars show different environmental variability risk levels.

In the Mid-Atlantic, total [recreational harvest](#) (retained fish presumed to be eaten) shows a long-term decline with recent harvest remaining near a time series low (Fig. 4). This pattern may indicate a shift towards catch-and-release strategies as opposed to catch for harvest. [Recreational shark landings](#) have generally decreased for most shark groups through 2024 (Fig 5). The recent low in pelagic shark landings is largely driven by regulatory changes implemented in 2018, followed by the closure of the shortfin mako fishery in 2022. These actions were intended to rebuild the North Atlantic shortfin mako stock and comply with binding recommendations by the International Commission for the Conservation of Atlantic Tunas (ICCAT).

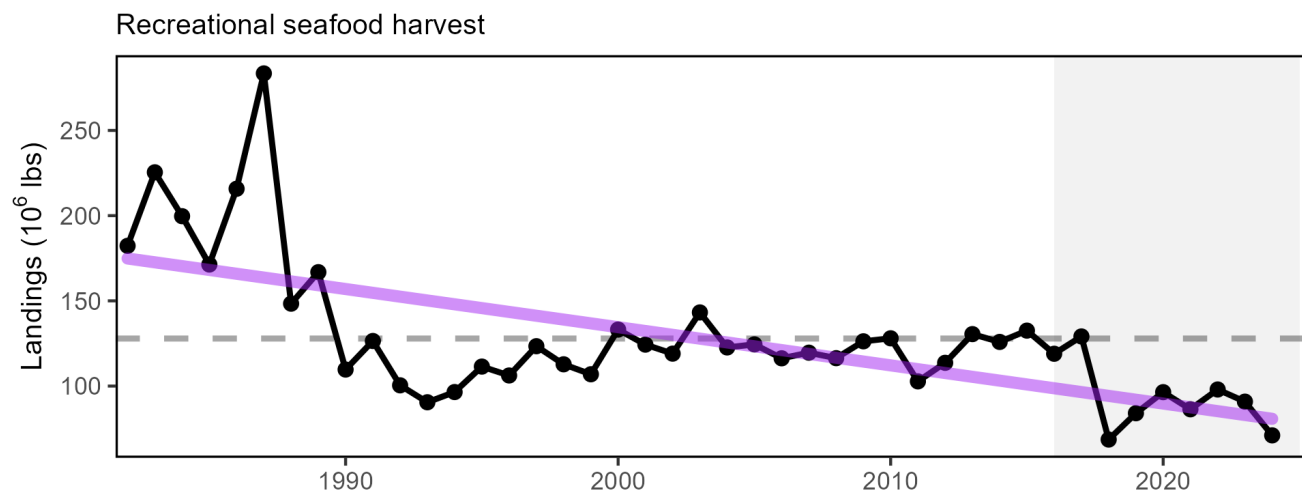


Figure 4: Total recreational seafood harvest (millions of pounds, black, significant decrease, purple) in the Mid-Atlantic region.



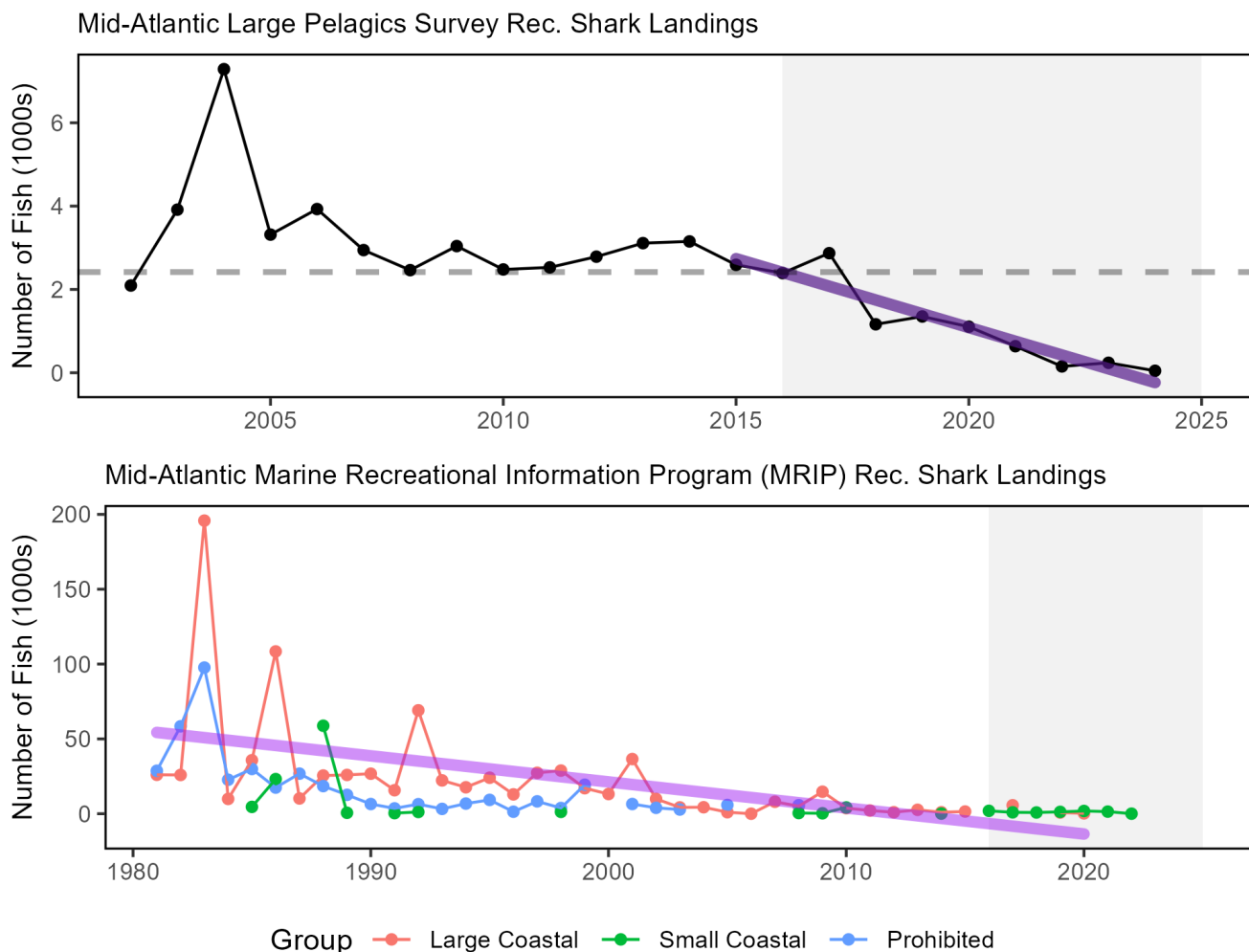


Figure 5: Recreational shark landings in the Mid-Atlantic region from NOAA Fisheries Large Pelagics Survey (top) and Marine Recreational Information Program (bottom) with declining trends (purple). Note that the trend line is associated with the Large Coastal group, which has had no reported MRIP landings after 2020.

[Aquaculture](#) can comprise a significant proportion of seafood production in specific communities, but not all aquaculture production is included in total seafood landings above. In 2022, the Northeast region produced approximately 6,300 metric tons of aquacultured shellfish, with revenue of \$133 million.

## Implications

Declining commercial landings (total and seafood) and recreational harvest can be attributed to many interacting factors, including combinations of ecosystem and stock production, management actions, market conditions, and environmental change. While we cannot evaluate all possible drivers at present, here we evaluate the extent to which stock status, management, and system biomass trends may play a role.

**Stock Status and Catch Limits** Single species [management objectives](#) (1. maintaining biomass above minimum thresholds and 2. maintaining fishing mortality below overfishing limits) are being met for all but two MAFMC-managed species (golden tilefish and Atlantic mackerel) (Fig. 6). However, the status of 5 stocks is unknown (northern shortfin squid, goosefish GOM/GB, goosefish southern GB/MAB, blueline tilefish, and chub mackerel).

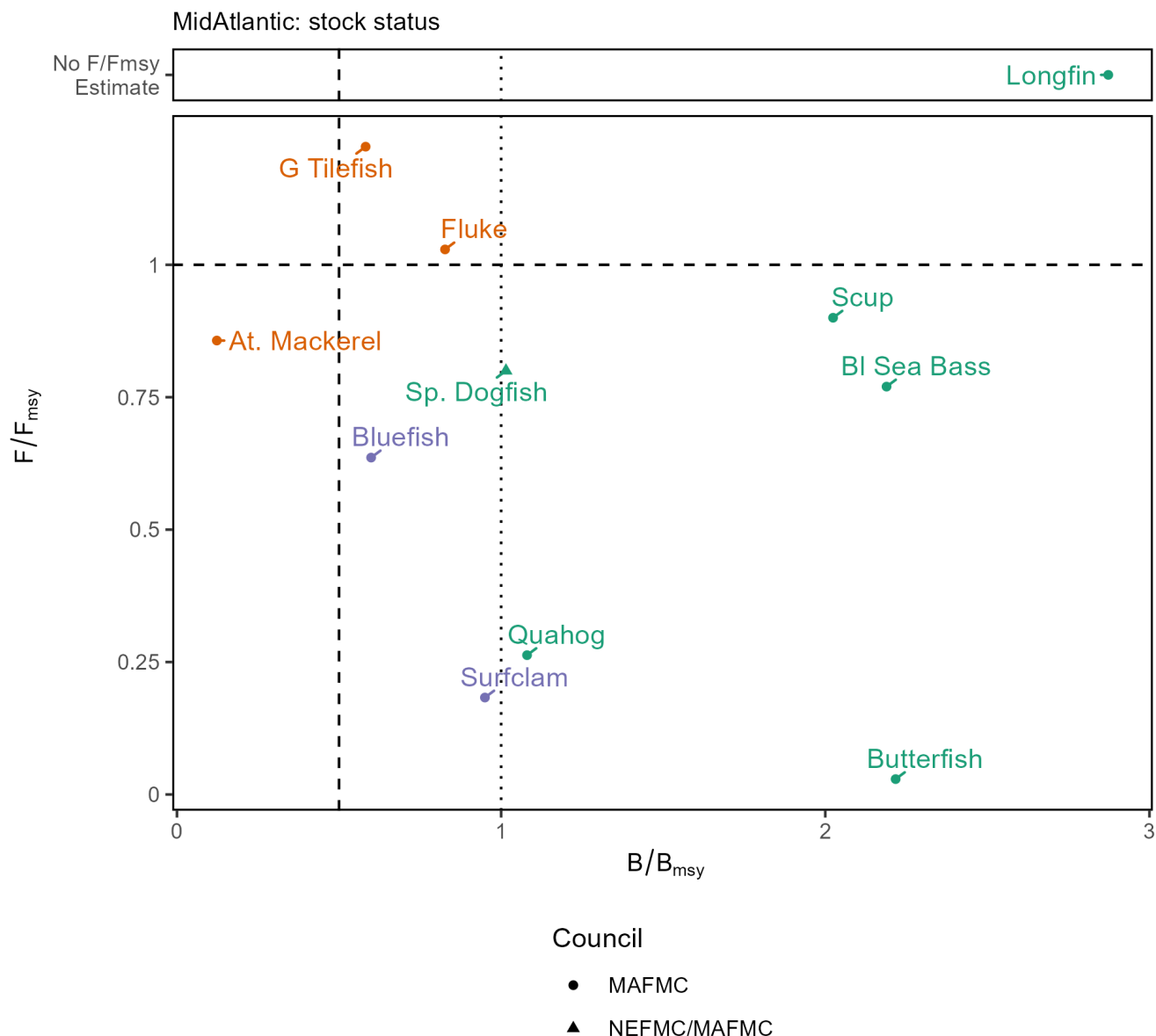


Figure 6: Summary of single species status for MAFMC and jointly federally managed stocks (spiny dogfish and both goosfish). The dotted vertical line is the target biomass reference point of  $B_{MSY}$ . The dashed lines are the management thresholds of one half  $B_{MSY}$  (vertical) or  $F_{MSY}$ . (horizontal). Stocks with a  $B/B_{MSY}$  estimate but without an  $F/F_{MSY}$  estimate are denoted in a separate box plot (top). Colors denote stocks with  $B/B_{MSY} < 0.5$  or  $F/F_{MSY} > 1$  (orange), stocks  $0.5 < B/B_{MSY} < 1$  (blue), and stocks  $B/B_{MSY} > 1$  (green).

Stock status and associated management constraints are unlikely to be driving decreased landings for some species, including quahog, surfclam, and northern shortfin squid. Most stocks are not fully utilizing their associated ABC or ACL, which includes landings and discards (Fig. 8). Quahog, surfclam, and northern shortfin squid have the largest ABC or ACLs but have low catch ratios, with less than 24% of ABC or ACL caught in 2024. All other species except chub mackerel and longfin squid have catch ratios over 80%, indicating that stock status and associated regulations are most likely constraining the landings of some species such as black sea bass, bluefish, and Atlantic mackerel. However, these management actions and regulations are enacted in response to biomass, such that less conservative regulations would not necessarily mean higher landings.

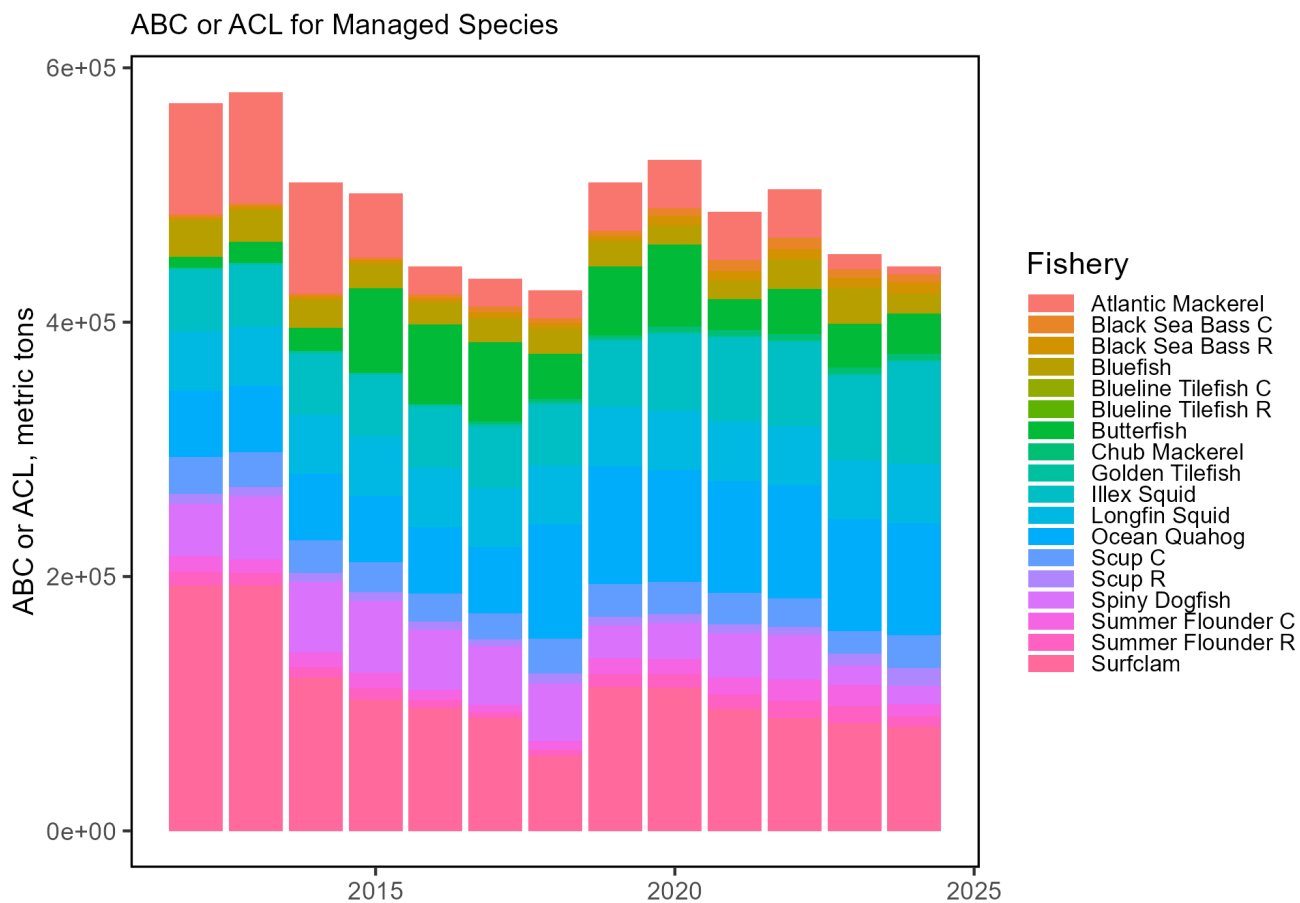


Figure 7: Sum of catch limits (in metric tons) across all MAFMC managed commercial (C) and recreational (R) fisheries

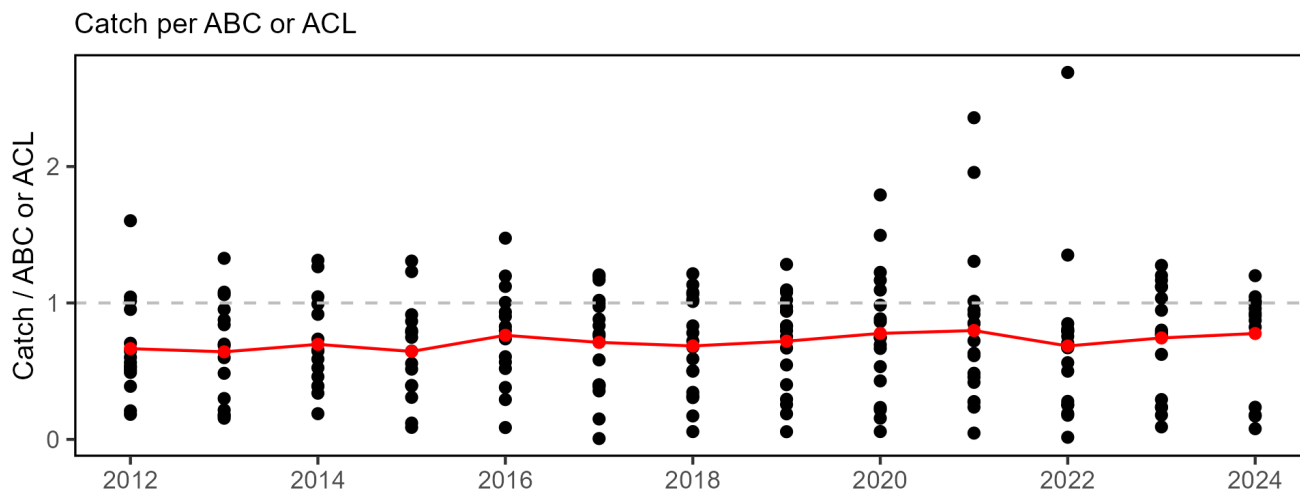


Figure 8: Total catch divided by ABC/ACL for MAFMC managed fisheries. High points are recreational black sea bass (2021) and scup (2022). Red line indicates the median ratio across all fisheries

**System Biomass** Major shifts in feeding guilds or ecosystem trophic structure are unlikely to be driving the decline in landings. **Aggregate biomass** trends derived from scientific resource surveys are mostly stable in the MAB, with long-term increases in spring piscivores, fall benthivores, and fall benthos (Fig. 9). While managed species make up varying proportions of aggregate biomass, trends in landings are not mirroring shifts in the overall trophic structure of survey-sampled fish and invertebrates. Future study should investigate whether shifts in the relative abundance of targeted species may be playing a role.

**Effect on Seafood Production** Stock status is not likely driving declining seafood production in the MAB, as all but two stocks are above the minimum threshold, and aggregate biomass trends appear stable or increasing. The decline in managed commercial seafood landings of surfclams and quahogs is most likely driven by market dynamics affecting landings, as the catch ratios have been relatively low for these species. The decrease in regional availability of scallops has contributed to the decline of benthos landings not managed by the MAFMC, with some of the most productive sea scallop fishing grounds closed through 2023 due to rotational management. The long-term declines in total and planktivore landings is driven in part by Atlantic menhaden fishery dynamics, including a consolidation of processors leading to reduced fishing capacity between the 1990s and mid-2000s.

Changes in the spatial distribution of surfclams and ocean quahogs may be constraining seafood production, as this results in areas with overlapping distributions and increased mixed landings. Mixed landings are currently prohibited by regulations, which could become problematic for harvesters. However, the MAFMC submitted an amendment in August 2025 to NOAA Fisheries to allow mixed surfclam and quahog trips; as of February 2026, the amendment remains under review by NOAA.

The recent decline in recreational seafood harvest is likely associated with a combination of targeted management actions, shifting social behaviors, and data collection changes. The decline in recreational shark landings can be attributed to management actions intended to reduce fishing mortality on mako sharks. The lower than average landings since 2018 for species other than sharks could be driven by either changes in fishing behavior or a change in NOAA Fisheries' Marine Recreational Information Program survey methodology in 2018. The decline in recreational seafood harvest may also be linked to decreases in sustenance fishing.

Future commercial and recreational landings are likely to be driven by environmental changes that require continued monitoring. Overall, the majority of landings from Mid-Atlantic ports is increasingly dependent on species with moderate environmental vulnerability. Fisheries and communities rely on different combinations of stocks, and individual stocks will respond differently to these drivers. Some key drivers include :

- **Unprecedented Climate Shifts:** Global ocean temperatures have reached record highs (see [2025 Highlights section](#)) and the Northeast US shelf has experienced long-term warming.
- **Distribution Shifts:** Stocks are shifting towards the northeast and into deeper waters throughout the Northeast US Large Marine Ecosystem (see [Climate Risks section](#)).
- **Ecosystem production:** Changes in ecosystem composition and biological production are impacting the stability of the **ecosystem** and pose **risks to setting catch limits**.
- **Community Risks:** Changes in the ecosystem can affect the **stability** of fisheries and pose risks to fishing communities.

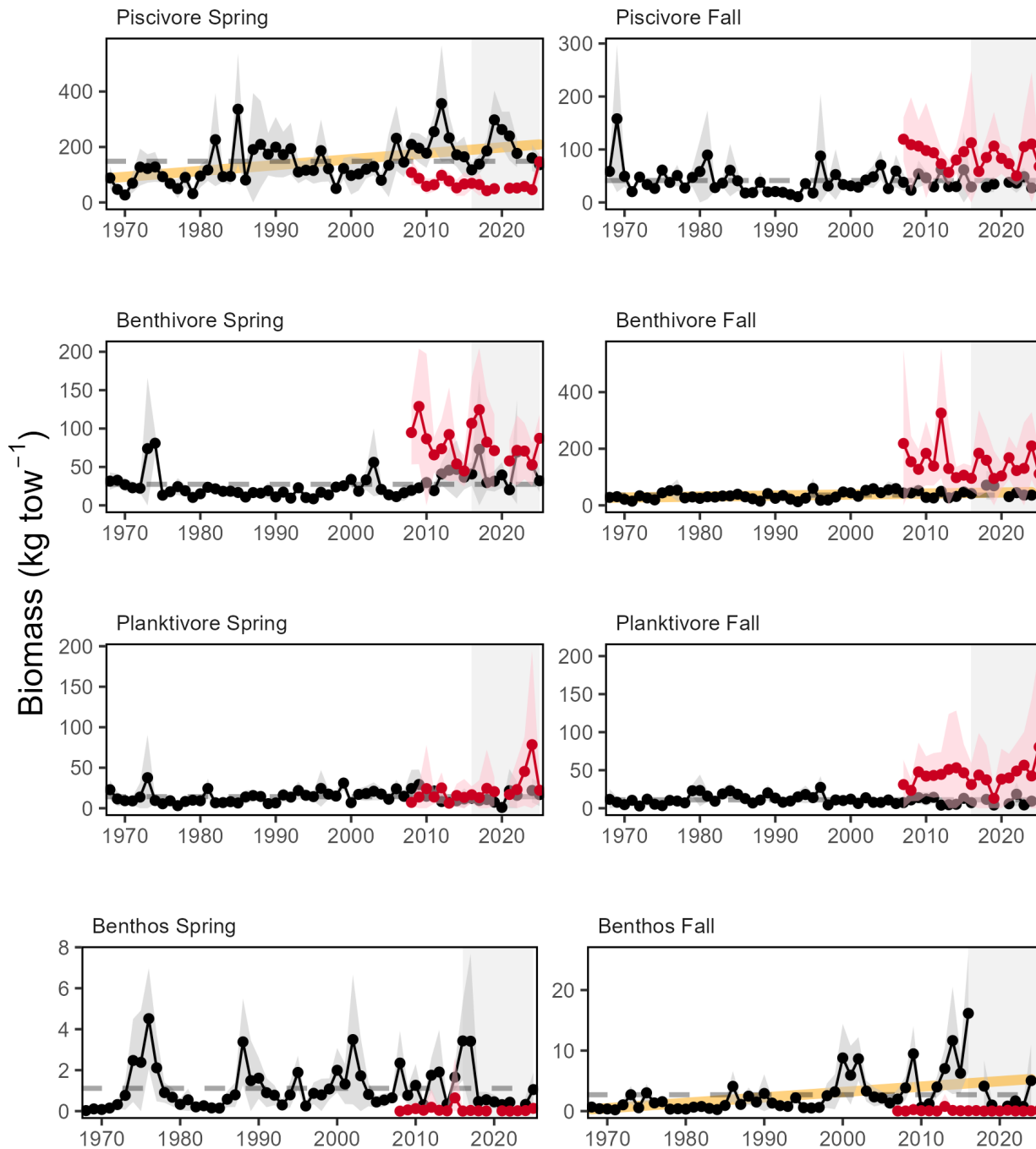


Figure 9: Spring (left) and fall (right) surveyed biomass in the Mid-Atlantic Bight. Data from the NEFSC Bottom Trawl Survey are shown in black, with the nearshore NEAMAP survey shown in red. Significant increases (orange lines) are present for spring piscivore and fall benthivore and benthos biomass. The shaded area around each annual mean represents 2 standard deviations from the mean.

## Commercial Profits

### Indicators: revenue & net revenue

Total [commercial revenue](#) and MAFMC managed species revenue (2024 USD) within the Mid-Atlantic Bight have declined over the past 20 years. In 2024, total revenue and MAFMC managed species revenue were both near an all-time low (Fig. 10).

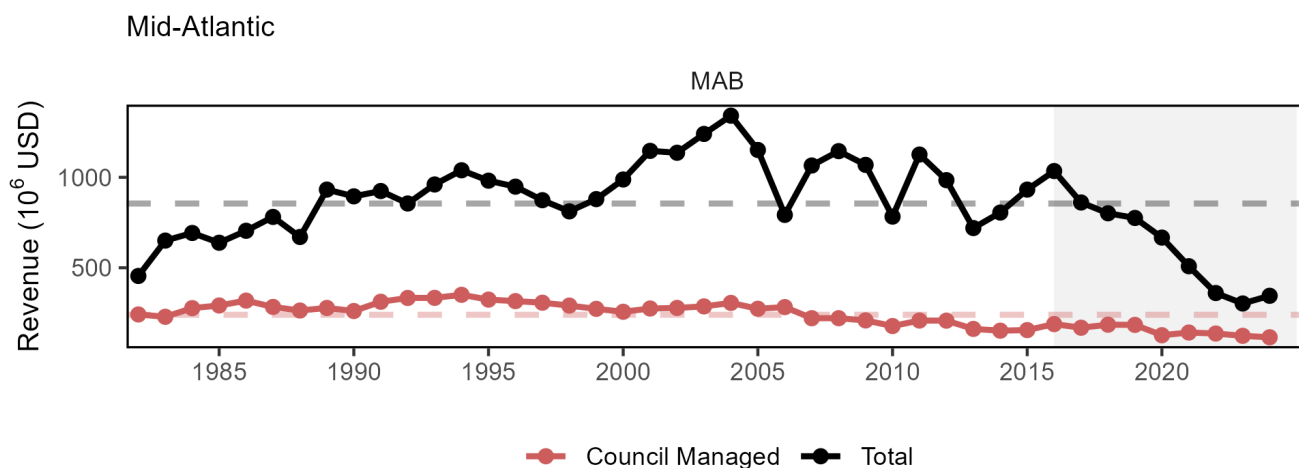


Figure 10: Commercial revenue (2024 USD) through 2024 for the Mid-Atlantic region: total (black) and from MAFMC managed species (red). Dashed lines represent the long-term annual mean.

Revenue earned by harvesting resources is a function of both the quantity landed of each species and the prices paid for landings. Therefore, total revenue patterns can be driven by harvest levels, the mix of species landed, price changes, or a combination of these. The [Bennet Indicator](#) (BI) decomposes revenue change into two parts, one driven by changing quantities (volumes), and a second driven by changing prices. All changes are in relation to a base year (1982). The 1982 base year was selected because that is the first year the relevant data is available and it allows for an extended period of time to evaluate market trends and dynamics. The BI results demonstrate that relatively high revenues in 2014-2016 were equally due to higher landings and prices (Fig. 11). In more recent years, both landings and prices have been closer to values from the reference year (1982). Low prices coupled with low volumes landed, led to low revenue in 2024. Recent lower than average revenues are partially due to declining prices of benthivores. Benthos prices increased from 2023, but overall benthos revenue remained low due to low volumes landed (Fig. 12).

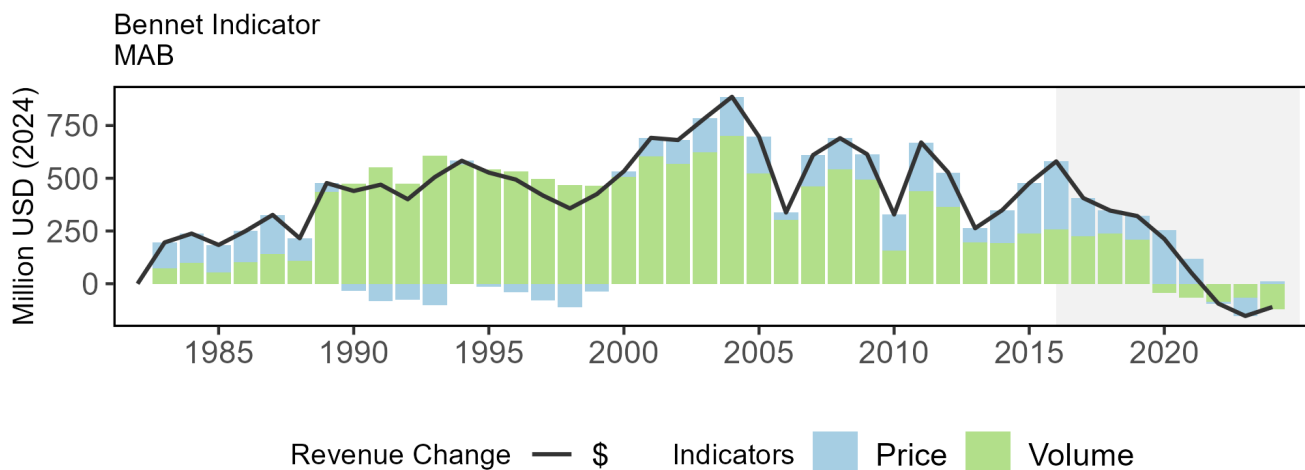


Figure 11: Revenue change from 1982 values in 2023 dollars (black); Price (blue), and Volume Indicators (green) for total commercial landings in the Mid-Atlantic Bight.

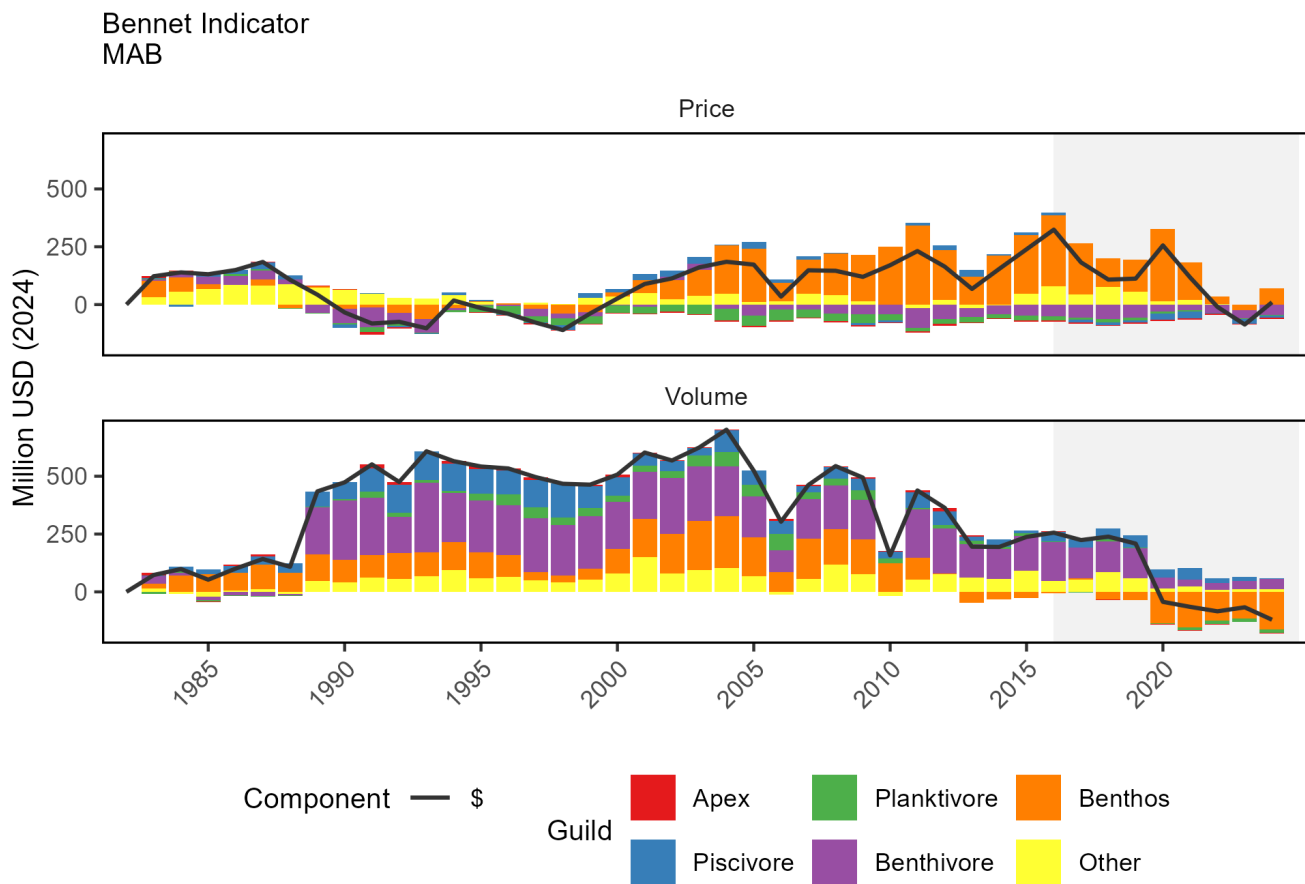


Figure 12: Total price (top) and volume (bottom) indicators in 2023 dollars (black) for commercial landings, and individual guild contributions to each indicator, in the Mid-Atlantic Bight.

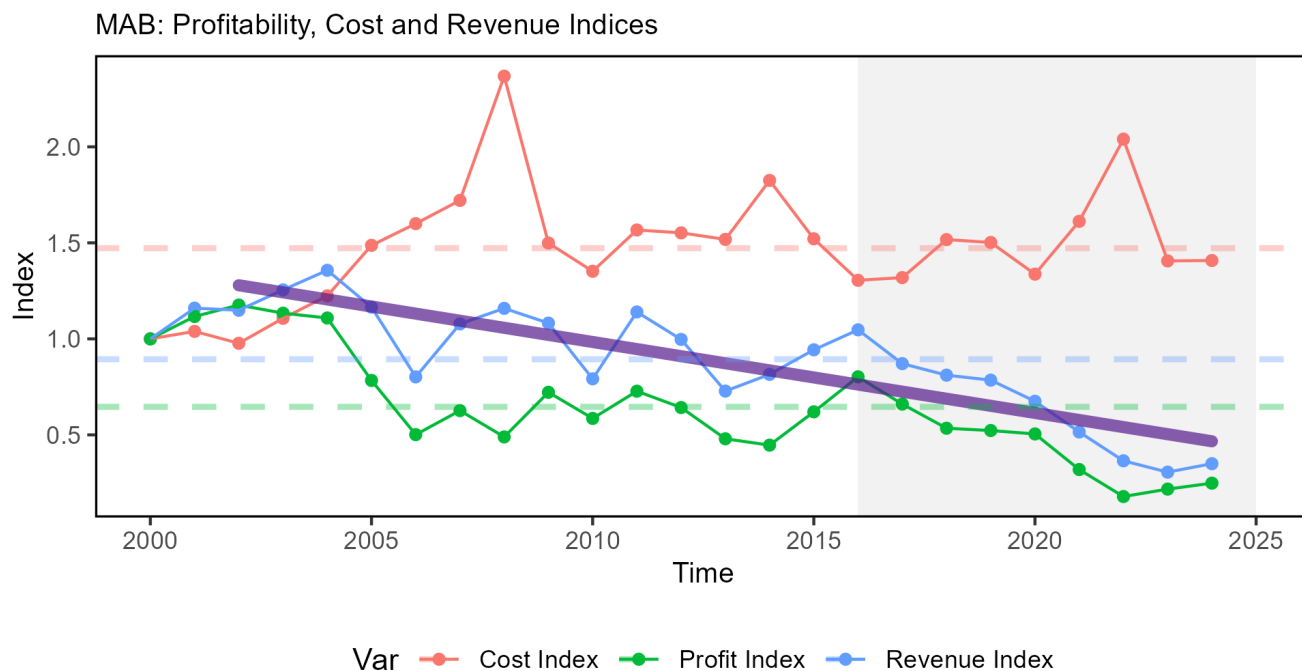


Figure 13: Profitability indices for Mid Atlantic federally managed species: cost index (red), profit index (green), and revenue index (blue). Dashed lines represent the long-term annual means for each index. Long-term declining trend associated with revenue index (blue).

This year, we present new indicators of [profitability](#): indices of cost, revenue, and profit based on trips catching federally-managed species. In this index, costs pertain to trip costs, excluding labor, estimated for all federal trips in the region. The profit indicator is net-revenue, determined as the difference between trip revenue and trip costs. Trips were spatially allocated to compile regional indices. Indices are presented as values relative to 2000, the first year in the dataset. In the Mid-Atlantic, costs have fluctuated, but overall remain near the time series mean, despite some high costs in 2022, 2014 and 2008. Revenue, however, has declined steadily since 2019 and is driving an overall decline in profits (Fig. 13).

For Mid-Atlantic ports, [total vulnerability](#) of revenue is high for the entire time series (2000-2024), with no long-term trend. This suggests that Mid-Atlantic port commercial fishing revenue is highly reliant on climate-sensitive species for the entire time period assessed (Fig. 14).



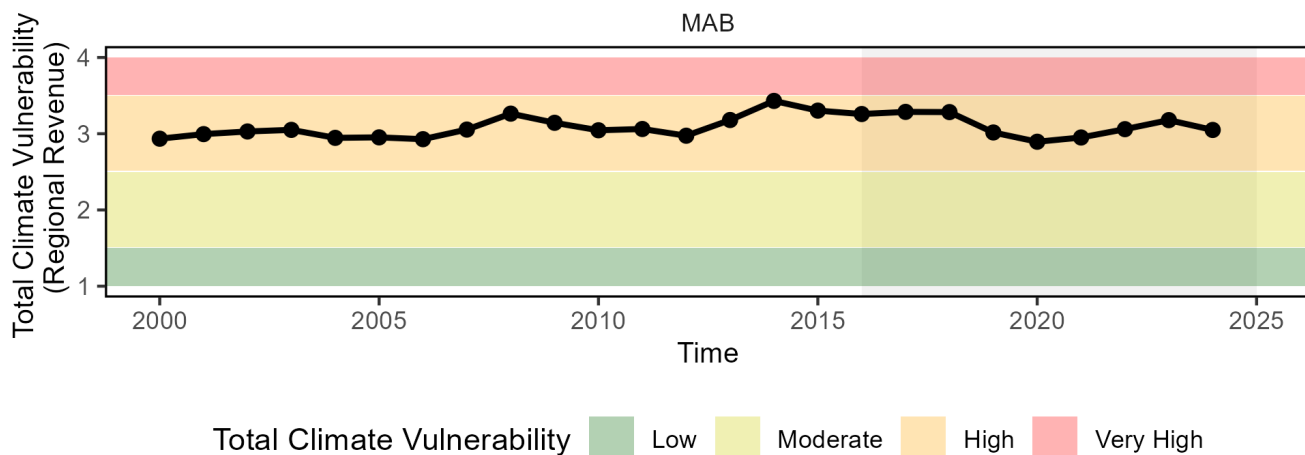


Figure 14: Mid-Atlantic region total climate vulnerability of commercial revenue (sum of Mid-Atlantic port revenue weighted by species climate vulnerability from Hare et al. 2016). Horizontal colored bars show different climate risk levels.

## Implications

Although the Mid-Atlantic region shows declining revenue since 2016, inflation-adjusted revenue from harvested species was still greater than 1982 levels until the past two years (Fig. 11). However, revenue from MAFMC-managed species has been below 2000 levels in several of the past 24 years (Fig. 13). The Bennet Index demonstrates that this decline is driven by lower volumes and there was no increase in price to compensate. Declines in landings of surfclams and ocean quahogs since 2012 are a result of decreased landings per unit effort over the same period, which may reflect changes in surfclam and quahog aggregation or distribution patterns. Changes in other indicators, particularly those driving landings and those related to climate change, require monitoring as they may become important drivers of revenue in the future. Multiple stressors including [warming](#) and [ocean acidification](#) are interacting in Mid-Atlantic shellfish habitats, particularly for surfclams, ocean quahogs, and scallops. This is reflected by the high environmental risk for landings from Mid-Atlantic ports.

## Recreational Opportunities

### Indicators: Angler trips, fleet diversity

**Recreational effort** (angler trips) in the MAB increased from 1982 to 2010, but has since declined to near the long-term average (Fig. 15). However, there is a long-term declining trend in recreational fleet diversity (i.e., effort by shoreside, private boat, and for-hire anglers) (Fig. 16). Billfish landings were notably high in 2025 (See 2025 Highlights Section), but long-term time series are in development.

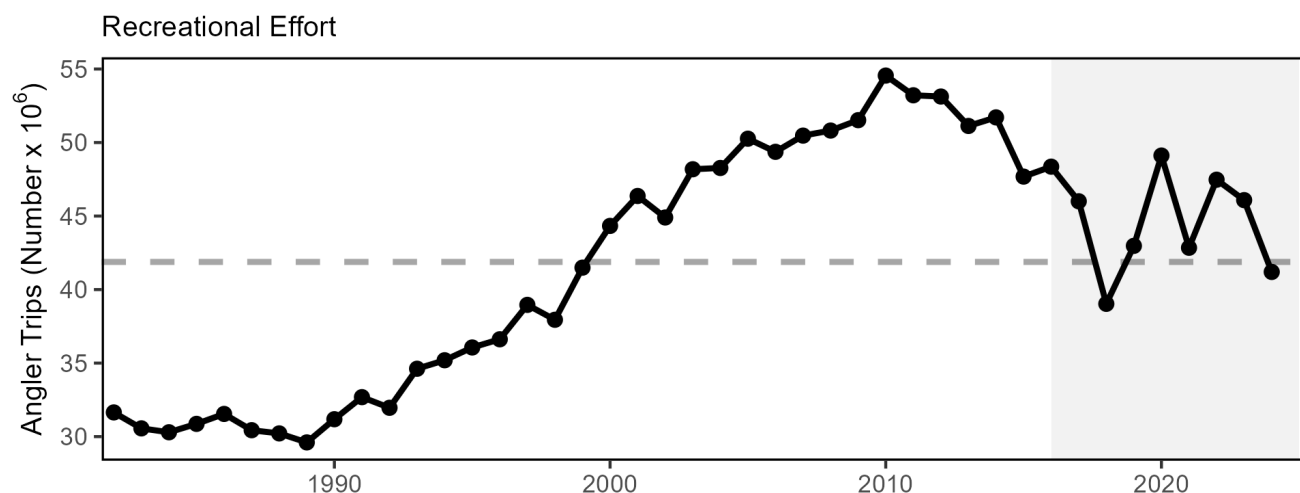


Figure 15: Recreational effort (total number of recreational angler trips from 1980-2023, black) in the Mid-Atlantic. Derived from MRIP's Effort Time Series Query.

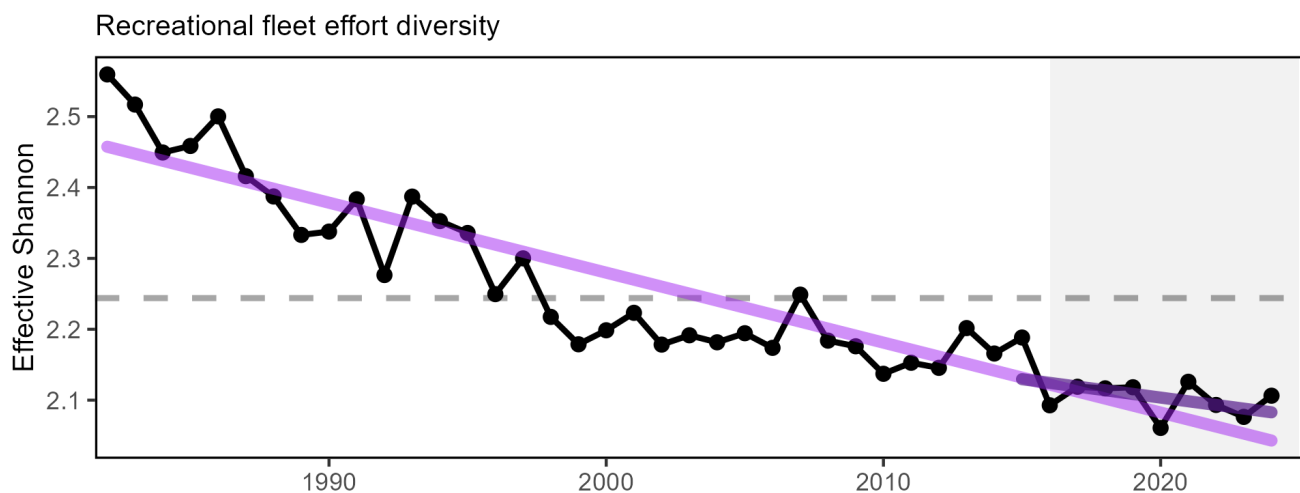


Figure 16: Recreational fleet effort diversity from 1980-2023 (black) in the Mid-Atlantic, with significant decrease in long-term (light purple) and short-term (dark purple) trends.

### Implications

While the overall number of recreational trips in the MAB is above the long-term average, the continuing decline in recreational fleet effort diversity suggests, at least in part, changes in angler behavior. Future study is required

to determine whether and to what extent the range and availability of recreational fishing options may drive these changes as well.

A contraction of party/charter trips (dropping from 2.2% of trips in 2021 to 1.3% in 2023) is the primary driver of the downward effort diversity trend, alongside a shift toward shorebased angling, which now makes up 60% of trips. Private boat effort has remained consistent to 2022 values.

Managers should consider the differing species and fish sizes for shore-based and vessel-based anglers. Some species use inshore regions as nursery grounds while other species only come inshore as adults. Many states have developed shore-based regulations where the minimum size is lower than in other areas and sectors to maintain opportunities in the shore angling sector. The MAFMC is currently considering recreational sector separation which might establish different options for managing the for-hire sector from other modes.

## Social and Community Risks

Fisheries management seeks to provide for sustained participation of fishing communities and to avoid adverse economic impacts to fishing communities. A new [composite indicator](#) (Port Commercial Fishing Activity Indicator or PCFA) utilizes NOAA data on dealers, fish landings, and commercial permits to explore trends in commercial fishing activity over time in top ports. This information can be used to understand how changes in fish stocks, regulations, and other social-ecological factors may have disparately impacted ports throughout the Greater Atlantic region.

The recreational engagement index has not been updated from last year and will be updated with similar methods as PCFAI in future reports. The recreational [engagement](#) index demonstrates participation levels in recreational fishing in a given community relative to other coastal communities in a region.

The Community Social Vulnerability Indicators (CSVI) utilize U.S. Census American Community Survey data to describe social characteristics at the municipality level (i.e., not just the fishing community) and provide context for the municipalities utilized by commercial fishing industry participants. Fishing industry participants that live in and/or utilize resources in municipalities with relatively concerning socio-demographic conditions may be more vulnerable to changes. The personal disruption index addresses factors that reduce adaptability to change such as unemployment or educational level. The poverty index is a composite index that indicates a community's financial standing relative to other communities. The population composition index characterizes groups within communities that may be more vulnerable to change. CSVI information for communities highlighted in the PCFA and recreational engagement index have been updated with the most recent census data.

Coastal fishing communities worldwide have or are likely to experience social, economic, and cultural impacts from climate change, both negative (e.g., loss of infrastructure, fish stock decline) and positive (e.g., increased abundance of valuable species). Changes in marine fisheries as a consequence of climate change will require adaptation by coastal fishing communities and fisheries managers alike. The Community Environmental Variability Risk Indicators (CEVRI) were developed to help examine trends in risk related to dependence on species vulnerable to climate and environmental changes.

### Indicators: Port Commercial Fishing Activity and Community Social Vulnerability

Six of the top ten 2024 communities are below in the [Port Commercial Fishing Activity Indicator \(PCFA\)](#) compared to their average scores from 2007-2011: Point Pleasant Beach, NJ; Ocean City, MD; Bronx, NY; Barnegat Light, NJ; Newport News, VA; Cape May, NJ. Of particular concern, Atlantic City, NJ and Newport News, VA both rank medium or higher for all three socio-demographic CSVIs, suggesting that fishing industry participants associated with this municipality may be more vulnerable to change. The other four top communities show higher port activity since 2007-2011; most notably Hampton Bays/Shinnecock, NY with an increase of 84%. Currently North Carolina communities are not presented due to data limitations.

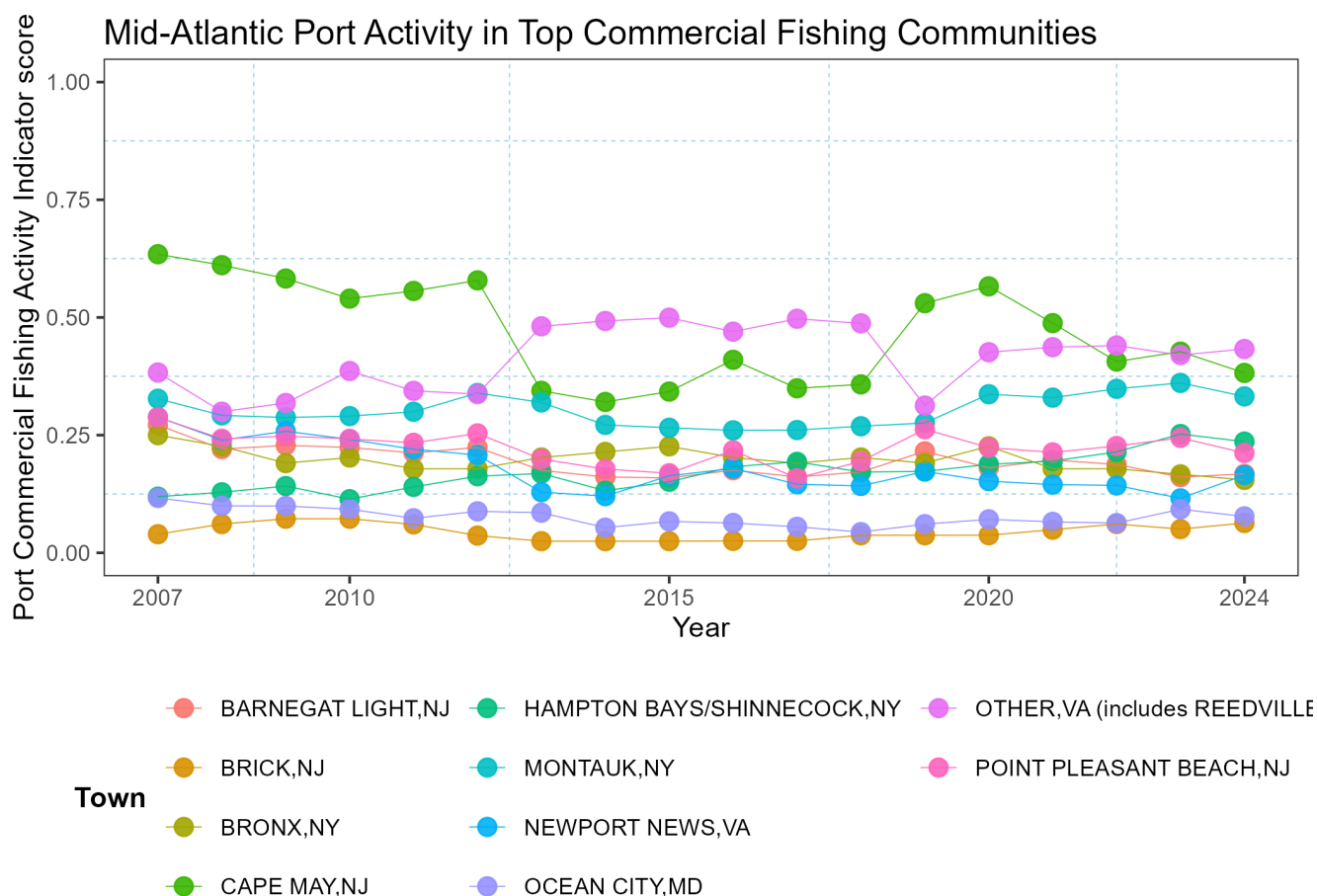


Figure 17: Port Commercial Fishing Activity Indicator scores over time with labels for the top commercially active fishing ports in the Mid-Atlantic.

Table 3: Socio-demographic indicator rankings (ranging from low = low vulnerability to high = high vulnerability) for Mid-Atlantic communities most engaged in commercial fishing, 2022. Blank spaces indicate no data available.

| Community                   | Personal Disruption | Population Composition | Poverty |
|-----------------------------|---------------------|------------------------|---------|
| Newport News, VA            | med                 | med high               | med     |
| Hampton Bays/Shinnecock, NY | low                 | med high               | low     |
| Ocean City, MD              | med                 | low                    | low     |
| Barnegat Light, NJ          | low                 | low                    | low     |
| Cape May, NJ                | low                 | low                    | low     |
| Point Pleasant Beach, NJ    | low                 | low                    | low     |
| Brick, NJ                   | low                 | low                    | low     |
| Montauk, NY                 | low                 | low                    | low     |

Of those included in the top-ranked recreational communities, both Morehead City, NC and Virginia Beach, VA had medium or higher ranks for at least one socio-demographic indicator (Table 4). This suggests that future changes

to recreational fishing conditions may disproportionately impact these places.

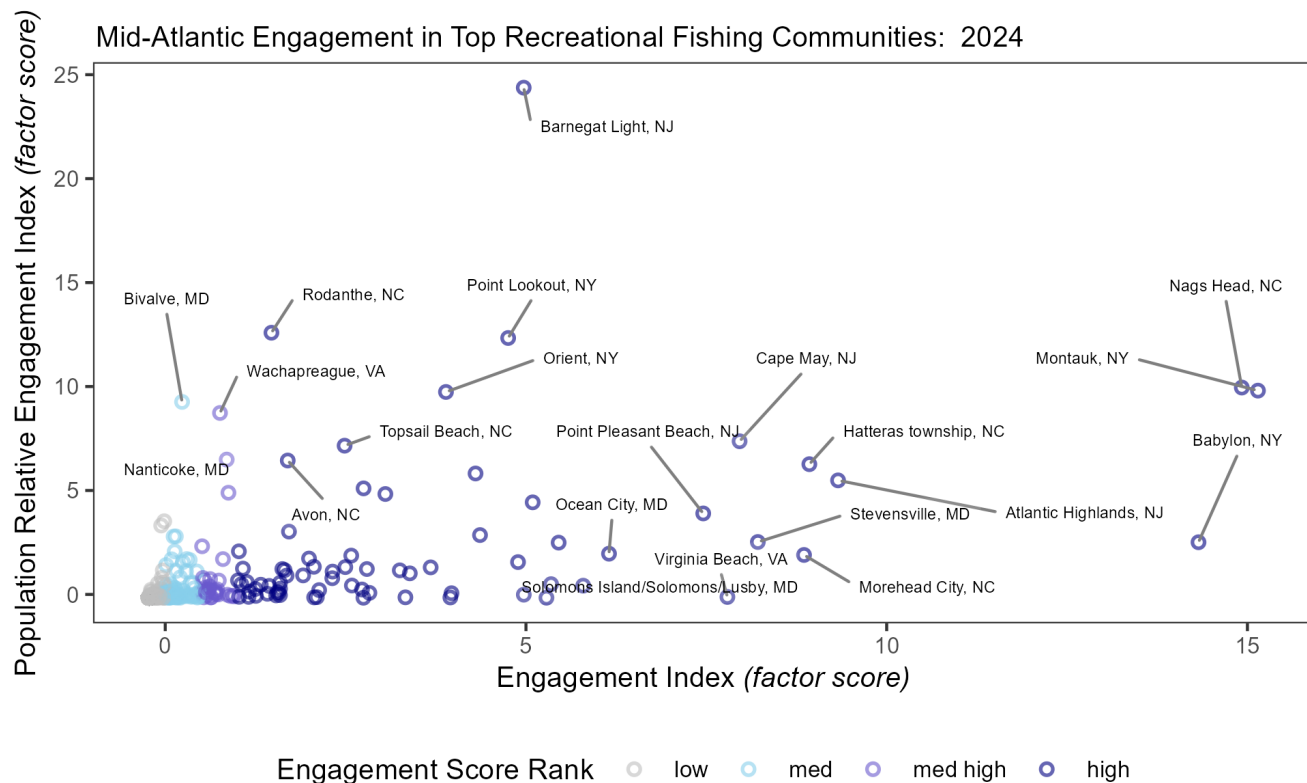


Figure 18: Recreational engagement and population relative engagement with labels for the top recreationally engaged fishing communities in the Mid-Atlantic (last updated 2023).

Table 4: Socio-demographic indicator rankings (ranging from low = low vulnerability to high = high vulnerability) for Mid-Atlantic communities most engaged in recreational fishing, 2022. Blank spaces indicate no data available.

| Community                | Personal Disruption | Population Composition | Poverty  |
|--------------------------|---------------------|------------------------|----------|
| Morehead City, NC        | med                 | low                    | med high |
| Virginia Beach, VA       | low                 | med                    | low      |
| Stevensville, MD         | low                 | low                    | low      |
| Nags Head, NC            | low                 | low                    | low      |
| Hatteras Township, NC    | low                 | low                    | low      |
| Atlantic Highlands, NJ   | low                 | low                    | low      |
| Cape May, NJ             | low                 | low                    | low      |
| Point Pleasant Beach, NJ | low                 | low                    | low      |
| Babylon, NY              | low                 | low                    | low      |
| Montauk, NY              | low                 | low                    | low      |

## Indicators: Community Environmental Variability Risk in the Mid-Atlantic

**Community Environmental Variability Risk Indicators** (CEVRI) measure risk by linking commercial landings and revenue to specific climate sensitivity factors, including temperature, ocean acidification, and stock status using the Climate Vulnerability Assessment (CVA) scores. These indicators calculate total sensitivity and vulnerability scores based on a community's dependence on species vulnerable to climate change. Risk scores range from low (1) to high (4), increasing as a community relies more heavily on species at higher risk from environmental shifts.

While long-term risk trends across the Mid-Atlantic remain stable, most individual fishing communities currently rank as high or very high risk. This high ranking demonstrates that a majority of regional communities depend on species that are highly vulnerable to changing ocean conditions for their commercial revenue. Strategies for management should account for this widespread reliance on climate-sensitive stocks.

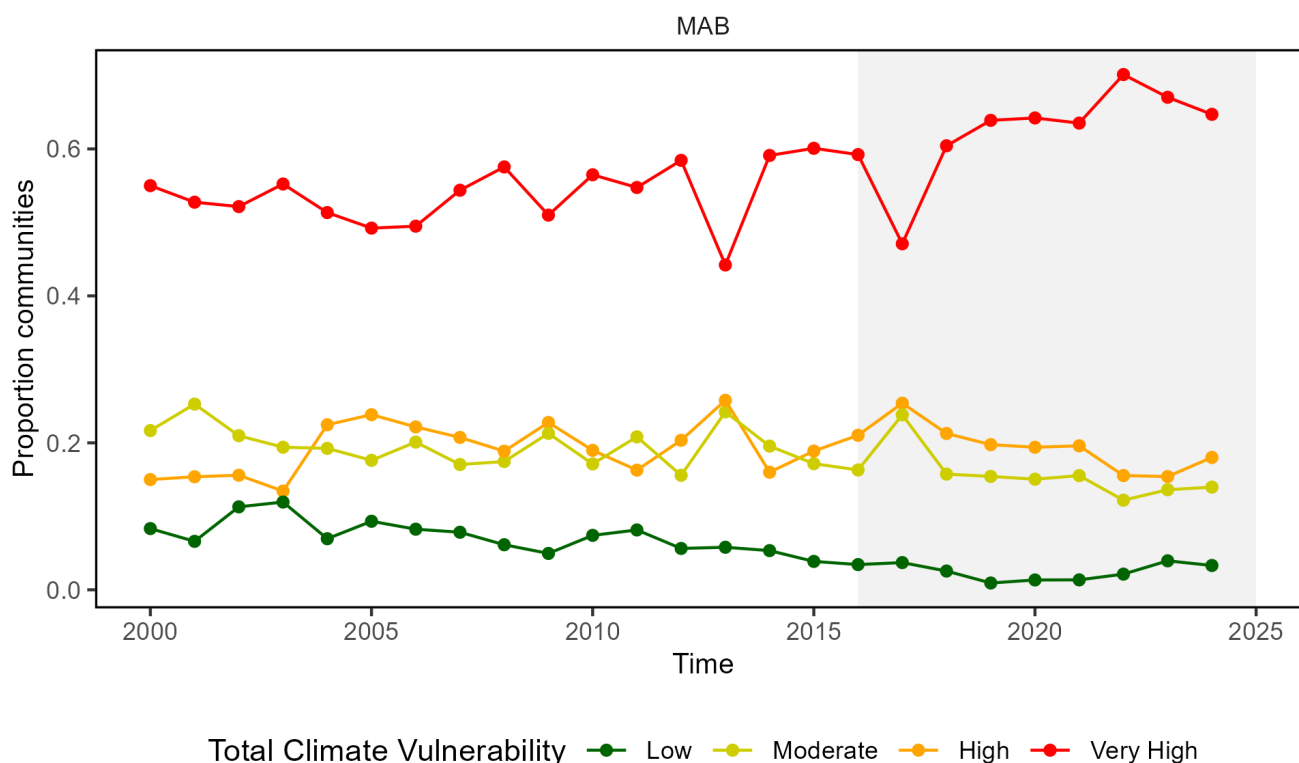


Figure 19: Proportion of Mid-Atlantic communities at each revenue climate vulnerability level over time. Total climate vulnerability ranges from low (green), moderate (yellow), high (orange), to very high (red).

## Implications

A range of socioeconomic and environmental variability risk concerns are found throughout Mid-Atlantic fishing communities, and the CSVI and CEVRI indicate socio-demographic concerns in the most highly active commercial Mid-Atlantic fishing ports. Fishing industry participants that utilize more vulnerable ports may be at increased relative risk to changes in fishing patterns due to regulations and/or ecosystem changes.

A majority of Mid-Atlantic communities have high to very high total environmental variability risk based on revenue. Coastal fishing communities are greatly affected by environmental change, both because of their physical location and because of their frequent social, cultural, and economic dependence on fishing. These impacts are expected to become more pressing as changes become more extensive. Changes in [ocean temperature](#) and [acidification](#) affecting marine life have the potential to directly impact fisheries and fishery dependent livelihoods.

## Protected Species

Fishery management objectives for protected species generally focus on reducing threats and on habitat conservation/restoration. Specific actions include managing bycatch to remain below potential biological removal (PBR) thresholds, recovering endangered populations, and monitoring unusual mortality events (UMEs). Protected species include marine mammals protected under the Marine Mammal Protection Act, endangered and threatened species protected under the Endangered Species Act, and migratory birds protected under the Migratory Bird Treaty Act. In the Northeast U.S., endangered/threatened species include Atlantic salmon, Atlantic and shortnose sturgeon, all sea turtle species, giant manta ray, oceanic whitetip shark, and five baleen whales. Here we report on performance relative to these objectives, as well as how observed and predicted ecosystem changes in the Northeast U.S may impact these objectives in the future.

### Indicators: bycatch, population (adult and juvenile) numbers, mortalities

The management objective for [harbor porpoise](#) has been met, as the average index (Fig. 20) remains below the current PBR threshold.

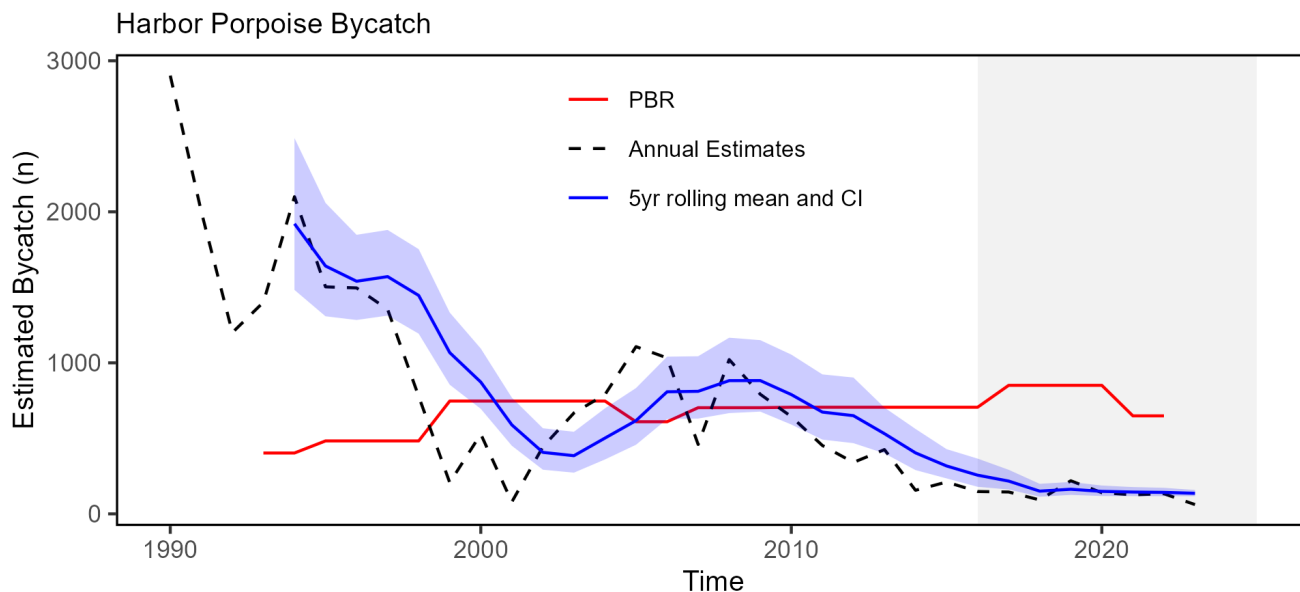


Figure 20: Harbor porpoise average bycatch estimate for Mid-Atlantic and New England gillnet fisheries (blue, confidence interval shaded) and the potential biological removal (red). The dashed line (black) represents the annual estimated bycatch.

The annual estimate for gray seal bycatch, most of which occurs in New England, has generally declined since 2019, in part driven by declining gillnet landings. Although, post-2019 estimates have greater uncertainty stemming from low observer coverage. The U.S. and Canadian range-wide PBR for [gray seals](#) is 12,052. Despite the PBR for the portion of this stock in U.S. waters being reduced to 756 animals bycatch (Fig. 21), but due to incomplete data on anthropogenic mortality and serious injury, bycatch for this stock is still considered unlikely to exceed the range-wide PBR. Thus the bycatch management objective gray seals has been met.



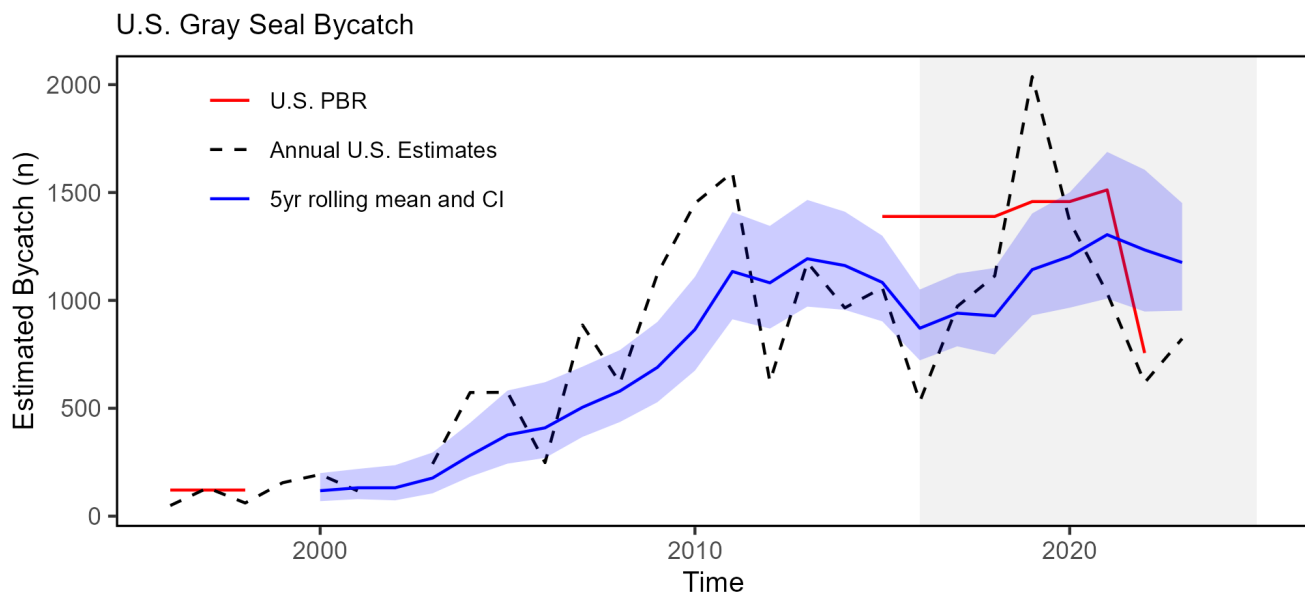


Figure 21: Gray seal five-year average bycatch estimate for New England and Mid-Atlantic U.S. gillnet fisheries (blue, with confidence interval shaded) and the potential U.S. biological removal (red). The range-wide PBR, including both U.S. and Canadian portions of the population, is 12,052 in the draft 2024 SAR. The dashed line (black) represents the annual estimated bycatch.

The [North Atlantic right whale population](#) was on a recovery trajectory until 2010, but has since declined (Fig. 22), slowing since 2020. The right whale population continues to experience annual mortalities above recovery thresholds. Reduced survival rates of adult females lead to diverging abundance trends between sexes. It is estimated that there are approximately 70 adult females remaining in the population.

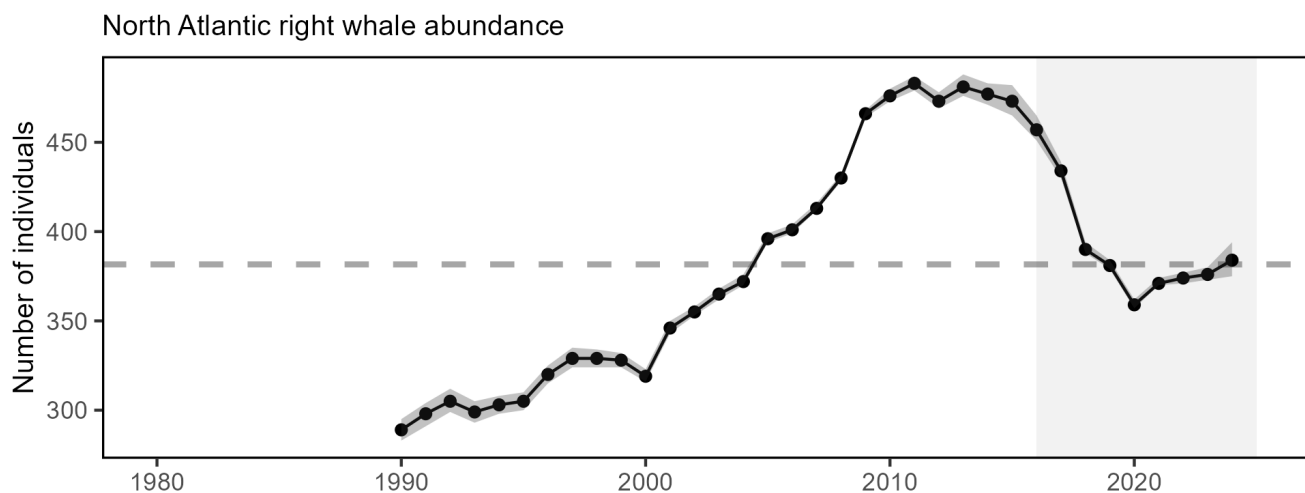


Figure 22: Estimated North Atlantic right whale abundance on the Northeast Shelf. 95% confidence interval shaded in gray around the line. Analysis is based on methods by Pace, Corkeron, and Kraus (2017), as documented most recently by Linden (2025).

North Atlantic right whale [calf counts](#) have generally declined after 2009 to the point of having zero new calves observed in 2018 (Fig. 23). However, since 2020, calf births have been closer to the long-term average, with 11 calves born in 2025.

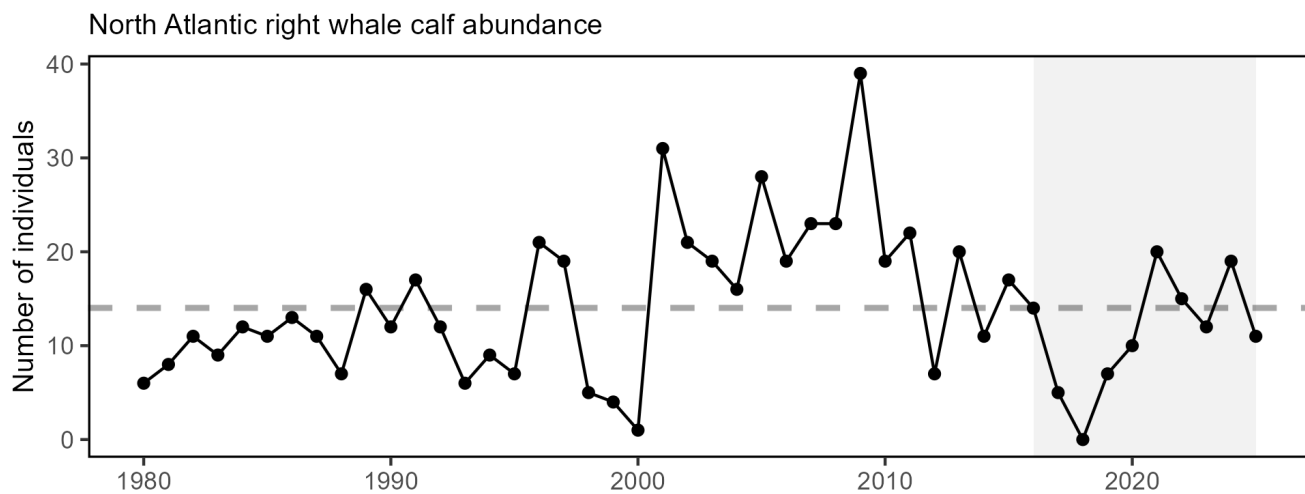


Figure 23: Number of North Atlantic right whale calf births since 1980. Calf birth estimates are available in Linden (2025).

Human interaction from entanglements or vessel strikes remains the primary cause of death in an ongoing North Atlantic right whale Unusual Mortality Event (UME) since 2017. As of 5 January 2026, the UME includes 168 individual whales: 41 confirmed mortalities (19 US; 22 Canada), 40 serious injuries, and 87 sublethal injuries or illnesses. Recent research suggests that many mortalities go unobserved and the true number of mortalities are about three times the count of the observed mortalities.

There is an ongoing UME for humpback whales (2016-present) and Atlantic minke whales (2018-present); suspected causes include human interactions. A UME for Northeast pinnipeds that began in 2018 for infectious disease is non-active pending closure as of February 2026.

## Implications

Bycatch management measures have been implemented to maintain bycatch below PBR thresholds. The downward trend in harbor porpoise bycatch could also be due to a decrease in harbor porpoise abundance in U.S. waters, reducing their overlap with fisheries, and a decrease in gillnet effort. The increasing trend in 5-year average gray seal bycatch may be related to an increase in the gray seal population ([U.S. pup counts](#)), supported by the dramatic rise over the last three decades in observed numbers of gray seal pups born at U.S. breeding sites plus an increase in adult seals at the breeding sites, some of which are supplemented by Canadian adults.

Strong evidence exists to suggest that interactions between right whales and both the fixed gear fisheries in the U.S. and Canada and vessel strikes in the U.S. are contributing substantially to the decline of the species. Further, right whale distribution has changed since 2010. [Recent research](#) suggests that recent climate driven changes in ocean circulation have resulted in right whale distribution changes driven by increased warm water influx through the Northeast Channel, which has reduced the [primary right whale prey](#) (the copepod *Calanus finmarchicus*) in the central and eastern portions of the Gulf of Maine. Additional potential stressors include offshore wind development, which overlaps with important habitat areas used year-round by right whales, including mother and calf migration corridors and foraging habitat. Additional information can be found in the [offshore wind risks section](#).

The UMEs are under investigation and are likely the result of multiple drivers. For all large whale UMEs, human interaction appears to have contributed to increased mortalities, although investigations are not complete.

A climate vulnerability assessment (Lettrich et al. 2023) is published for Atlantic and Gulf marine mammal populations.

## Stability

This year, we have updated the definition of stability for fisheries and ecosystems as a measure of how consistent we expect the system to be over time. Three components of stability are considered for the purpose of this report: volatility, adaptive capacity, and a shift from baseline. Volatility is a measure of predictability, where volatile conditions indicate that future years are more likely to be different than the recent past. Adaptive capacity refers to a system's ability to respond to changes without fundamentally changing its composition or structure. A shift from baseline refers to a systemic shift in a system towards a new status, where prior conditions may no longer be the norm. Measures of volatility are currently being developed. Therefore, we assess fisheries and ecosystem stability as “stable” if there is no notable change in adaptive capacity or shifts from a historic baseline, and “not stable” if there are changes in either of these components.

**Fishery Stability** Indicators suggest that Mid-Atlantic fisheries have broadly shifted from the historic baseline. Commercial fishery fleet count has declined while fleet revenue diversity has been stable over time in the MAB, but current values are above the long-term average (Fig. 24). [Revenue per unit effort](#) remains steady or increasing over time for most gear types, indicating financial viability of current fishing operations. This indicates that the commercial fleet composition has changed, but the portfolio of species targeted is similar over time (Fig. 25). Target species such as Atlantic mackerel and quahog have had reduced catch limits in recent years, resulting in reduced landings in these fisheries, and a decline in scallop catch within the MAB has severely reduced the total revenue generated in the region. Because non-MAFMC managed landings and revenue have declined, a larger share of the regional landings and revenue come from Council-managed fisheries.

The [Crew Survey](#) shows that specific aspects pertaining to sustainability and resilience of the fishing lifestyle are declining: predictability of earnings, the amount of time away from home, the physical fatigue of the job, and the personal health impacts have all been cited as dissatisfaction rates increase. Overall job satisfaction remains relatively stable over time, but unveils vulnerability as additional survey results show an aging population, particularly an increase in the 55+ crew cohort, and fewer individuals entering the fishery. This suggests a reduced capacity for Mid-Atlantic commercial fisheries to adapt to future uncertainties and change. New [Communities at Sea](#) indicators that assess fishing communities' ability to adapt to change are in development and will provide additional fishing industry indicators in future reports.

Despite reduced [recreational landings](#) (Fig. 4), the number of recreational trips is near average (Fig. 15), suggesting a shift to catch-and-release fishing. Billfish (i.e., white marlin) catch-and-release was especially high, possibly due to shifting effort due to the closure of the recreational bluefin tuna fishery in August 2025. Shark and large sport fish regulations, the right environmental conditions, and other circumstances may also contribute to reduced recreational landings. As noted above, [recreational fleet effort diversity](#) is declining (Fig. 16), suggesting a shift in recreational fishing opportunities. The Mid-Atlantic has experienced a contraction of the party and charter sectors, with more recreational angling occurring from shore. Recreational species catch diversity has no long-term trend and has been at or above the long-term average since 2016 (Fig. 26), indicating that anglers continue to catch a mix of species.

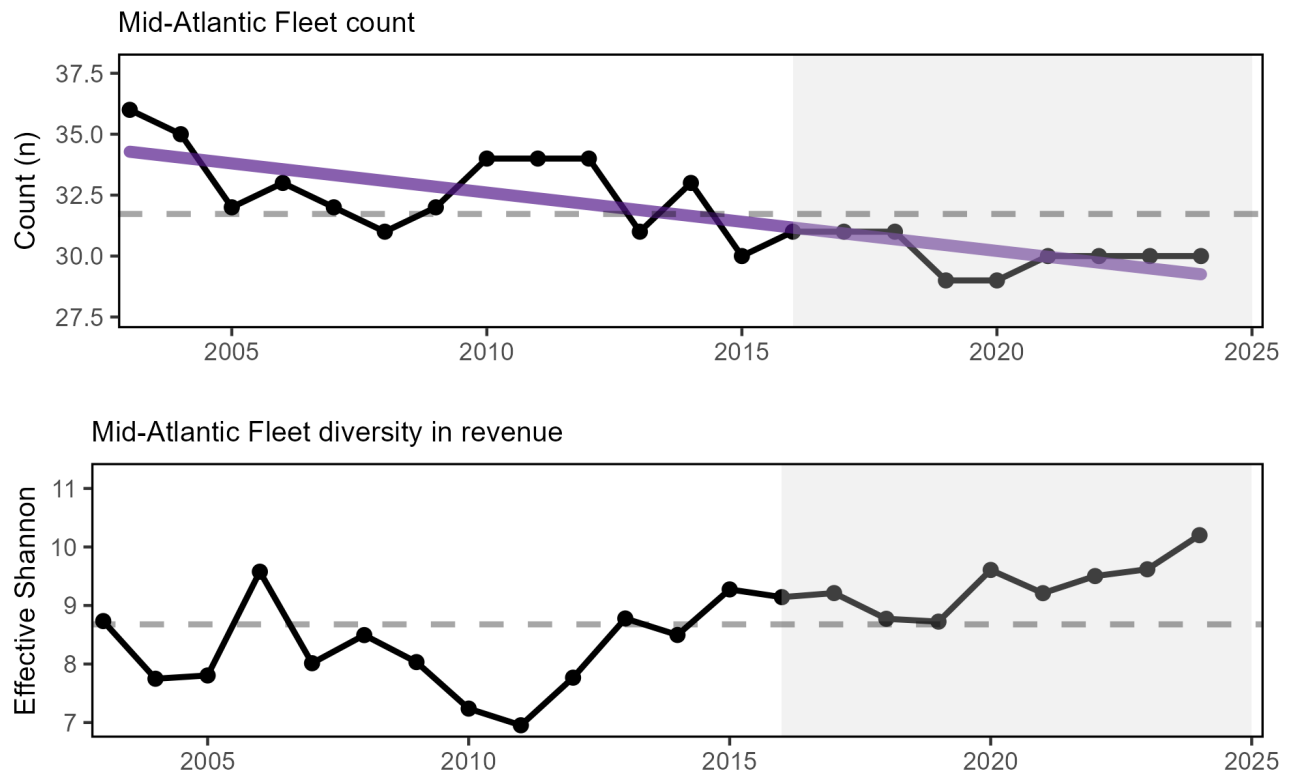


Figure 24: Commercial fleet count (top) and fleet diversity in revenue (bottom) in the Mid-Atlantic (black) with significant decline in fleet count (purple line).

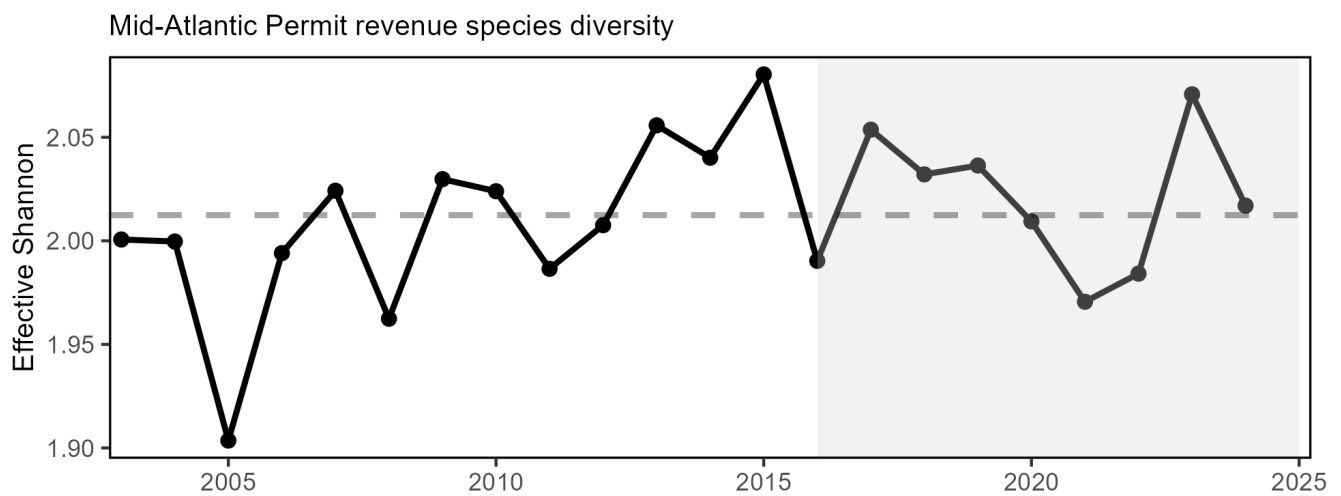


Figure 25: Species revenue diversity (permit-level species effective Shannon index) in the Mid Atlantic.

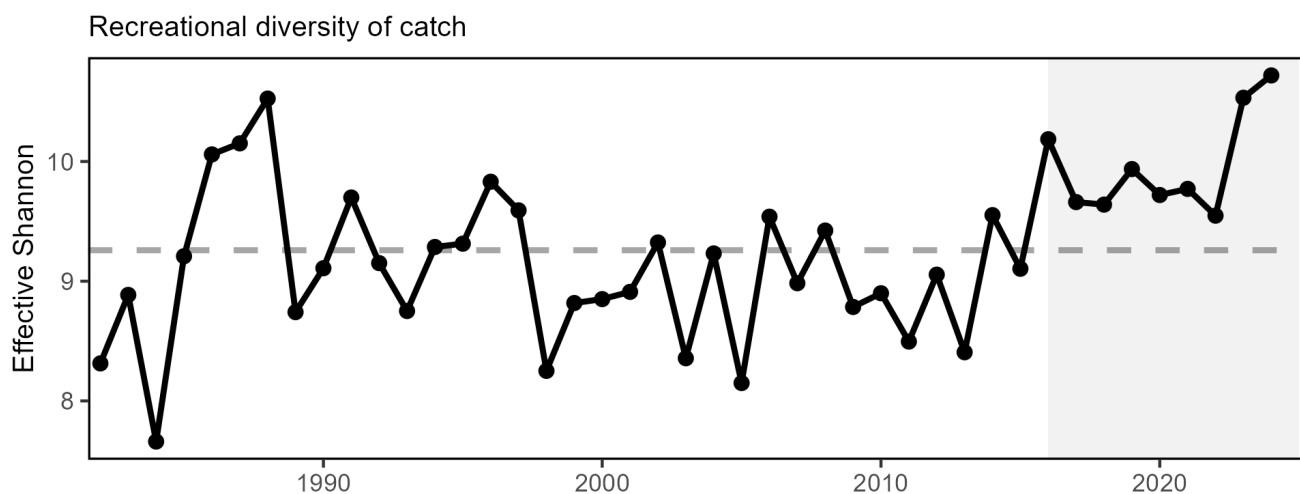


Figure 26: Diversity of recreational catch in the Mid Atlantic. Derived from MRIP's Catch Time Series Query.

**Ecological Stability** Long-term changes in biological processes suggest the Mid-Atlantic ecosystem is experiencing a systemic shift. Total annual [primary production](#), a measure of the total amount of carbon (i.e., energy) produced by phytoplankton per year, has no clear trend (Fig. 27), suggesting stability in energy at the base of the food web. However, we are monitoring for shifts in the phytoplankton community, which can affect the amount of primary production available to higher trophic levels. [Zooplankton diversity](#) is increasing in the MAB, and measures of zooplankton community composition also indicate a long-term shift in [zooplankton communities](#). Together, these indicators show a gradual but systemic change in lower trophic levels towards a community with a higher proportion of euphausiids and less dominated by copepods, which would not be expected in a stable ecosystem.

There are long-term increases in the biomass of the euphausiid, benthivore, and benthos [guilds](#) (Fig. 9). These lower trophic groups have similar roles within the ecosystem and these changes indicate a shift towards an ecosystem with a higher representation of those functional groups. [Adult fish diversity](#), the expected number of species in a standard number of individuals sampled from the NEFSC bottom trawl survey, appears stable over time, with current values within one standard deviation from most historic estimates (Fig. 28). This suggests that biomass increases in some guilds is due to an overall productivity increase rather than an influx of new species.

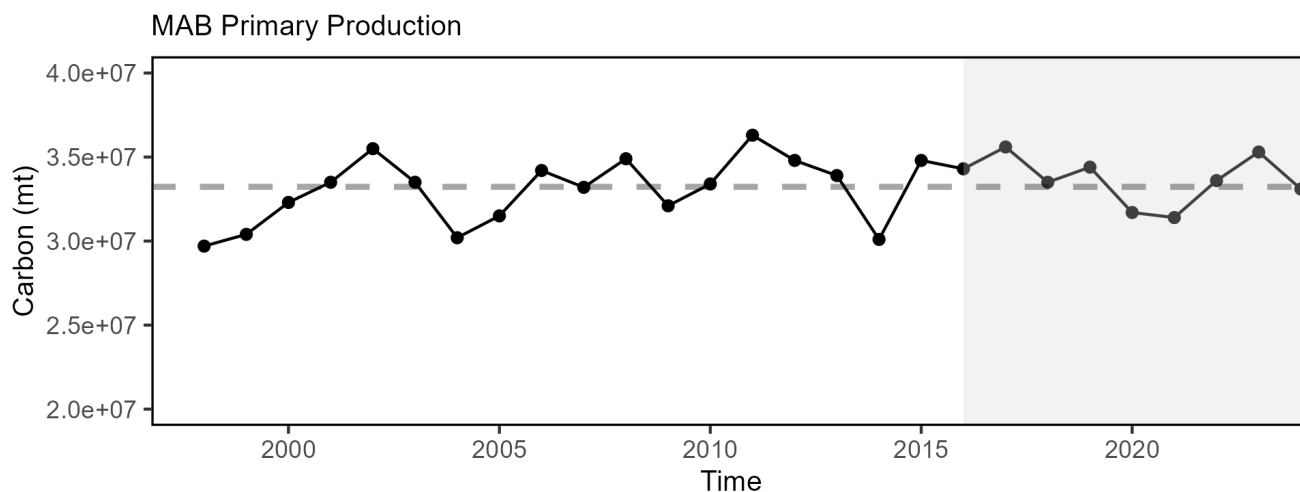


Figure 27: Total areal annual primary production for the MAB. The dashed line represents the long-term (1998-2024) annual mean.

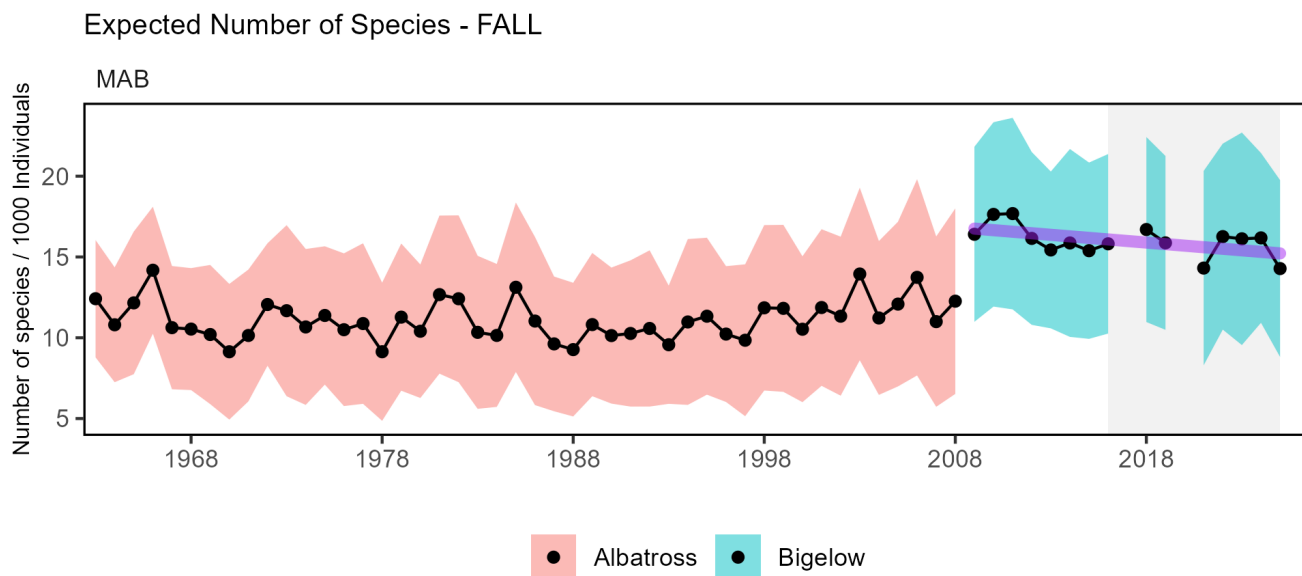


Figure 28: Adult fish diversity in the Mid-Atlantic Bight, based on expected number of species in a standard number of individuals. Results from survey vessels Albatross (red) and Bigelow (blue) are reported separately due to catchability differences.

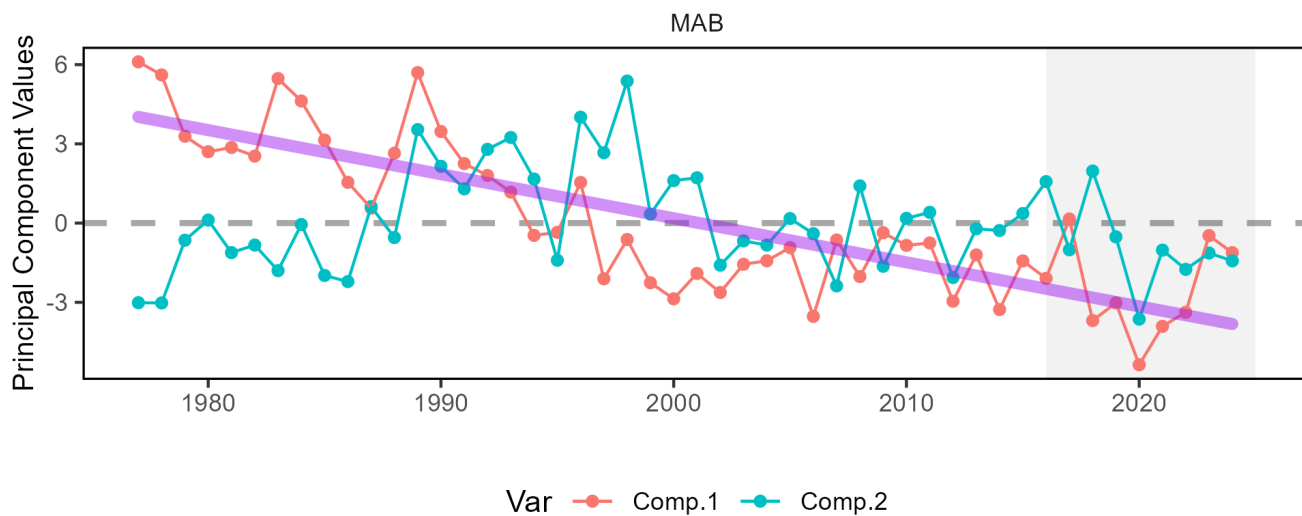


Figure 29: Principal component analysis of zooplankton community composition in the MAB. Lines show the first two principal components (colors). The declining trendline is associated with the first principal component. This trend is driven by an increasing abundance of sea butterflies, hyperiid amphipods, echinoderm larvae, arrow worms, and the copepod *Calanus minor*, and a decreasing abundance of the copepods *Pseudocalanus* spp., *Centropages hamatus*, *Acartia* spp., and *Temora longicornis*.

**Functional traits**, such as length at maturity, maximum body size, or fecundity, serve to synthesize change in complex, diverse communities by looking beyond species-specific trends. Furthermore, shifts in functional trait distributions for the fish community can indicate changes in ecosystem-scale resilience. There is evidence of long-term change in trait distributions in the MAB, particularly in the fall season (Fig. 30) (Fig. 31). The fall finfish community in the MAB is showing long-term shifts towards faster life history strategies with lower trophic levels, smaller offspring, younger age and shorter length at maturity, and faster growth rates. This indicates shifts in a system increasingly composed of smaller, fast-growing species. The long-term trends in the spring season are, however, more equivocal, with some evidence in shifts towards slower life history strategies, including larger length-at-maturity and offspring size. The lack of trend in finfish diversity suggests that these changes in fish communities are not due to a change in the total number of species.

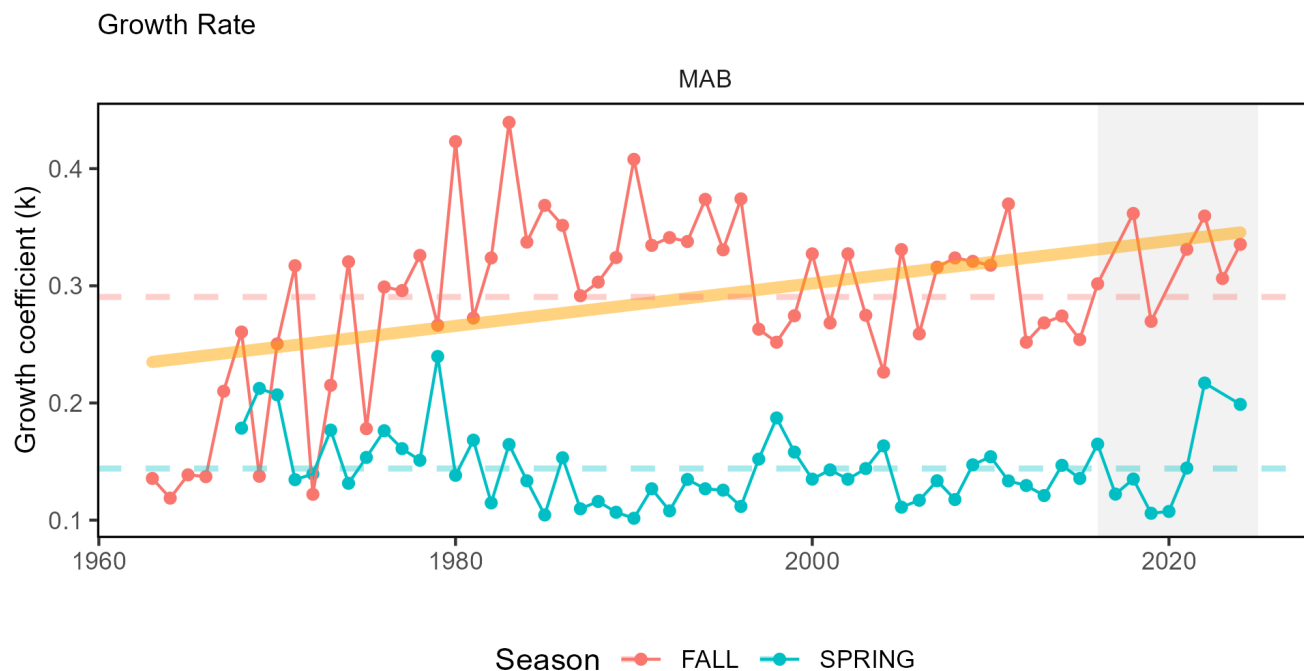


Figure 30: Fish community functional traits (growth rate) in the Mid Atlantic Bight based on Fall (red) and Spring (blue) survey data. Dashed lines represent the long-term annual mean for each survey.

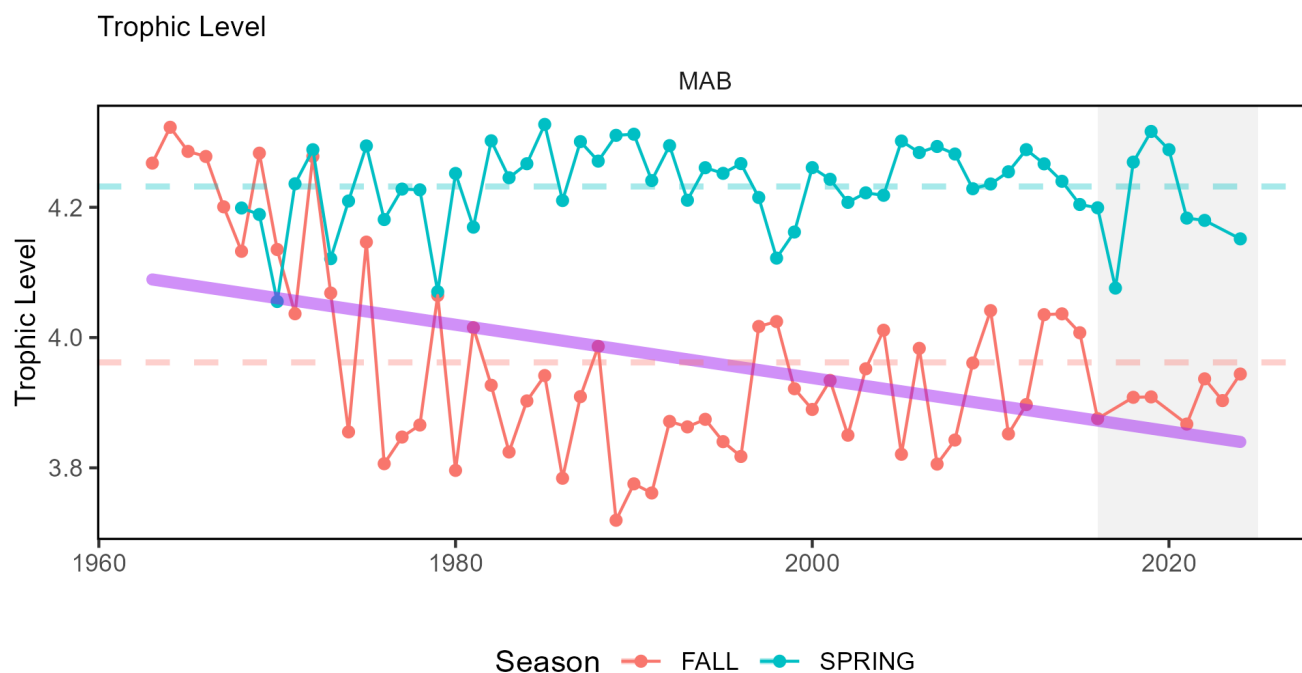


Figure 31: Fish community functional traits (trophic level) in the Mid Atlantic Bight based on Fall (red) and Spring (blue) survey data. Dashed lines represent the long-term annual mean for each survey.

## Implications

Fleet diversity indices are used by the MAFMC in their EAFM risk assessment to evaluate stability objectives, as well as risks to fishery resilience and maintaining equity in access to fishery resources. Instability in the commercial fleet count metric suggests lower capacity to respond to the current range of fishing opportunities. Commercial species permit revenue diversity is relatively stable (Fig. 25) but comparisons are limited by missing historical (pre-2003) clam fishery data.

Declining recreational fleet effort diversity indicates that the party/charter boat sector continues to contract, with shoreside angling becoming a greater percentage of recreational angler trips. Stability in recreational species catch diversity has been maintained by a different set of species over time. A recent increase in Atlantic States Marine Fisheries Commission (ASMFC) and South Atlantic Fishery Management Council (SAFMC) managed species in recreational catch is helping to maintain diversity in the same range that MAFMC and New England Fishery Management Council (NEFMC) managed species supported in the 1990s. These changes in effort and species trends may necessitate new or changing management considerations to ensure effective tools and opportunities are in place to support recreational fisheries.

Production at the base of the food web is variable, but stable over time. Mid-Atlantic species composition is changing, shifting towards a higher proportion of benthic and demersal fish. Stable adult fish diversity indicates the same overall number and evenness over time, but doesn't rule out species substitutions (e.g., warm-water species replacing cold-water species). There is evidence for long-term change in finfish trait distributions in the Mid-Atlantic.

In the MAB, both the fisheries and ecosystem are exhibiting long-term systemic shifts away from historical norms. While these changes don't appear abrupt like one would expect during a regime shift, they do indicate a potential change in baseline conditions.



## Risks to Meeting Fishery Management Objectives

### Climate and Ecosystem Change

#### Risks to managing spatially

Shifting species distributions, or (changes in spatial extent or center of distribution), alter both species and fishery interactions. In particular, shifting species distributions can affect expected management outcomes when spatial allocations and bycatch measures are based on historical fish and protected species distributions. Species availability to surveys can also change as distributions shift within or outside of survey footprints, complicating the interpretation of survey trends.

Coastwide indicators are reviewed in this section to evaluate spatial change throughout the Northeast US shelf. Indicators are identical between the Mid-Atlantic and New England reports.

**Indicators: Fish and protected species distribution shifts** As noted in the [Seafood Production Implications section](#), the combined center of [distribution](#) for 48 Northeast Shelf commercially or ecologically important fish species continues to show movement towards the northeast and generally into deeper water (Fig. 32). An analysis of recreational landings data from 2002 to 2019 found evidence of distribution shifts for several [highly migratory species](#), including sharks, billfish and tunas.

[Habitat model-based species richness](#) suggests shifts of both cooler and warmer water species to the northeast. Similar patterns have been found for [marine mammals](#), with multiple species shifting northeast between 2010 and 2017 in most seasons (Fig. 33).

Megabenthos center of gravity shows a short-term northward and eastward trend in spring (Fig. 34). Megabenthos are large, non-federally-managed benthic invertebrates sampled by scallop dredge, otter trawl, and the Campbell grab. These include crabs, decapods, and sea stars, which are often prey for many managed species.

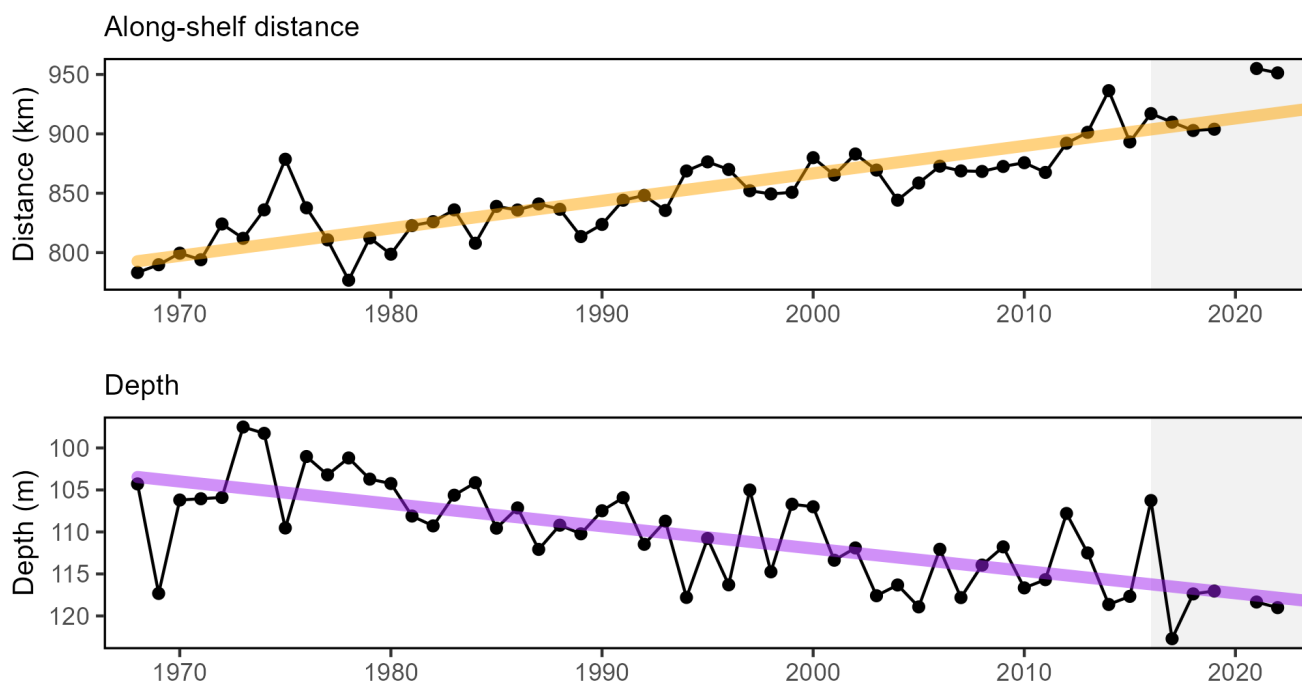


Figure 32: Aggregate species distribution metrics for species in the Northeast Large Marine Ecosystem: along shelf distance with increasing trend (orange), and depth with decreasing trend indicating deeper water (purple).

## Whale and Dolphin Distribution Shifts

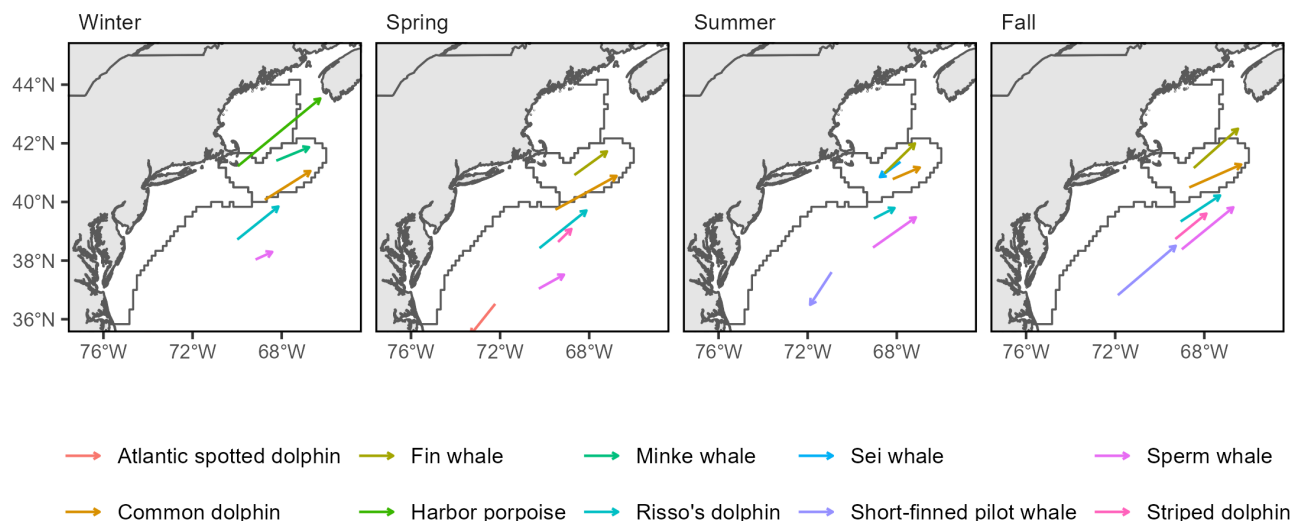


Figure 33: Direction and magnitude of core habitat shifts, represented by the length of the line of the seasonal weighted centroid for species with more than 70 km difference between 2010 and 2017 (tip of arrow).

## Northeast U.S. Megabenthos Distribution

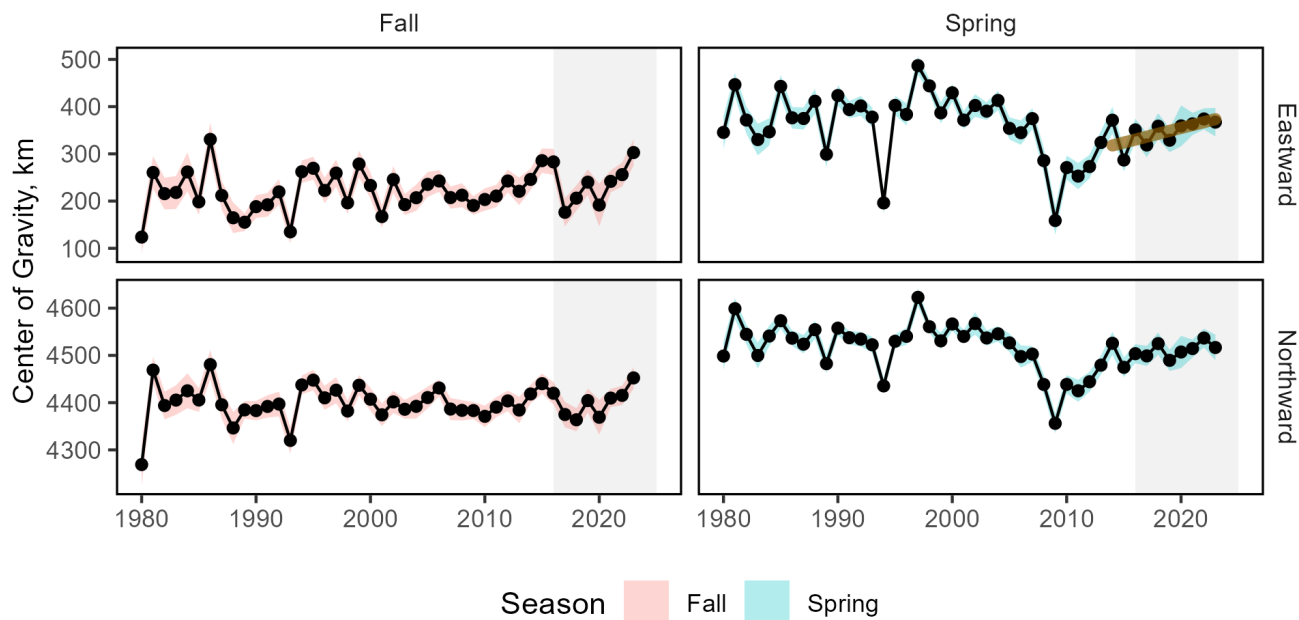


Figure 34: Eastward (top) and northward (bottom) shifts in the center of gravity for megabenthos species on the Northeast U.S. Shelf in fall (left) and spring (right), with recent increasing trend (orange) for spring eastward center of gravity.

**Drivers:** Mobile populations shift distributions to maintain suitable habitat and prey fields, possibly expanding ranges if new suitable habitat exists. Changes in managed species distribution is partially related to the [distribution of forage biomass](#). Since 1982, the fall center of gravity of forage fish (20 species combined) has moved to the north and east (Fig. 35). Spring forage fish center of gravity has moved northward but without an eastward trend. Some of the whale and dolphin distribution shifts (Fig. 33) are likely in response to these forage fish shifts. [Small copepods](#), widespread prey of many larval and juvenile fish, show a similar shift in center of gravity as forage fish, to the north and east in the fall, as well as northward in spring.

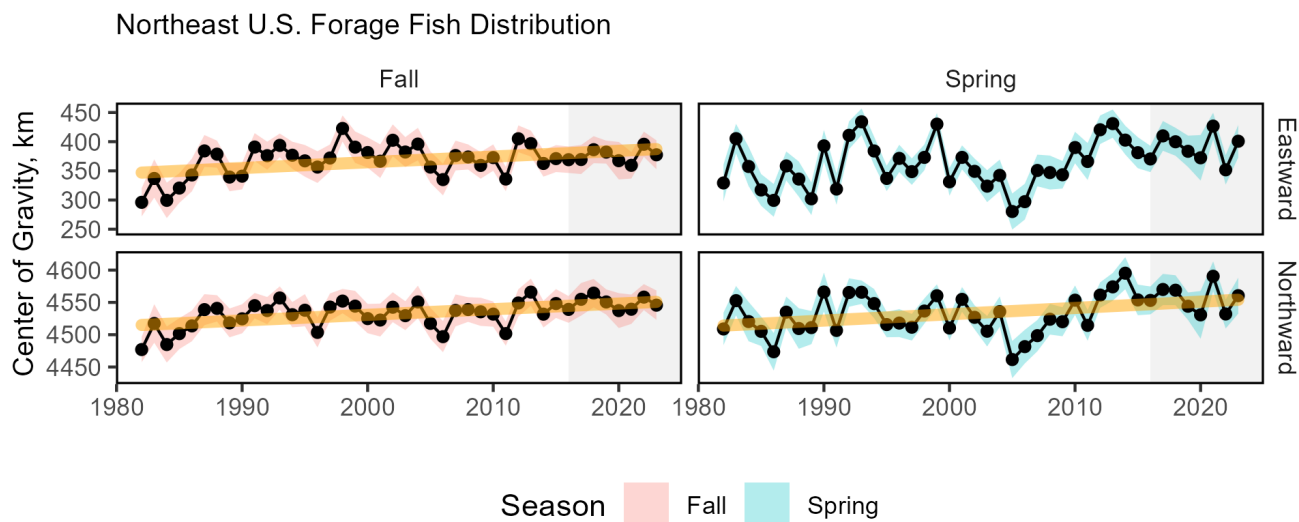


Figure 35: Eastward (top) and northward (bottom) shifts in the center of gravity for 20 forage fish species on the Northeast U.S. Shelf in fall (left) and spring (right), with increasing trend (orange) for fall eastward and northward and spring northward center of gravity.

In contrast, **macrobenthos** center of gravity has shifted west and south in the spring (Fig. 36). Macrobenthos are small bottom-dwelling invertebrates including polychaete worms, small crustaceans, bivalves (non-commercial), gastropods, nemerteans, tunicates, cnidarians, brittle stars, sea cucumbers, and sand dollars, and are prey for many managed species. **Large copepods** (including *Calanus finmarchicus*) and euphausiids do not have long-term trends in their centers of gravity (Fig. 37) (Fig. 38), but small copepods show shifts eastward and northward. Some targeted species distributions may shift in response to these shifts in forage, copepod, and macrobenthos distributions.

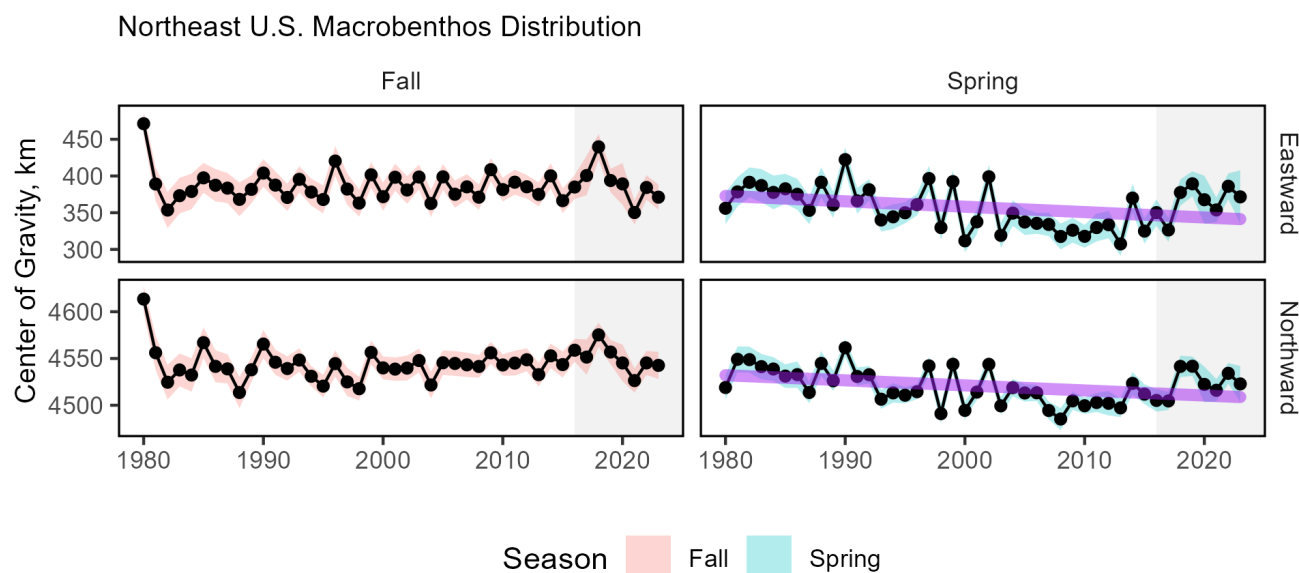


Figure 36: Eastward (top) and northward (bottom) shifts in the center of gravity for macrobenthos species on the Northeast U.S. Shelf in fall (left) and spring (right), with decreasing trend (purple) for spring eastward and northward center of gravity.

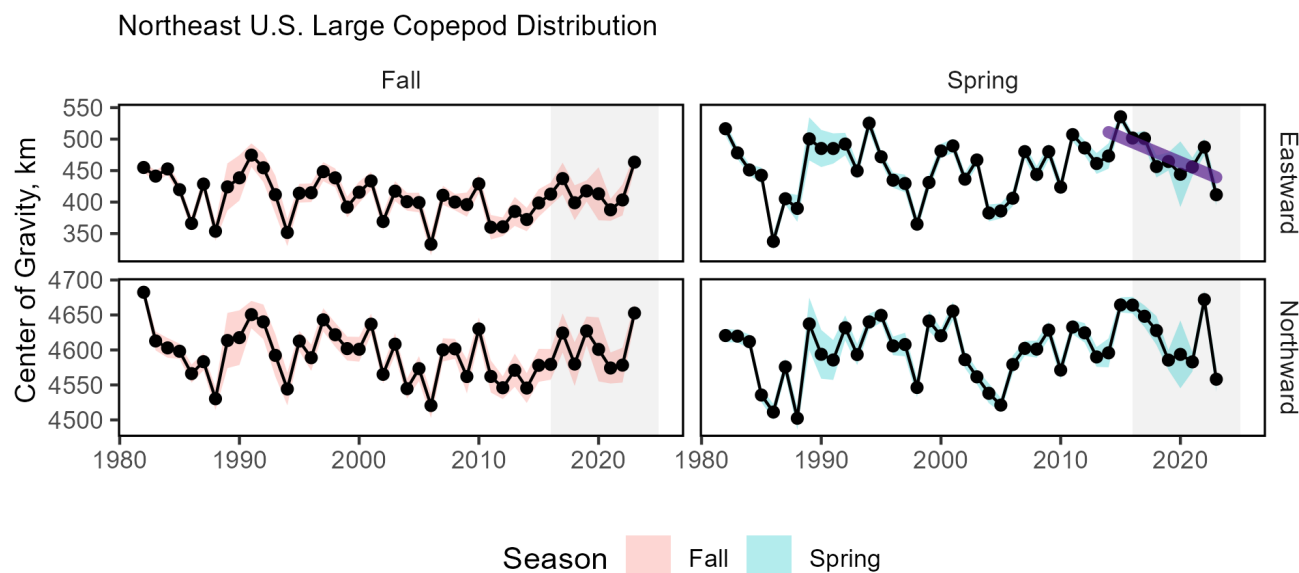


Figure 37: Eastward (top) and northward (bottom) shifts in the center of gravity for large copepod species on the Northeast U.S. Shelf in fall (left) and spring (right).

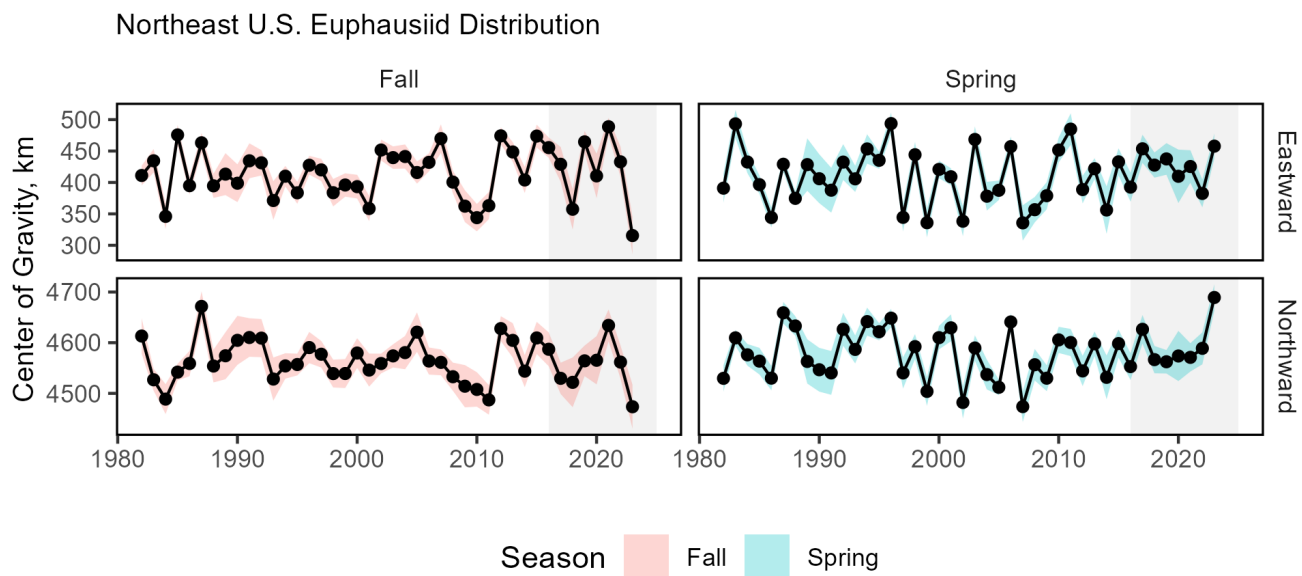


Figure 38: Eastward (top) and northward (bottom) shifts in the center of gravity for Euphausiid species on the Northeast U.S. Shelf in fall (left) and spring (right).

Ocean temperatures influence the distribution, seasonal timing, and productivity of managed species (see sections below). The Northeast US shelf, including the Mid-Atlantic, has experienced a continued warming trend for both the [long term annual sea surface](#) (Fig. 41) and [seasonal surface](#) and [bottom temperature](#). However, 2025 surface and bottom temperatures were near normal to cooler than normal conditions in all seasons in the MAB (see also the [2025 Highlights section](#)).

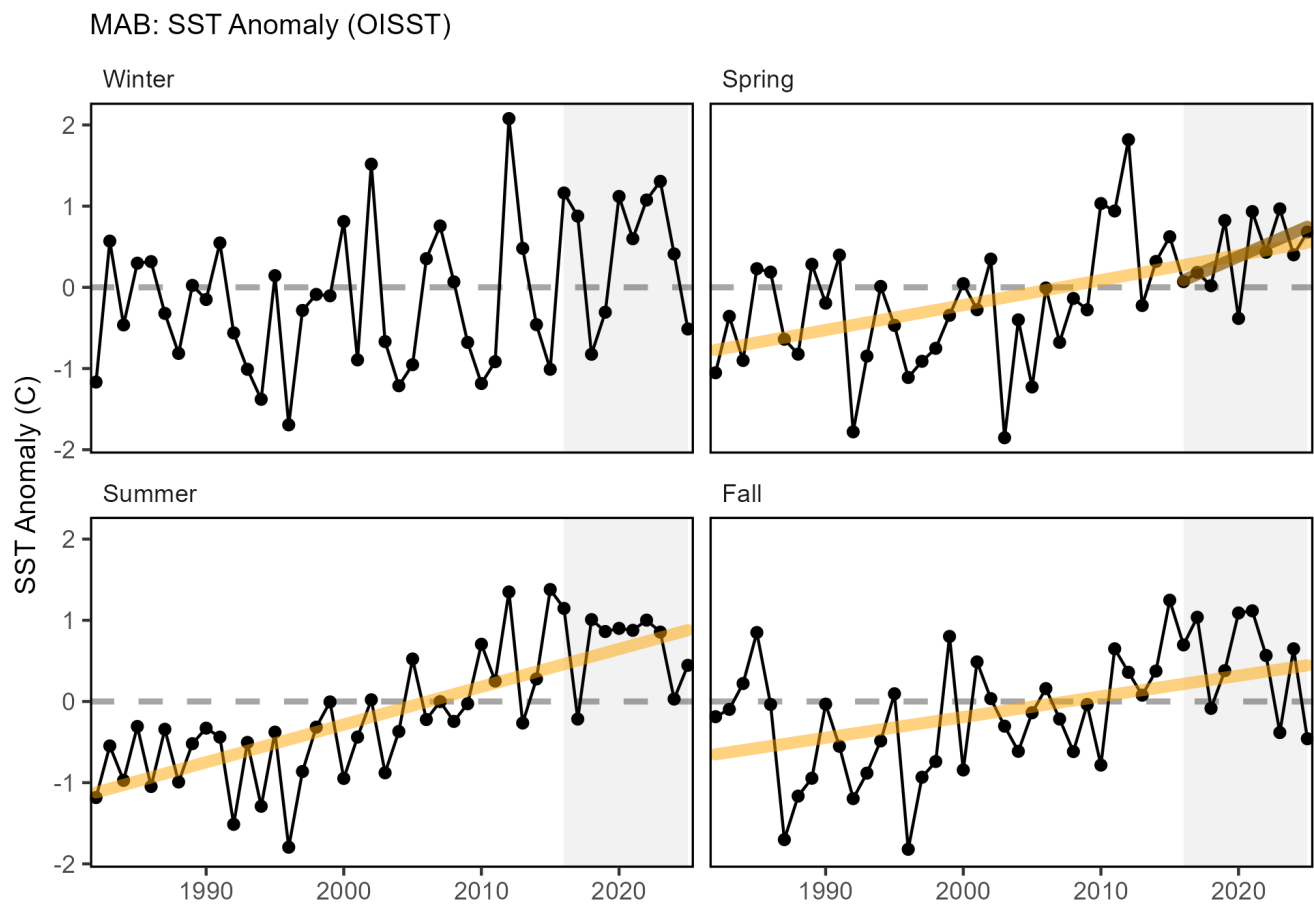


Figure 39: Seasonal Optimum Interpolation Sea Surface Temperature (OISST) anomaly by season for the MAB, with increasing trends (orange).

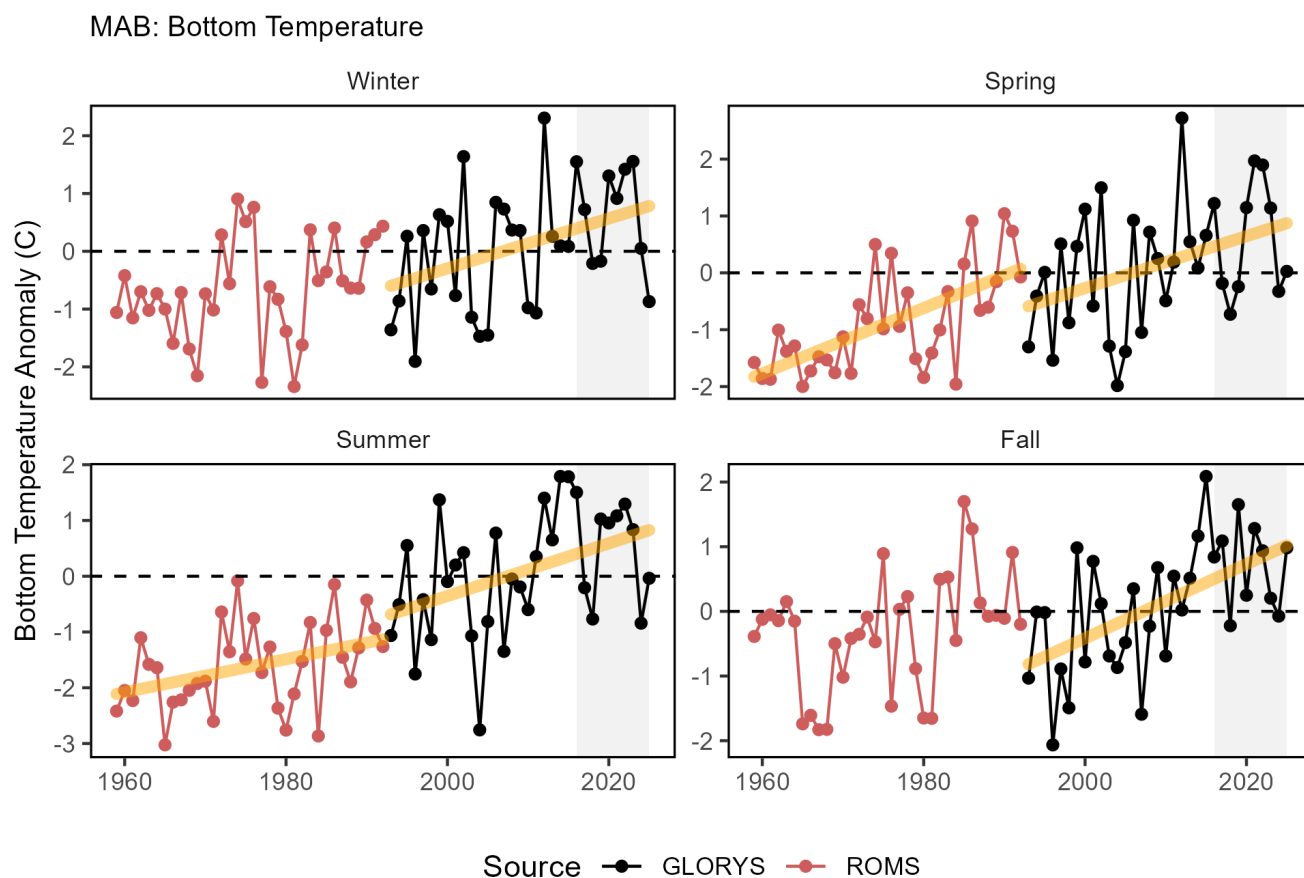


Figure 40: GLORYS (black) and debiased ROMS (red) seasonal bottom temperature anomaly in the MAB, with increasing trends (orange).

Species suitable habitat can expand or contract when changes in temperature and major oceanographic conditions alter distinct water mass habitats. The variability of the Gulf Stream is a major driver of the predominant oceanographic conditions of the Northeast U.S. continental shelf. As the [Gulf Stream](#) had become less stable and was shifting northward until a recent shift in 2023. Since then, the Gulf Stream has been closer to the long-term average, and the supply of Labrador Slope Water to the Northwest Atlantic Shelf has increased. These changes are linked to some of the cooler water temperatures observed in 2024 and 2025 and the composition of the source water entering the Gulf of Maine through the Northeast Channel (see [2025 Highlights](#)).

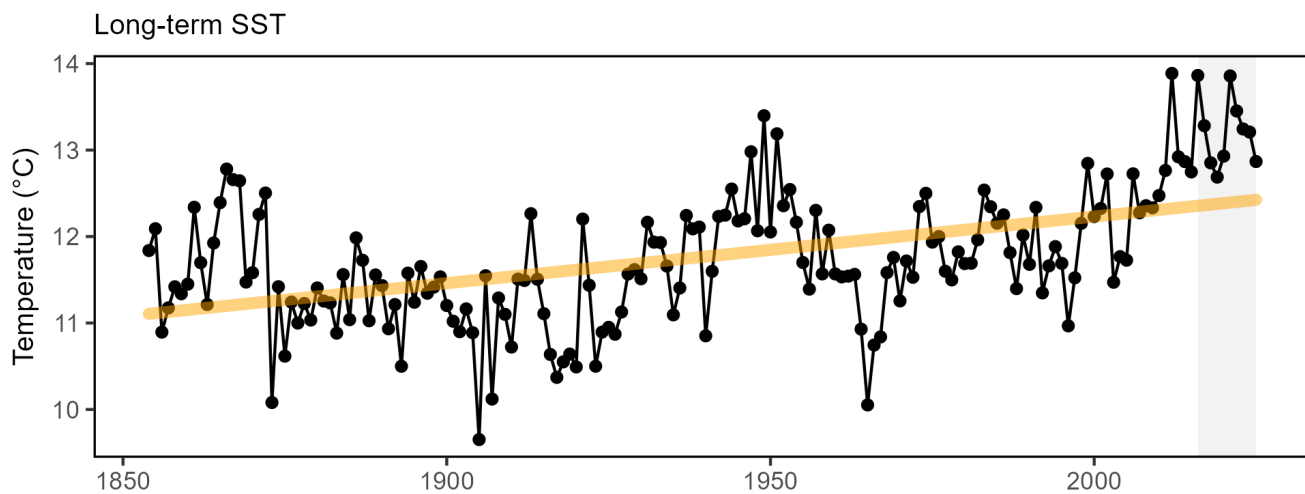


Figure 41: Northeast US annual sea surface temperature (SST, black), with increasing trend (orange).

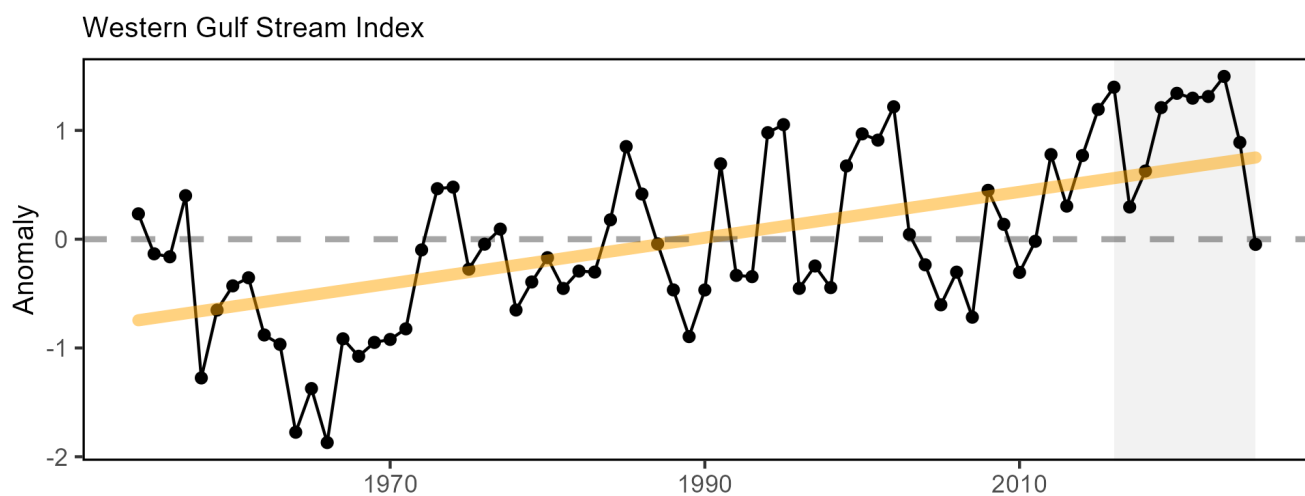


Figure 42: Index representing changes in the location of the western (between 64 and 55 degrees W) Gulf Stream north wall (black). Positive values represent a more northerly Gulf Stream position, with increasing trend (orange).

Changes in ocean temperature and circulation alter habitat features such as the Mid-Atlantic Bight [Cold Pool](#), a band of relatively cold near-bottom water present from spring to fall over the northern MAB. The cold pool represents essential fish spawning and nursery habitat, and affects fish distribution and behavior. The cold pool has been getting warmer and its areal extent has been shrinking over time (Fig. [43](#)). In 2025, however, the cold pool temperature index and extent were above the long-term average, likely due to the influx of Labrador Slope and Scotian Shelf waters into the system. Mobile target species that track a preferred temperature range can show increased interannual variability in their distributions as regional temperatures fluctuate from record warms to average over short periods of time.



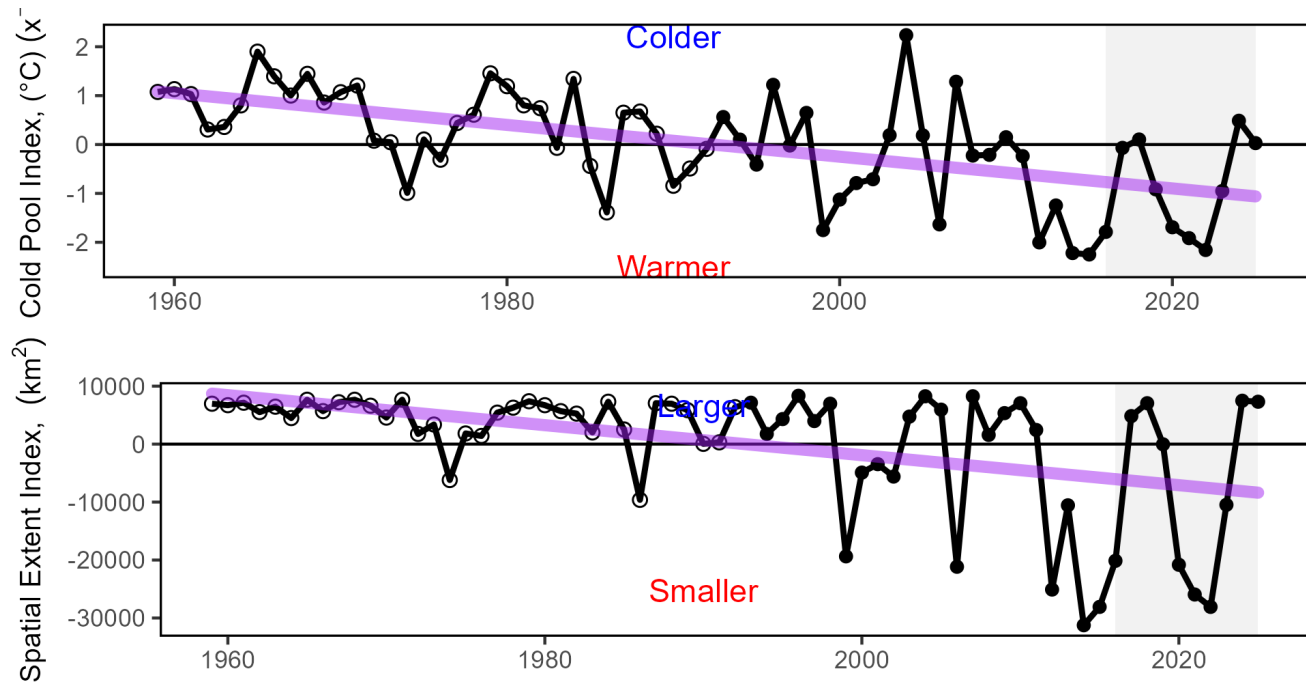


Figure 43: Seasonal cold pool mean temperature (top) and spatial extent index (bottom), based on bias-corrected ROMS-NWA (open circles) and GLORYS (closed circles), with declining trends (purple).

**Future Considerations** Distribution shifts caused by changes in thermal habitat and ocean circulation are likely to continue as long as long-term trends persist. Episodic and short-term events (see [2024 Highlights](#) and [2025 Highlights](#)) may increase variability in the trends, however species distributions are unlikely to reverse to historical ranges in the short term. Increased mechanistic understanding of distribution drivers is needed to better understand future distribution shifts: species with high mobility or short lifespans react differently from immobile or long-lived species.

[MOM6 decadal oceanographic forecasts](#) suggest a tendency towards near-normal temperatures over the next decade due to decadal variability in regional circulation. 2026 seasonal forecasts show a high probability of below average surface and bottom temperatures in the winter months. Forecast uncertainty is higher during the spring and summer seasons, and above average conditions are predicted for the fall. These forecasts will continue to be evaluated to determine how well they are able to predict episodic and anomalous events that are outside of the long-term patterns.

Adapting management to changing stock distributions and dynamic ocean processes will require continued monitoring of populations in space and time while evaluating management measures against a range of possible future spatial distributions. The upcoming Climate Vulnerability Assessment 2.0 will also be incorporating MOM6 output and forecasts to help predict changes in species distributions and quantify species exposure to predicted future change. Processes like the [East Coast Coordination Group](#) and the HMS Climate Vulnerability Assessment can help coordinate management.

## Risks to managing seasonally

The effectiveness of seasonal management actions (fishing seasons or area opening/closing periods) depends on a proper alignment with the seasonal life cycle events (phenology) of fish stocks (e.g., migration and spawning timing). If not accounted for, changes in the timing of these biological cycles can reduce the effectiveness of seasonal management measures. The timing of seasonal patterns can also change the interactions between fisheries and non-target species thus influencing bycatch and the availability of species to surveys.

**Indicators: Timing shifts** Indicators of phenological changes in fish populations require regular sampling and observations, and therefore a limited number of these indicators are currently available. One indicator shows shifts in [spawning timing](#) of haddock and yellowtail flounder. Spawning of both haddock stocks occurred earlier in the year, as indicated by more resting (post-spawning) stage fish in recent years compared to earlier in the time series (Fig. 44). The high percentage of northern stock (Cape Cod/GOM) yellowtail flounder females in the resting maturity stage shown earlier in the time series is reflective of spring surveys sampling them well before spawning, which peaks in June for the northern stock. More recently, the females are much closer to spawning, indicating that yellowtail flounder are spawning earlier in the year. Similarly, increased catch of post-spawning fish in Southern New England, indicates that the peak spawning of the southern stock has also shifted to earlier in the year. Yellowtail flounder spawning is related to bottom temperature, week of year, and decade sampled for each of the three stocks. Changes to spawning times could impact the survival of early life stages of fish, subsequently affecting the larger population size, health, and market value.

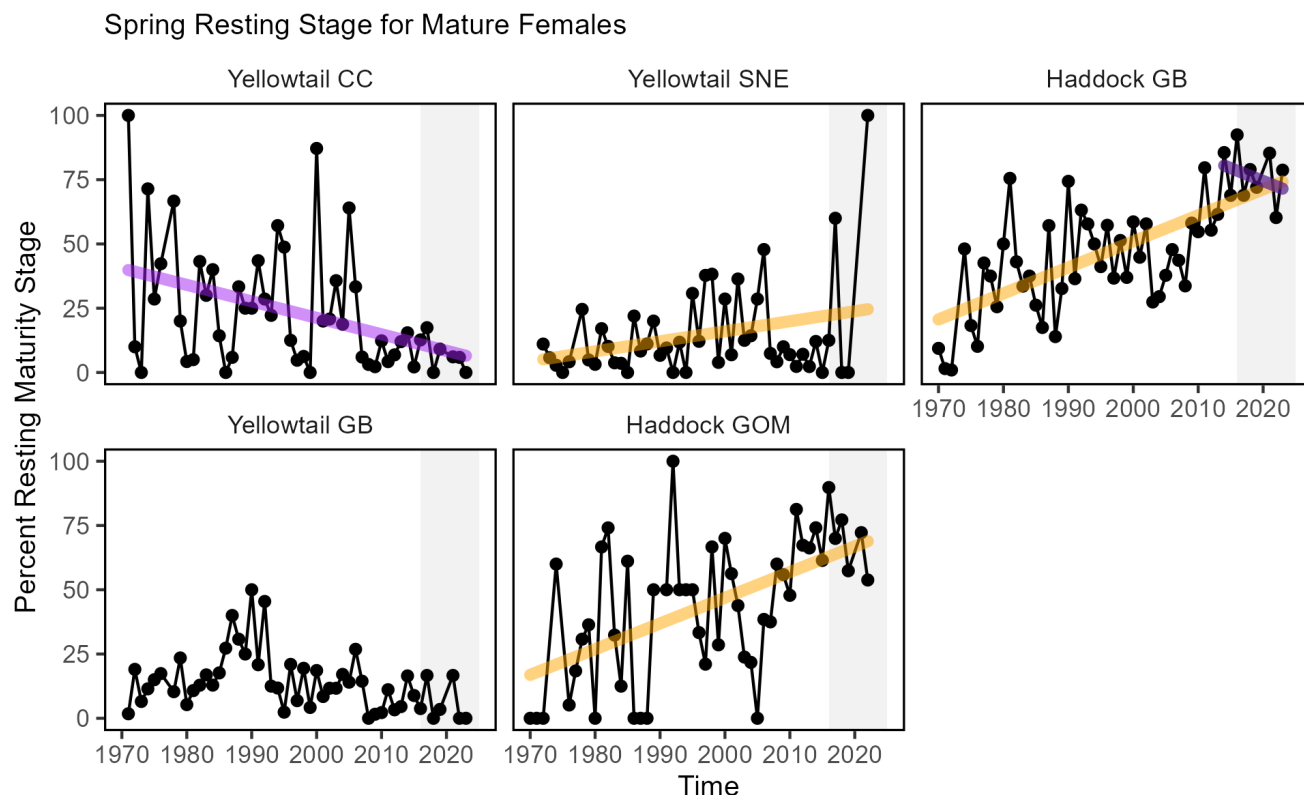


Figure 44: Percent resting stage (non-spawning) mature female fish (black) from spring NEFSC bottom trawl survey with significant increases (orange) and decreases (purple) from two haddock and three yellowtail flounder stocks: CC = Cape Cod Gulf of Maine, GOM = Gulf of Maine, GB = Georges Bank, SNE = Southern New England.

**Migration timing** of some tuna and large whale migrations has changed. An analysis of recreational fishing data between 2019 and 2022 identified multiple shifts in important HMS species. For example, Bigeye tuna were caught 50 days earlier; small and large bluefin tuna were caught 38 and 80 days earlier respectively in Massachusetts; and blue marlin in New York were caught 27 days earlier. A separate analysis of acoustic telemetry data predicted delayed departure of southward-migrating sharks from the northeast region under future sea surface temperatures. These results are further supported by the Atlantic Highly Migratory Species Climate Vulnerability Assessment, which found that 57 of 58 highly migratory species and stocks have high or very high potential to shift distributions. In Cape Cod Bay, peak spring habitat use by right and humpback whales has shifted 18-19 days later over time.

Understanding whether seasonal patterns are changing for stocks requires regular observations throughout the year. For example, baseline work on [cetacean presence in Southern New England](#) shows different seasonal use patterns for whale and dolphin species. Despite the importance of understanding seasonal patterns, we have few indicators that directly assess timing shifts of species. We plan on incorporating more indicators of timing shifts and phenology in future reports.

**Drivers:** The drivers of timing shifts in managed stocks are generally coupled to shifts in environmental or biological conditions, since these can result in changes in habitat quality or food availability within the year. Changes in the timing of fall phytoplankton blooms and seasonal shifts in zooplankton communities are indicators of changes in seasonal food availability to stocks.

Along with the overall warming trends in the Mid-Atlantic Bight, ocean summer conditions have been lasting longer (Fig. 45) due to the earlier [transition](#) from cool spring conditions to warm summer conditions and the later transition from warm summer conditions to cooler fall temperatures. These transition dates relate how daily temperatures compare to the seasonal norm. Changes in the timing of seasonal environmental cycles can alter biological processes (migrations, spawning, etc.) that are triggered by seasonal events.

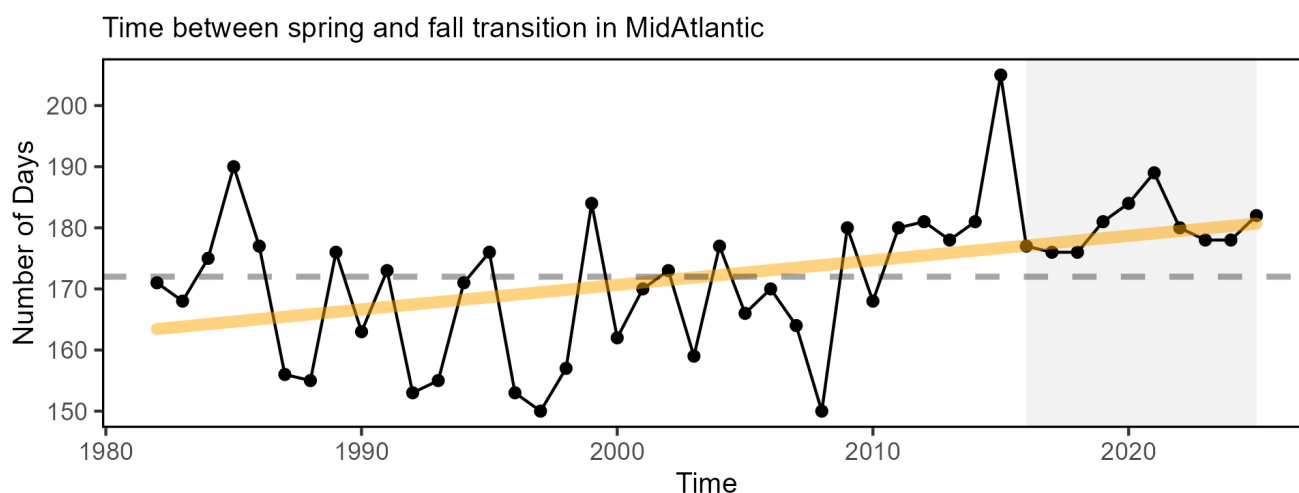


Figure 45: Ocean summer length in the MAB: the annual total number of days between the spring thermal transition date and the fall thermal transition date (black), with an increasing trend (orange). Transition dates are based on sea surface temperatures.

As noted above, the Mid-Atlantic Cold Pool is a summer to early fall feature that creates seasonally suitable habitat for some species. Cold pool persistence has decreased indicating that the duration of the cold pool habitat is shorter compared to the 1960s (Fig. 46). However, all cold pool indices were near or above the long-term average in 2025 and likely related to the influx of northern waters into the system (see 2024 Highlights ). A change in the timing of the autumn breakdown of the Cold Pool may impact the recruitment of species that rely on it for seasonal cues and habitat. Southern New England-Mid Atlantic yellowtail flounder recruitment and settlement are related to the strength of the MAB Cold Pool (a factor of extent and persistence). The correlation of pre-recruit settlers to the Cold Pool is thought to represent a bottleneck in yellowtail flounder life history, whereby a local and temporary increase in bottom temperature can negatively impact the survival of settlers. Including the effect of Cold Pool variations on yellowtail recruitment reduced retrospective patterns and improved predictive skill in a stock assessment model. This connection is especially important given the long-term decline in the duration of the Cold Pool.

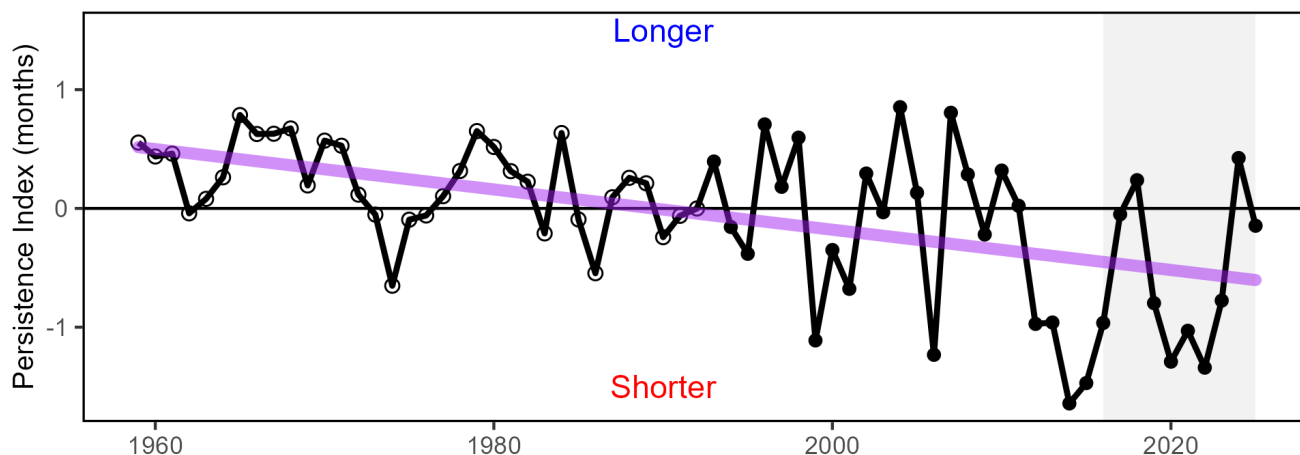


Figure 46: The Mid Atlantic Bight Cold Pool persistence index based on bias-corrected ROMS-NWA (open circles) and GLORYS (closed circles), with significant long-term decline (purple).

The seasonal timing of Mid-Atlantic phytoplankton blooms shows high interannual variability during the fall bloom period (October-December, Fig. 47). The significant increase in January chlorophyll suggests that the fall bloom period is continuing into the winter, with higher phytoplankton concentrations now than in the late 1990s. The significant decrease of chlorophyll in September could be related to warmer temperatures persisting into early fall and nutrient limitation causing a delay in the fall bloom. Changes to bloom timing can create a mismatch with the timing of larval fish development and may impact recruitment.

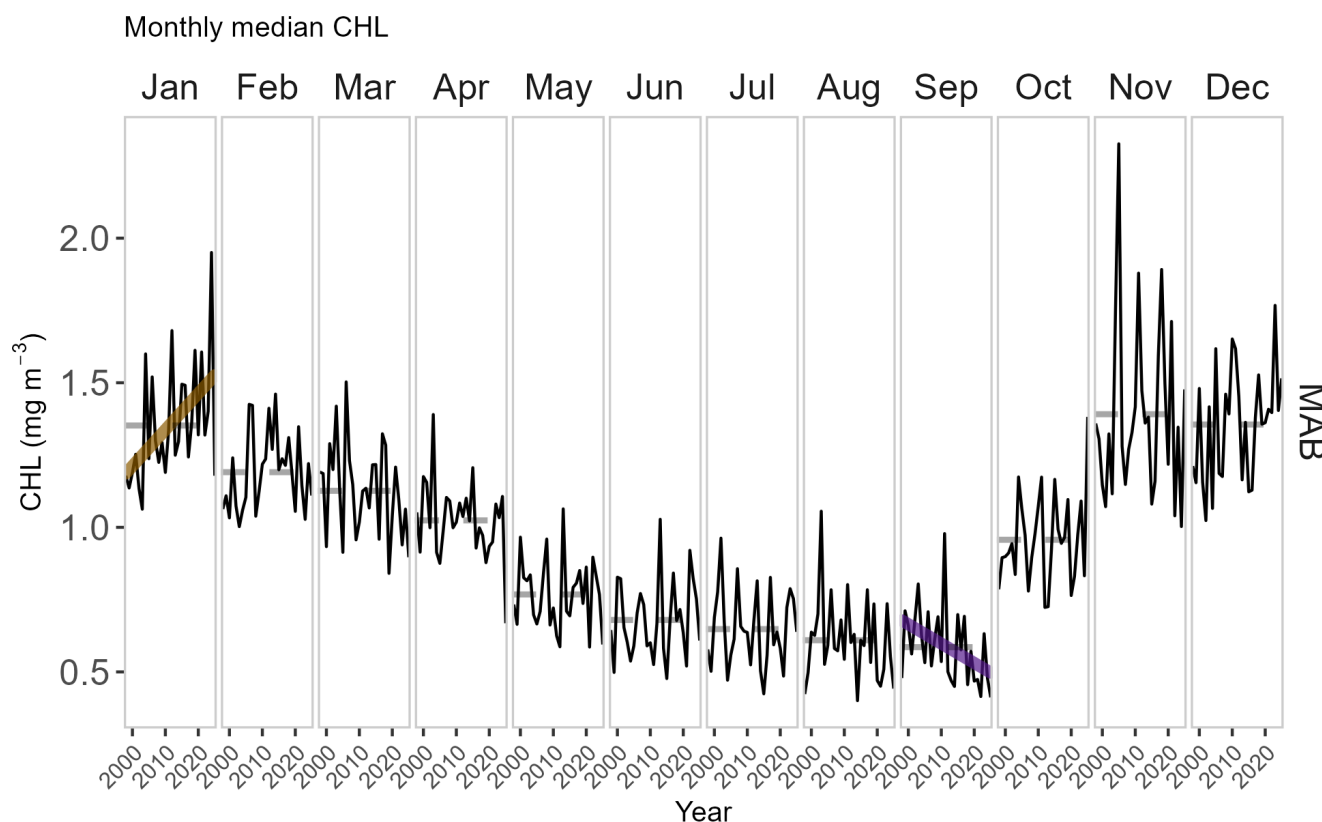


Figure 47: Monthly median chlorophyll a concentration in the MAB (black).

**Future Considerations** Species are reliant on environmental processes to dictate the timing of their behavior (e.g., phytoplankton bloom timing, thermal transition, or the duration of the cold pool). Some changes are episodic and have interannual variability, while others may be shifting away from a historic baseline on the scales of years to decades. Other species may rely on the general seasonal succession of environmental drivers (e.g., the timing of the fall turnover) to cue biological processes, and long-term trends in seasonal transitions are unlikely to reverse in coming years. Thus, timing shifts in migration or spawning may continue. Management actions that rely on effective alignment of fisheries availability and biological processes should continue to evaluate whether prior assumptions on seasonal timings still hold, and new indicators should be developed to monitor timing shifts for stocks.

### Risks to setting catch limits

The efficacy of short-term stock projections and rebuilding plans rely on accurate understanding of processes affecting stock growth, reproduction, and natural mortality. These biological processes are often driven by underlying environmental change. If ignored, environmental change may increase the risk that established stock-level biological reference points no longer reflect the current population and increase projection uncertainty, both of which can contribute to quota misspecification.

**Indicators: Fish productivity and condition shifts** Indicators of [fish productivity](#) are derived from observations (surveys) or models (stock assessments). Fish productivity declined during the 1990's and 2000's with declining production of summer flounder and has been variable since, as described by the small-fish-per-large-fish anomaly indicator (derived from NEFSC bottom trawl survey) (Fig. 48). Bluefish, black sea bass, and goosefish have sporadic years with large positive anomalies, but most years have small negative anomalies. This decline in fish productivity is also shown by a similar analysis based on stock assessment model outputs (recruitment per spawning stock biomass anomaly). Most species had positive recruitment anomalies in the 1990s and 1990s and are currently

showing negative anomalies, indicating a decline in productivity. Fish productivity can be affected by parental condition, environmental conditions, timing and availability of prey for recruits, as well as retention of recruits within favorable habitat. High offshore advection during spawning seasons can reduce recruitment success and affect overall fish productivity.

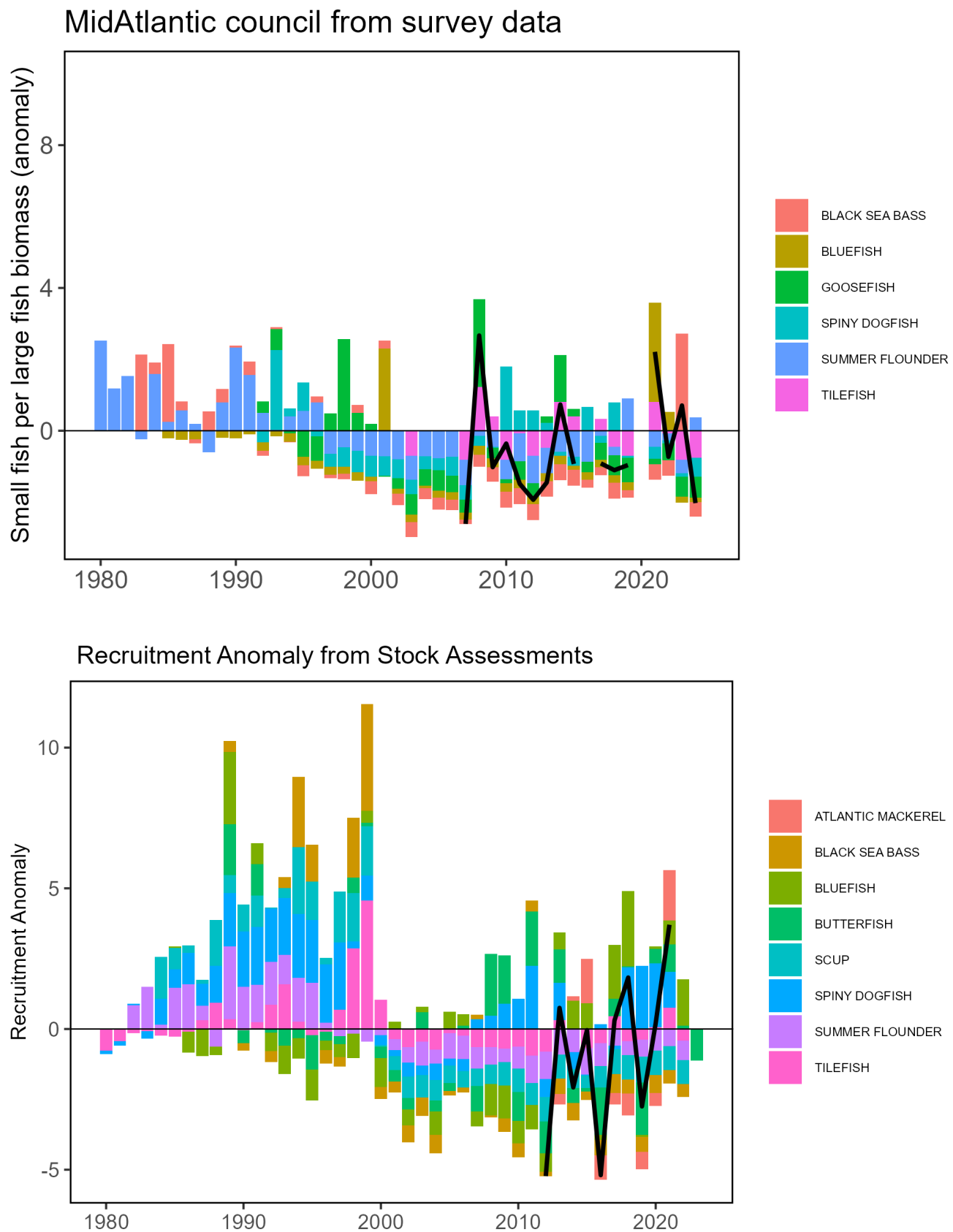


Figure 48: Fish productivity measures. Top: Small fish per large fish survey biomass anomaly of Mid Atlantic Fishery Management Council managed species in the Mid-Atlantic Bight. Bottom: assessment recruitment per spawning stock biomass anomaly for stocks managed by the Mid Atlantic Fishery Management Council. The summed anomaly across species is shown by the black line, drawn across all years with the same number of stocks analyzed.

The health of individual fish (i.e., fish condition, measured as weight for a given length) can contribute to population productivity through improved growth, reproduction and survival. Mid-Atlantic fish condition was generally high to very high prior to 2000, low to very low from 2001-2010 (concurrent with declines in productivity, Fig. 48)), and mixed since 2011. In 2025, condition continued to be mixed, with general improvement since a relatively low condition year in 2021 (Fig. 49). Preliminary analyses show that years dominated by small copepods and warmer spring temperatures may improve fish condition for Atlantic mackerel and butterfish. Similar environmental drivers may be important to other species.

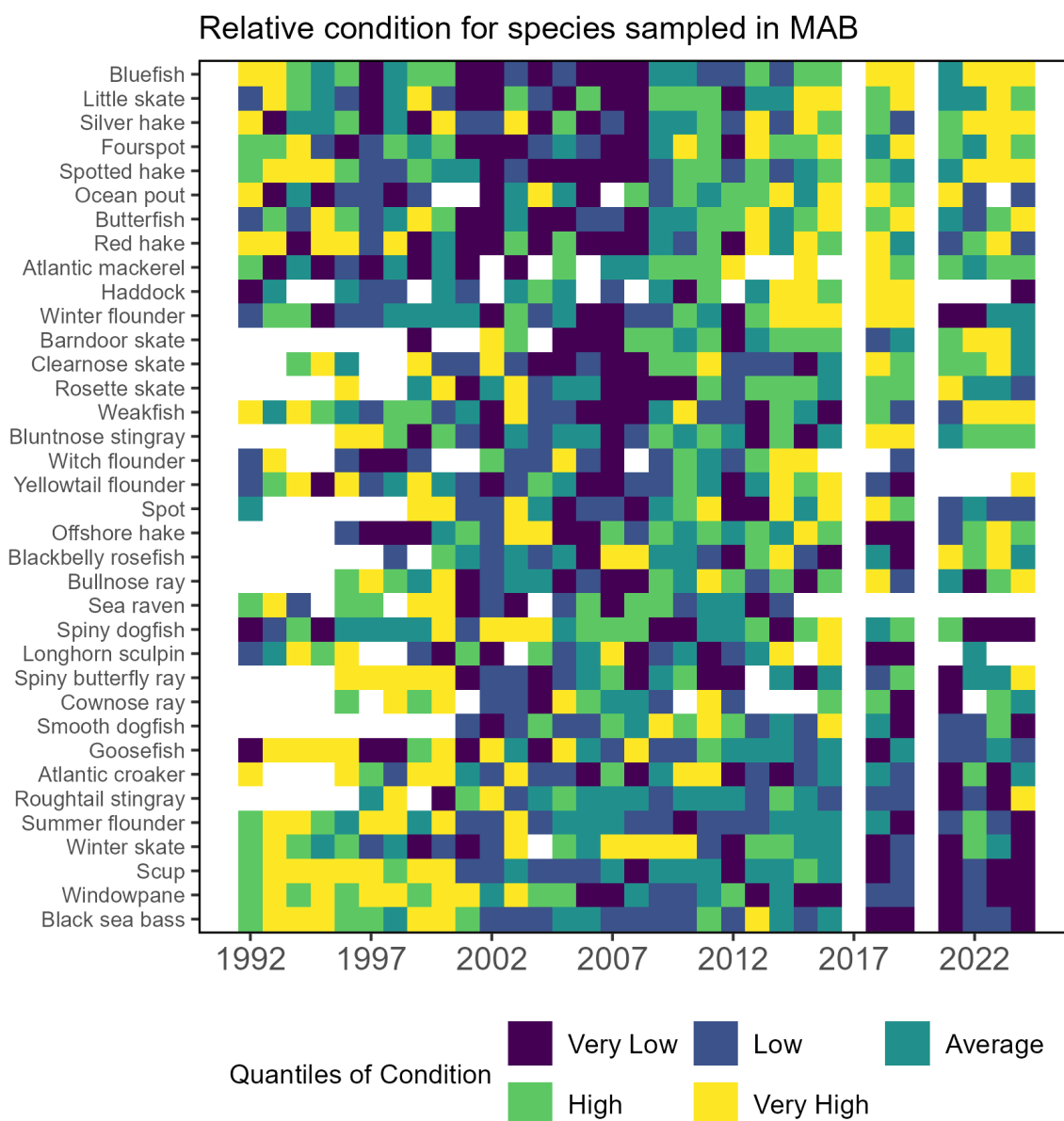


Figure 49: Condition factor for fish species in the MAB based on fall NEFSC bottom trawl survey data. MAB data are missing for 2017 due to survey delays, and no survey was conducted in 2020.

**Drivers:** Fish productivity and condition are the cumulative effects of physiological, ecological, and environmental factors. Major factors include increased metabolic demands from increasing temperature and changes in the availability and quality of prey. Long-term environmental trends and episodic extreme temperatures, ocean acidification,



and low oxygen events represent multiple stressors that can affect growth rates, reproductive success, recruitment, and cause mortality.

**Biological Drivers: Forage quality and abundance** The energy density (ED) of prey, in conjunction with its mass, indicates the total amount of energy available to higher trophic level predators. The quality and abundance of this forage base directly impact the productivity and movement of managed and protected species. Management should consider these energetic links, as shifts in forage quality can alter the health of individual stocks and the entire ecosystem.

Forage fish [energy content](#) fluctuates based on growth, reproduction, environmental conditions, and ecosystem productivity. In the Mid-Atlantic Bight, butterfish are the most abundant high ED forage species (Fig. 50), though their fall ED have recently declined toward lower spring averages. Atlantic herring and Atlantic mackerel also serve as high-energy prey, but herring show recently low ED and declining abundance, while mackerel are most abundant in the spring despite having higher ED in the fall. Moderate energy forage species (longfin squid, northern shortfin squid, and silver hake) are of intermediate abundance and show minimal annual and seasonal variation in ED. Other species have high ED but lower abundance decreasing their reliability as a food source.

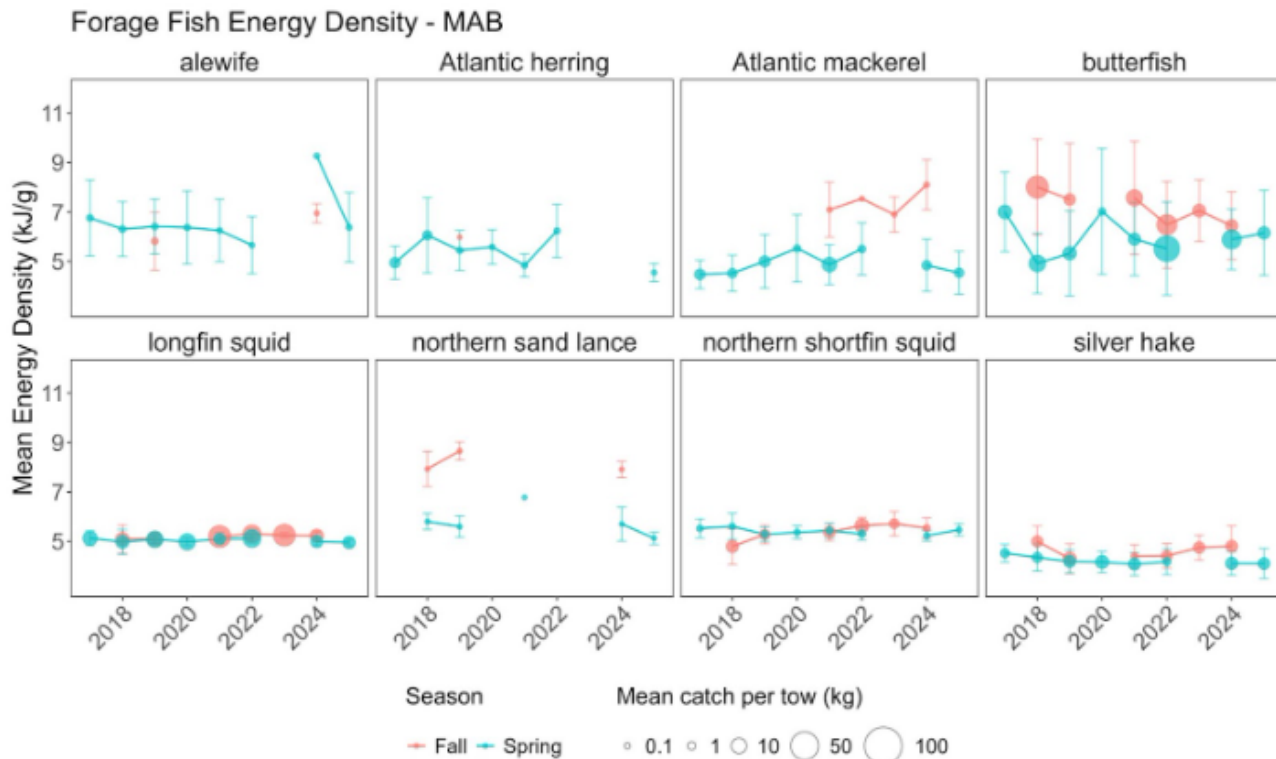


Figure 50: Energy density (mean and standard deviation) of eight forage species from NEFSC bottom trawl surveys by season and year for the MAB. Symbol size represents abundance (mean kg/tow) estimated from bottom trawl survey tows in the MAB.

Changes in the overall abundance of forage fish can influence managed species productivity as it relates to changes in food availability. A spatially-explicit [forage index](#) for the Mid-Atlantic shows a long term declining trend in fall, with higher forage biomass in fall than spring (Fig. 51). Forage biomass was highest during fall in the early-1980s. The decrease of fall forage biomass in the Mid-Atlantic may reduce the health and reproductive output of fish species. Additionally, this may be exacerbated by lower energy densities of prey, especially in years of higher water temperatures when metabolic demands are higher.

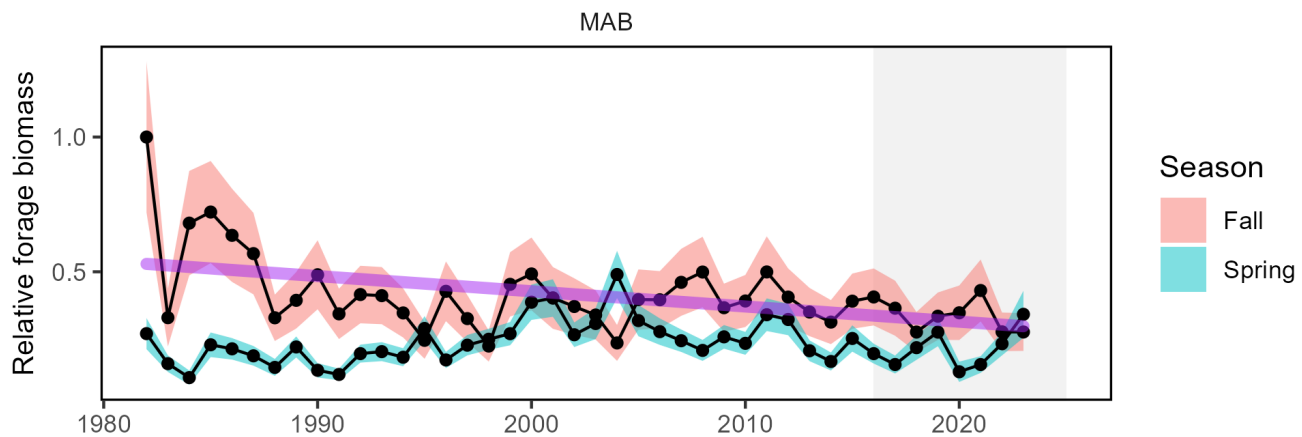


Figure 51: Forage fish index in the MAB for spring (blue) and fall (red) surveys, with a decline (purple) in fall. Index values are relative to the maximum observation within a region across surveys.

[Benthic invertebrates](#) are extremely important forage for some managed species (e.g., black sea bass). Macrobenthos are small benthic organisms that tend to be prey for larger benthos and benthivores. Macrobenthos indices show long-term declines in spring (Fig. 52), indicating a potential decrease in food availability for their predators. In contrast, Mid-Atlantic megabenthos indices show long-term increases in spring. Fish productivity may be positively impacted in recent years for juvenile fish that target macrobenthos, such as small crustaceans and polychaetes, and negatively impacted for fish such as black sea bass and striped bass that target megabenthos such as crabs. Other species that are generalist feeders such as scup and skates may not be as impacted by offsetting trends in the benthic community.

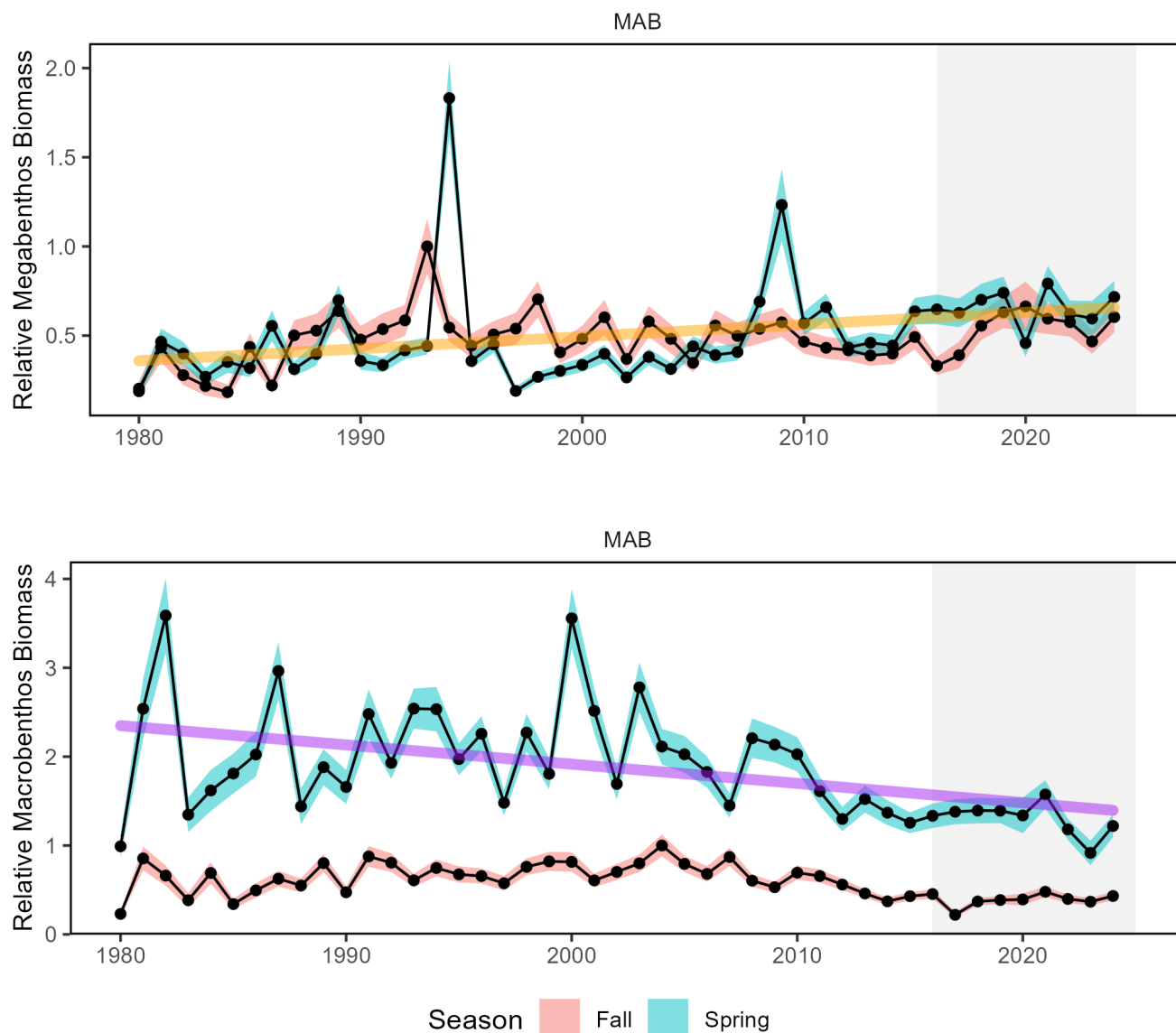


Figure 52: Changes in spring (blue) and fall (red) benthos abundance in the MAB for megabenthos (top) and macrobenthos (bottom), with significant long-term increasing (orange) and decreasing (purple) trends.

**Biological Drivers: Lower trophic levels** **Phytoplankton** are the foundation of the food web and are the primary food source for zooplankton and filter feeders such as shellfish. Multiple environmental and oceanographic drivers affect the abundance, **composition**, spatial distribution, and productivity of phytoplankton. While changes in phytoplankton productivity could affect fish productivity (including the productivity of forage fish), there is no clear long-term trend in Mid-Atlantic total primary production (Fig. 27).

Changing **zooplankton abundance** may impact forage fish energy content and abundance, as well as the prey field of filter feeding whales, and managed species through food web impacts. Mid-Atlantic indices show high variability without a clear trend for large copepods, while small-bodied copepods (*Calanus finmarchicus*) show long-term and recent decreases, and krill (Euphausiids) show increasing trends (Fig. 53). Energy density varies by season and location, with high-energy large copepods most abundant on the Northeast shelf from April through June. The community is undergoing a systemic shift away from copepod dominance and toward increased krill

presence.

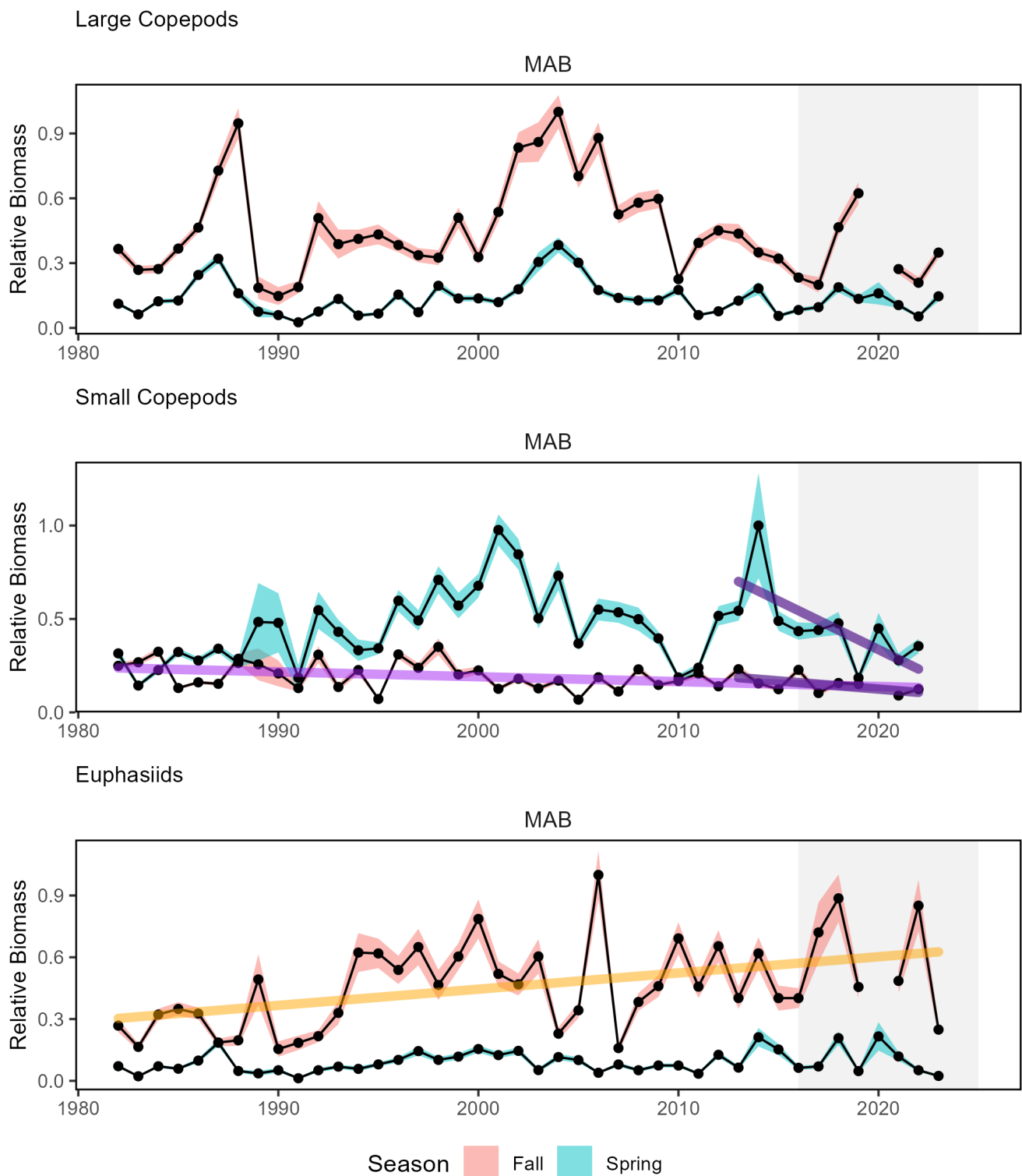
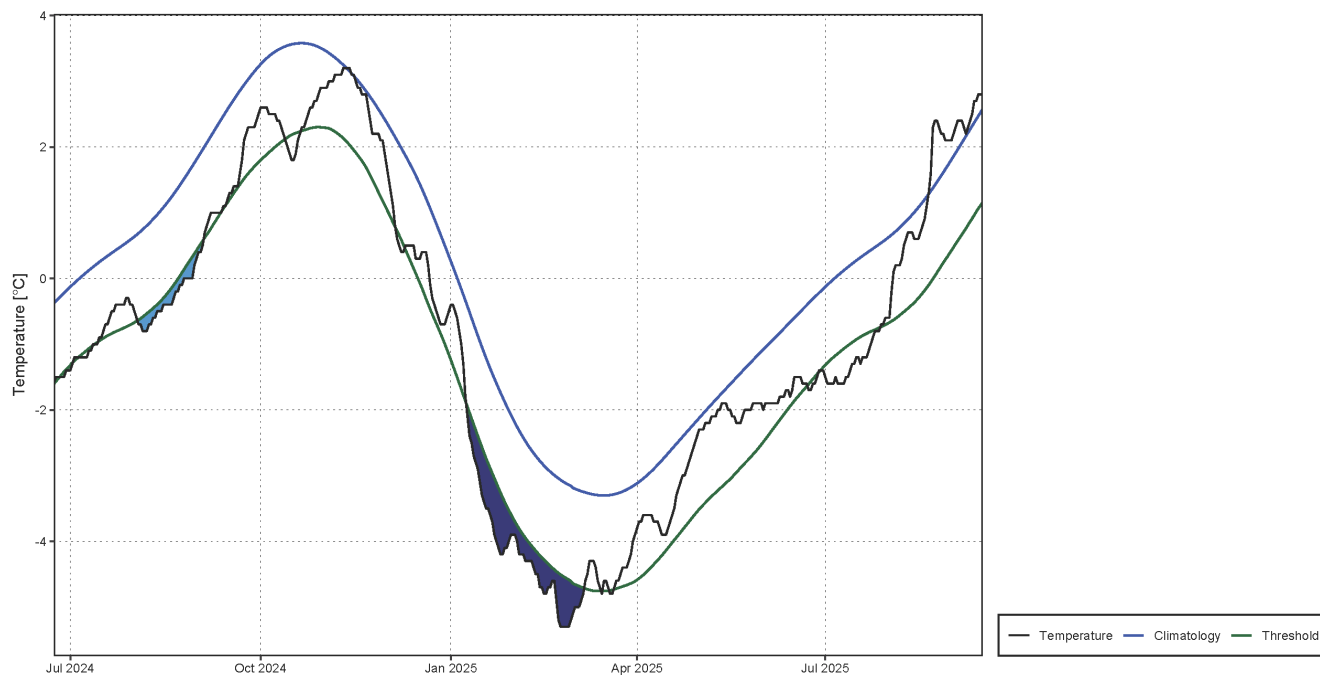


Figure 53: Changes in three dominant zooplankton (*Calanus finmarchicus*, *Calanus typicus*, and *Pseudocalanus spp*) abundance anomalies for in the MAB for large (top) and small (middle) copepods, and Euphausiids (bottom), with significant decreases (short-term, dark purple; long-term, light purple) in small copepods and and long-term increases (orange) in Euphausiids.

**Environmental Drivers** Fish production can also be directly related to the prevailing environmental conditions by altering metabolism (growth), reproductive processes, and survival. Marine species possess thermal tolerances and can experience stressful or lethal conditions if water temperatures exceed certain levels. We have observed in past years extreme temperatures at both the [surface](#) and [bottom](#) that exceed [thermal tolerance](#) limits for some fish and shellfish. However, in 2025, Mid-Atlantic surface and bottom temperatures were near or below the long-term average and the amount of habitat exceeding a 24 °C thermal tolerance was limited to the southern MAB, where those conditions occurred for fewer than 30 days (Fig. 54).

A single surface [marine heatwave](#) occurred in the Mid-Atlantic Bight in 2025, starting July 15th and lasting seven days. This brief event was the only heatwave recorded across the entire Continental Shelf for the year. The MAB experienced six surface marine [cold spells](#) in 2025, including an event in February that ranked as the 8th strongest on record. Additionally, a significant bottom cold spell occurred in January, lasting 57 days and ranking as the 5th strongest on record. During this period, bottom temperatures averaged 7.2 °C, nearly 2 °C lower than the historical average.

Lower ocean temperatures near long-term averages will affect species differently across the region. While cold-water species like cod may benefit from these conditions, warm-water species such as black sea bass are unlikely to see positive effects. This variability in regional cooling highlights the need for management to account for shifting species distributions and productivity.



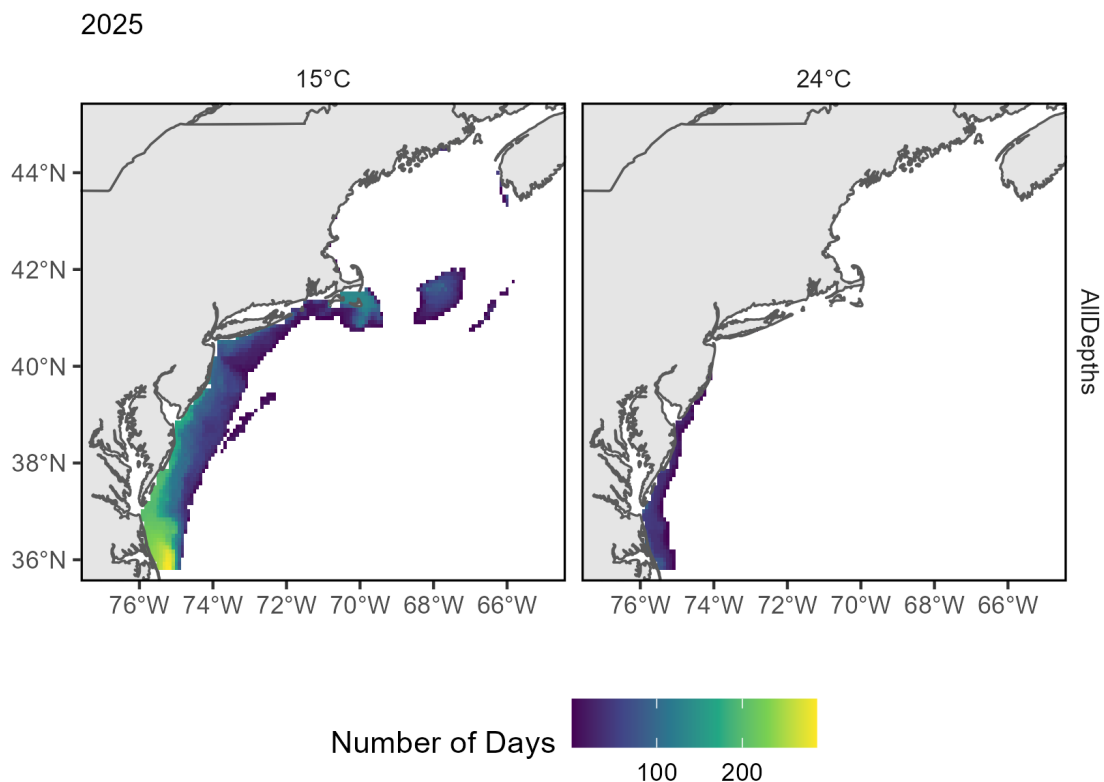
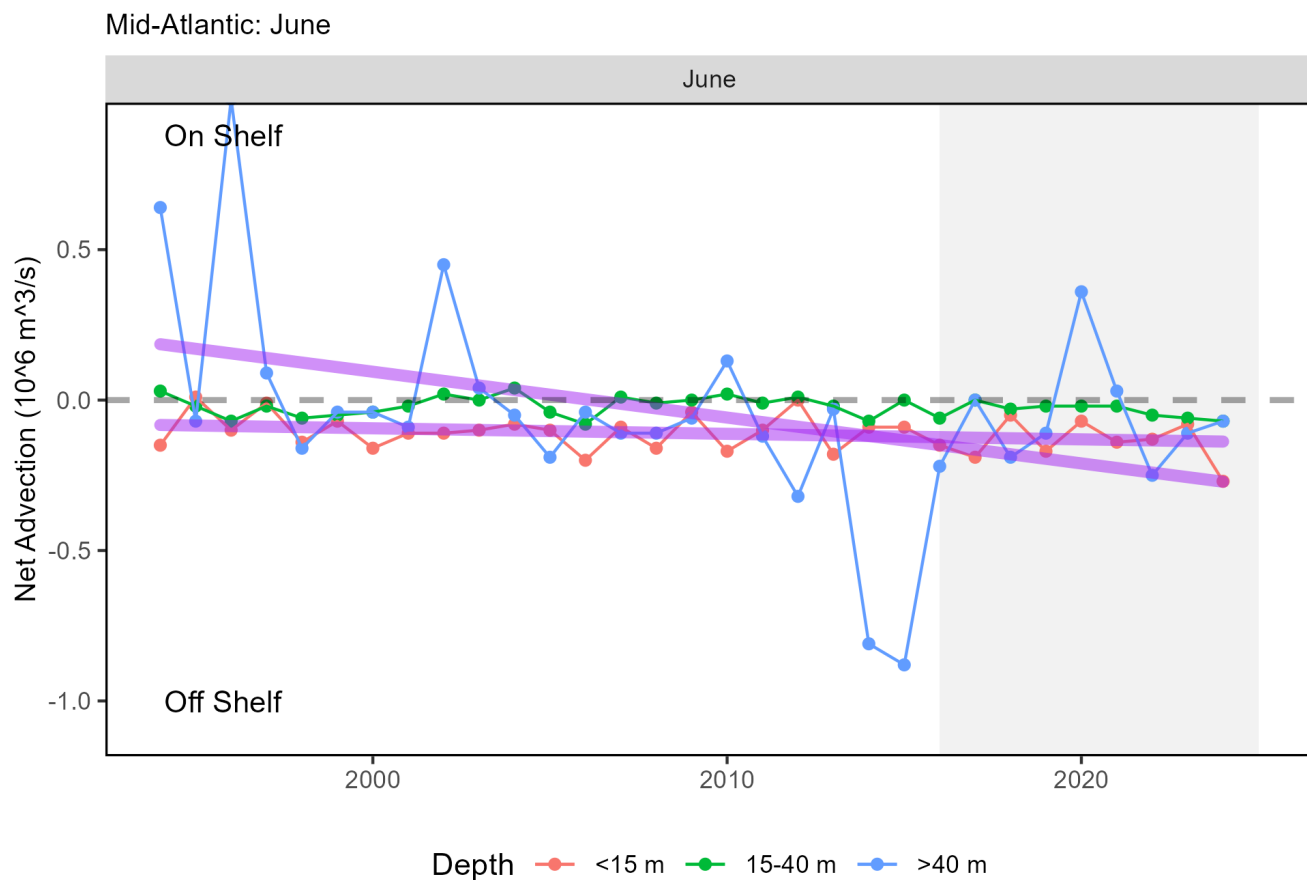


Figure 54: The number of days in 2024 where bottom temperature exceeds 15 degrees (left) and 24 degrees (right) based on the GLORYS 1/12 degree grid.

The newly-developed [advection index](#) (Fig. ??) shows total transport of water onto and off the continental shelf which can impact the survival of early life stages of fish and invertebrates. Long-term trends in the Mid-Atlantic show increased offshore movement of surface and bottom waters in June, which could decrease retention of some species. Further species level studies are needed to link spawning timing and larval periods to advection trends at the corresponding spatial and temporal scalesdepth and month.



**Ocean acidification** (OA) risks vary among species and include reduced survival, growth, reproduction, and productivity, where high OA risk indicates potential negative effects to species. OA risk can also be heightened during colder conditions due to increased CO<sub>2</sub> absorption by colder water or by transport of high CO<sub>2</sub> water masses as was suggested to have occurred in 2024 (see [2024 Highlights](#)). The OA indicator observed on the Mid-Atlantic coastal shelf during summer 2024 was the most extreme recorded when compared to all of the years sampled (since 2007). In 2025, however, OA risk conditions were less than those observed in 2023 and 2024. High OA conditions in 2025 were limited to a few outer shelf coastal New Jersey (NJ) observations in spring, where sensitivity levels for Atlantic sea scallops were exceeded (not shown, see [ocean acidification](#)), and in nearshore NJ waters in summer, where sensitivity levels for Longfin squid were reached (Fig. 55). Although relatively cool bottom seawater temperatures in 2025 were similar to 2024, salinity was higher in 2025, which suggests a different composition of oceanographic properties and water masses between the two years and as a result, different OA risk conditions.



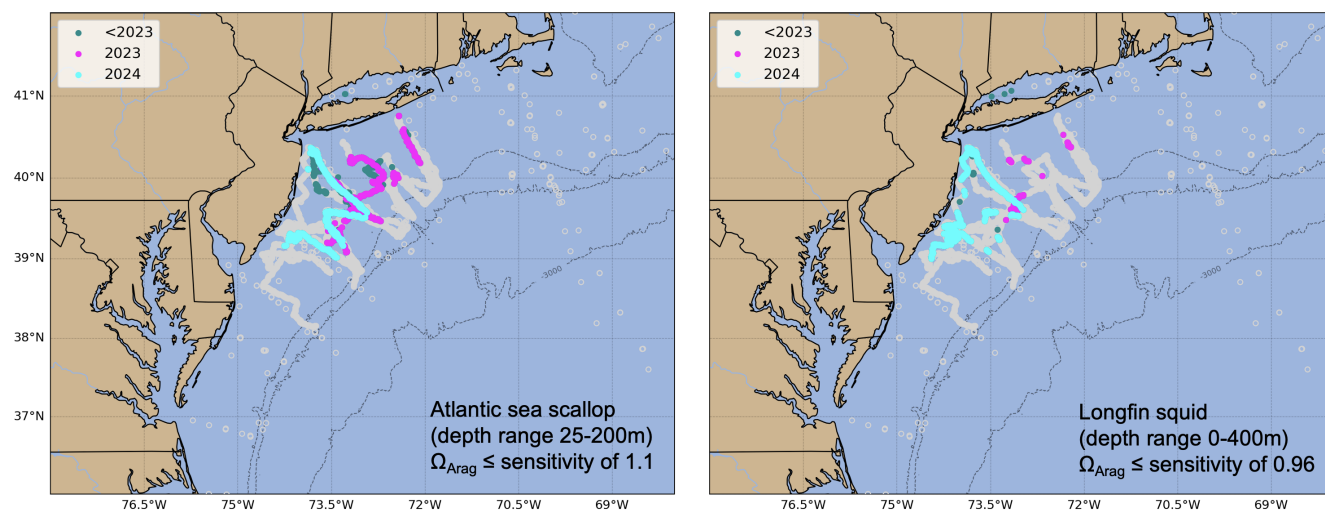


Figure 55: Locations where bottom aragonite saturation state ( $\Omega_{Arag}$ ; summer only: June-August) were at or below the laboratory-derived sensitivity level for Atlantic sea scallop (left panel) and longfin squid (right panel) for the time periods 2007-2022 (dark cyan), 2023 only (magenta) and 2024 only (cyan). Gray circles indicate locations where bottom  $\Omega_{Arag}$  values were above the species specific sensitivity values..

**Low dissolved oxygen levels** ( $< 5$  mg/L) remained localized and brief on the MAB shelf in 2025, resulting in no industry-reported mass mortality events despite the potential for hypoxia to reduce species growth or cause death. Localized hypoxia ( $< 2$  mg/L) occurred nearshore east of Point Pleasant, NJ, southwest of Newport, RI, and at the western end of the Cape Cod Canal, while broader shelf-wide levels below 5 mg/L were not widespread. This contrast follows previous years where hypoxic events in Cape Cod Bay (2019, 2020) and off New Jersey (2023) potentially caused fish, lobster, and crab mortality. While shelf-wide monitoring data is currently limited, biological and oceanographic drivers of oxygen levels continue to be tracked to assess the duration and extent of future events.

**Drivers: Predation** The abundance and distribution of marine mammal, shark predators, and other Atlantic Highly Migratory Species (HMS), can affect both the productivity and mortality rates on managed stocks. Predators can consume managed species or compete for the same resources, resulting in increased natural mortality or decreased productivity. The northeast shift in whales and dolphins (Fig. 33) indicates a change in the overlap between marine mammals and managed fishes. Since we also observe distribution shifts in managed species as well as forage species, the effect of changing predator distributions alone is difficult to quantify.

Indicators for shark populations, combined with information on gray seals (see Protected Species Implications section, above), suggests predator populations range from stable (sharks) to increasing (gray seals) in the MAB. Stock status is mixed for HMS stocks (including sharks, swordfish, billfish, and tunas) occurring throughout the Northeast U.S. shelf. While there are several HMS species considered to be overfished or that have unknown stock status, the population status for some managed Atlantic sharks and tunas is at or above the biomass target, suggesting the potential for robust (or rebuilt) predator populations and subsequent predation pressure on managed species. Increasing predator populations or changing distribution of predators may result in increased predation pressure.

## Other Ocean Uses: Offshore Wind

Offshore wind development is active and dynamic throughout the region. The following section reflects the status of projects as of 5 February, 2026.

### **Indicators: development timeline, revenue in lease areas, coastal community vulnerability**

All reported potential offshore wind development status and data are based on BOEM's Offshore Renewable Activities page and projects' Final Environmental Impact Statements. In 2025, the Presidential Memorandum 90 FR 8363 removed existing planning areas and excluded the establishment of additional lease areas.

As of January 2026, 38 offshore [wind development](#) leases are under different stages of development in the Northeast (Fig. 56). One project (South Fork Wind Farm) is fully operational and another (Vineyard Wind 1) is partly operational while construction finishes. The southern New England region has two other projects currently under construction (Revolution Wind and Sunrise Wind). Empire Wind and Coastal Virginia Offshore Wind (CVOW) are currently under construction in the New York Bight and Mid-Atlantic Region, respectively, with CVOW expected to start generating power in early 2026.

Construction of these projects during 2025 affected fisheries managed by the [Mid-Atlantic] [New England] Fishery Management Council. There are eight additional projects that have Construction and Operations Plan (COP) approvals (three in Southern New England and five in the Mid-Atlantic/New York Bight) that could begin construction in 2026, however, construction schedules are highly uncertain at this time. Seven additional projects have submitted COPs and are pending approval, while the remaining projects are under the site assessment phase and have not submitted COPs to date (Fig. 56).

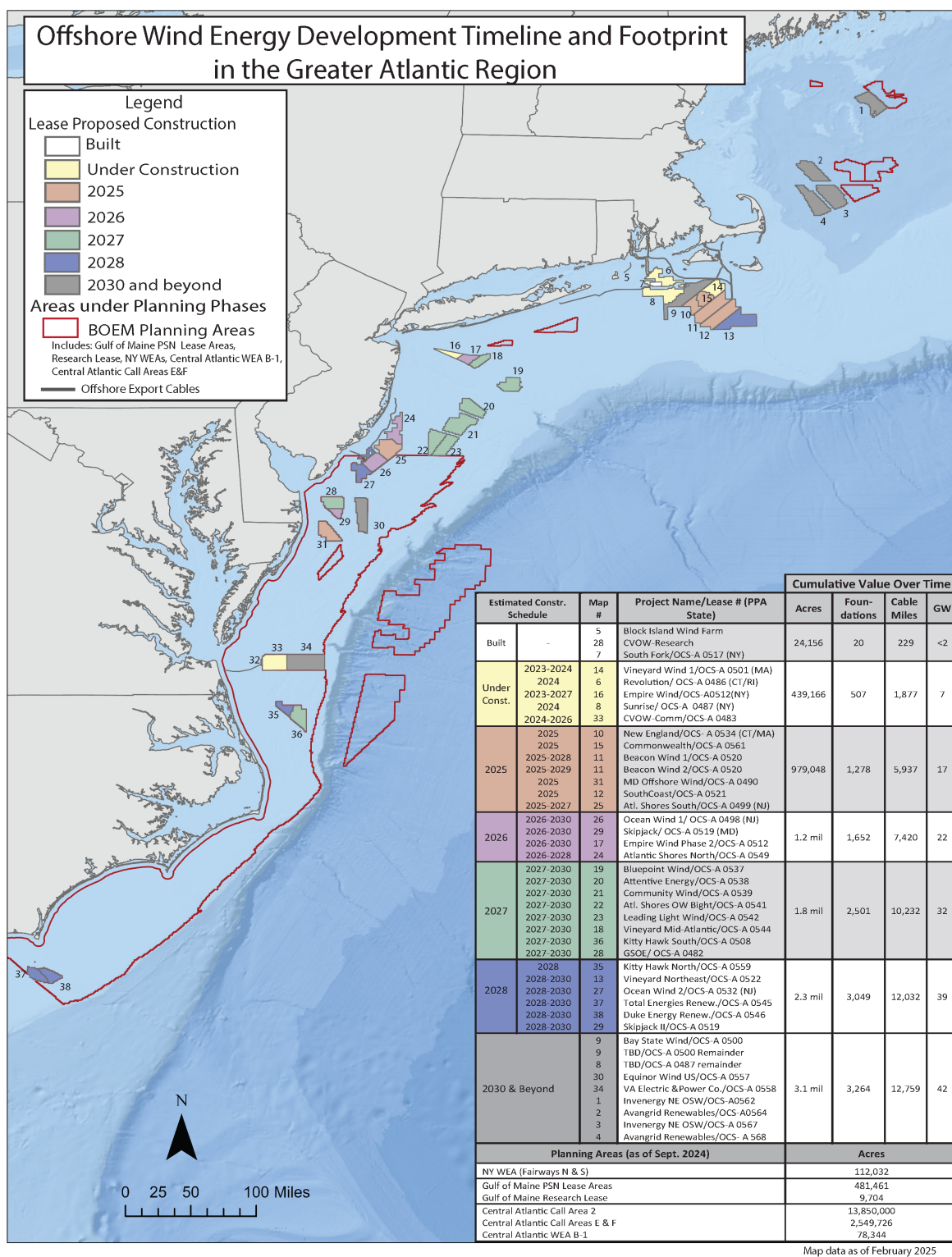


Figure 56: All Northeast Project areas by year construction ends (each project has a 2 year construction period).

With the first offshore wind energy projects now under construction and operation, all indicator analyses in this section follow a different reporting format than in previous years. Where previous years reported data for all lease

areas, this year we investigate impacts of the six commercial scale projects currently under construction or operation, (i.e., Active Projects: South Fork Wind Farm, Revolution Wind, Sunrise Wind, Empire Wind 1, Vineyard Wind 1, and CVOW-Commercial).

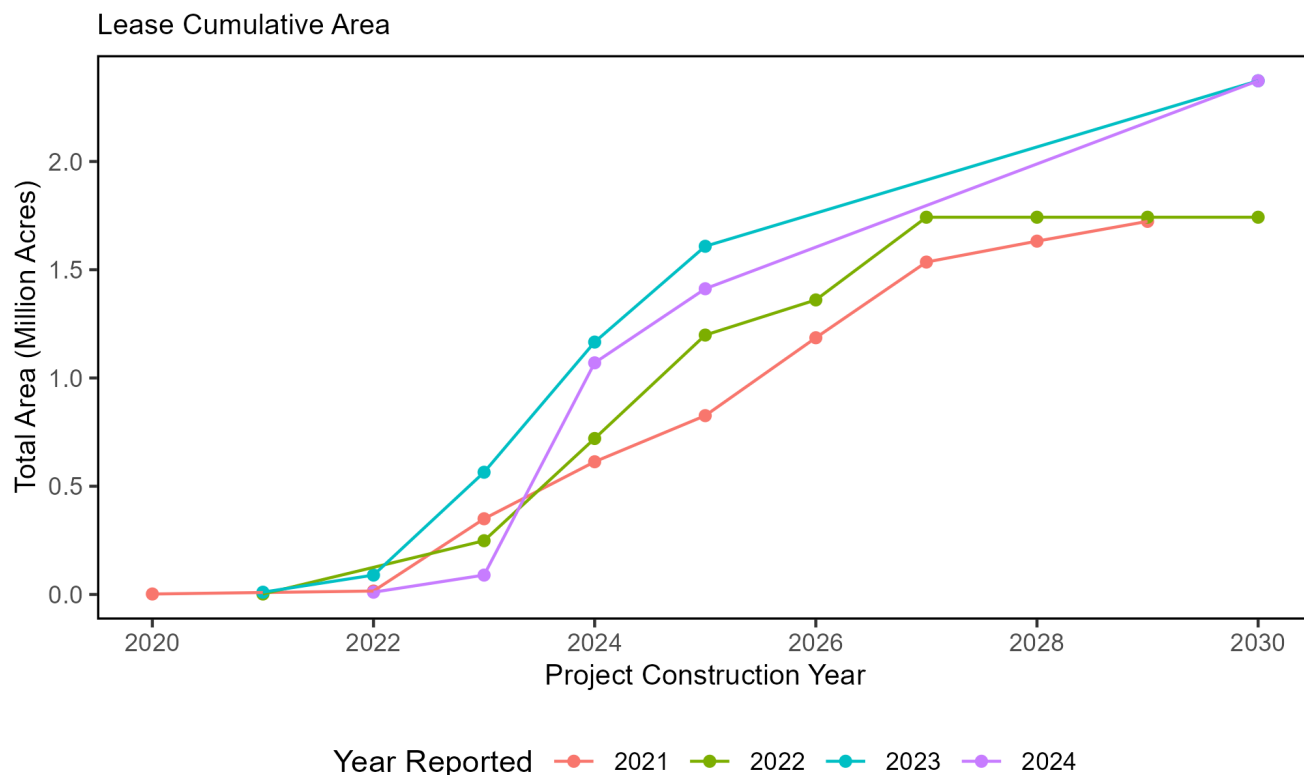


Figure 57: Total area proposed for wind development on the Northeast Shelf through 2030.

Offshore wind indicators are based on federal logbook data and do not include all data for all fisheries; therefore a complete evaluation of potential offshore wind energy development impacts would need to be supplemented by other data sources. For further information on the utility of the data, see the [socioeconomic impacts of offshore wind development data reports page](#).

Based on federal vessel logbook data, [commercial fishery revenue](#) from trips within Active Projects have varied annually from 2008-2024, with less than \$500,000 in maximum annual revenue overlapping with these areas for most fisheries with the exception of the longfin squid (\$2 million), monkfish (\$1.1 million), ocean quahog (\$783,000), and summer flounder (\$556,000) in specific years (Fig. 58).

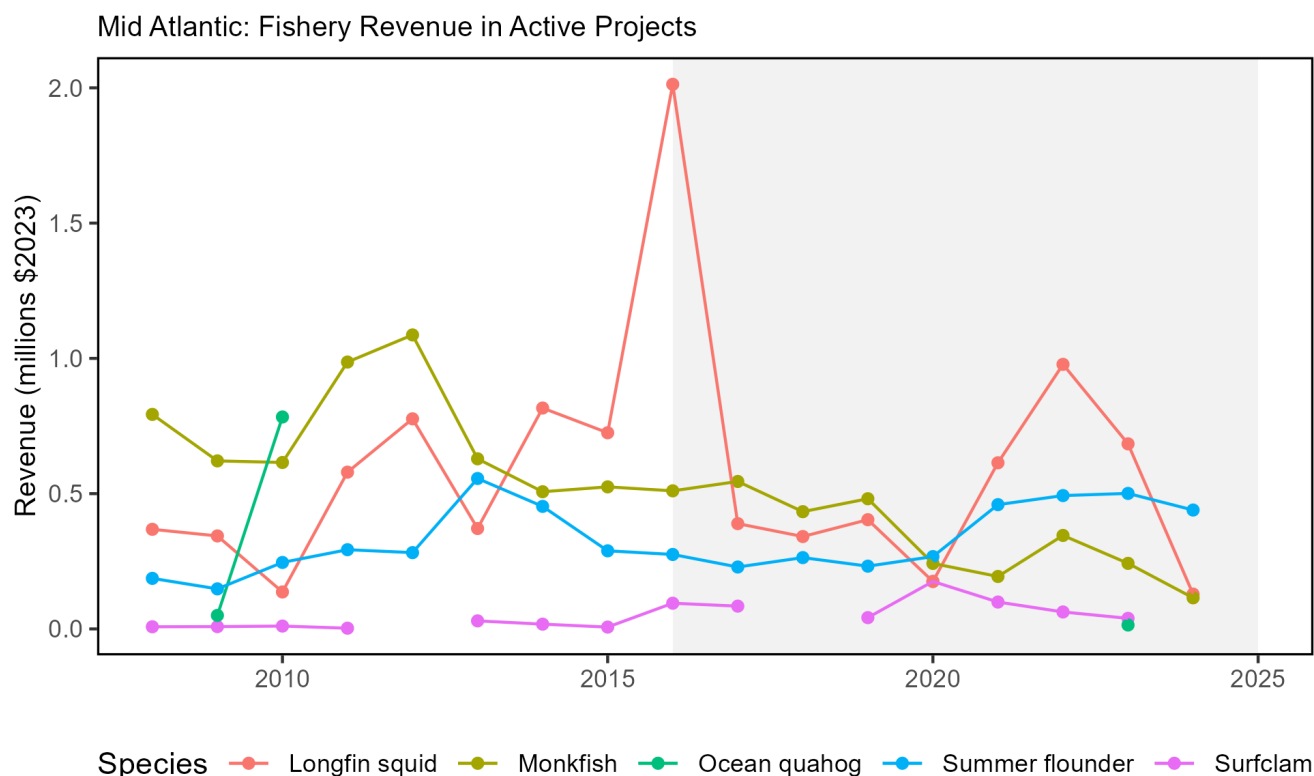


Figure 58: Revenue of species managed by the Mid-Atlantic Fishery Management Council within Active Projects.

Within Active Projects, the MAFMC managed fisheries most affected based on historic landings include longfin squid, monkfish, scup, Atlantic mackerel, and summer flounder, with a maximum of 6% of annual regional fishery revenue for chub mackerel occurring within Active Projects during 2008-2024, 5% for bluefish, 4% for butterfish, and 3% each for monkfish, scup, black sea bass, and longfin squid, respectively (see Table 5). Future offshore wind development may increase effects on these and additional species if more projects begin construction. Future fishery resource overlap with wind leases, especially surfclams and ocean quahogs, may change due to species distribution shifts attributable to climate change and recruitment and larval dispersion pattern changes caused by hydrodynamic flow disruptions from turbine foundations, which could also affect fishery landings/revenue.

Table 5: Mid-Atlantic managed species Landings and Revenue from Wind Energy Areas. \*Less than a maximum of 50,000 lb was reported landed annually in wind energy lease areas for these species.

| NEFMC, MAFMC, and ASMFC<br>Managed Species | Maximum Percent Total Annual<br>Regional Species Landings | Maximum Percent Total Annual<br>Regional Species Revenue |
|--------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| Longfin Squid                              | 3.31                                                      | 3.25                                                     |
| Monkfish                                   | 4.66                                                      | 3.23                                                     |
| Ocean Quahog                               | 2.21                                                      | 2.34                                                     |
| Summer Flounder                            | 1.78                                                      | 2.01                                                     |
| Scup                                       | 3.32                                                      | 3.26                                                     |
| Atlantic Mackerel                          | 2.93                                                      | 2.40                                                     |
| Black Sea Bass                             | 2.70                                                      | 3.01                                                     |
| Atlantic Surfclam                          | 0.69                                                      | 0.65                                                     |
| Butterfish                                 | 4.37                                                      | 3.88                                                     |
| Spiny Dogfish                              | 1.66                                                      | 1.77                                                     |
| Illex Squid                                | 0.24                                                      | 0.42                                                     |

Table 5: Mid-Atlantic managed species Landings and Revenue from Wind Energy Areas. \*Less than a maximum of 50,000 lb was reported landed annually in wind energy lease areas for these species.

| NEFMC, MAFMC, and ASMFC<br>Managed Species | Maximum Percent Total Annual<br>Regional Species Landings | Maximum Percent Total Annual<br>Regional Species Revenue |
|--------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| Bluefish                                   | 3.60                                                      | 4.85                                                     |
| Golden Tilefish                            | 0.22                                                      | 0.24                                                     |
| Chub Mackerel                              | 5.32                                                      | 5.80                                                     |
| Blueline Tilefish                          | 0.14                                                      | 0.11                                                     |

The socio-demographic conditions, and resultant vulnerabilities, of some communities may further exacerbate the impacts of offshore wind development in the Northeast such that the impacts of offshore wind development are expected to differentially [impact specific coastal communities](#) (Fig. 59)

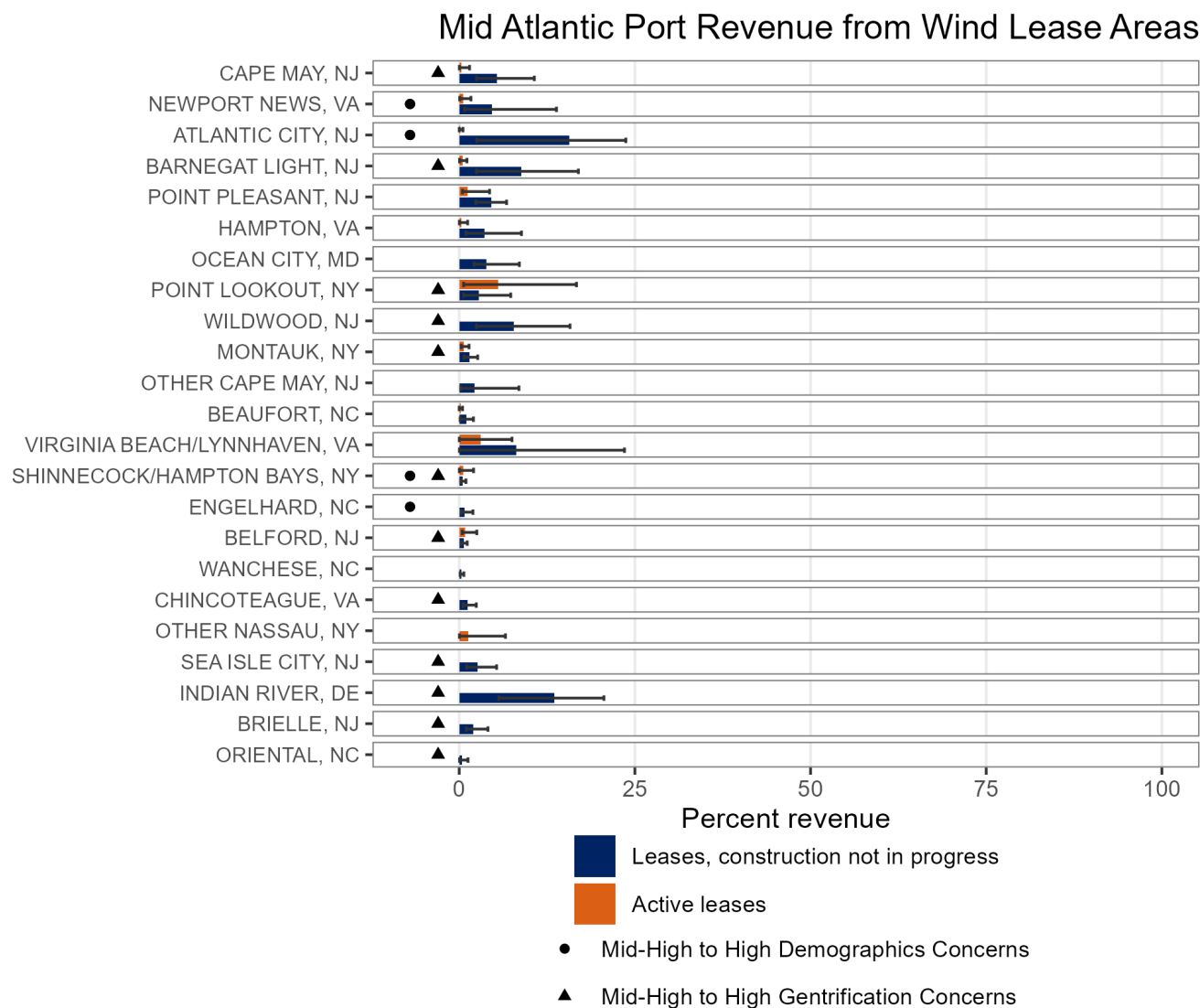


Figure 59: Percent of Mid-Atlantic port revenue from Wind Energy Areas (WEA) from all leases (green), leases not under construction leases (blue), and active leases (orange). Note that North Carolina fisheries management is split between the Northeast and Southeast, and this plot only includes data reported to the Northeast Fisheries Science Center.

Based on federal vessel logbook data, Point Lookout, NY (5.5% average, 17% maximum) and Virginia Beach, VA (3% average, 7.5% maximum) have the highest potential revenue loss from the Active Projects based on 2008-2024 total port fisheries revenue. Fewer Mid-Atlantic ports are affected by the Active Projects to date, as most are

in the southern New England region, with the exception of CVOW and Empire Wind 1 (Fig. 59). Additional fishing revenue may be lost as more areas historically used for fishing are developed for offshore wind energy. In seven New England ports, Mid-Atlantic managed species account for at least 50% of landings from the Active Project areas by value or weight (Fig. 60). Furthermore, impacts of offshore wind development may unevenly affect individual operators, with some permit holders deriving a much higher proportion of revenue from wind areas than the port-based mean.

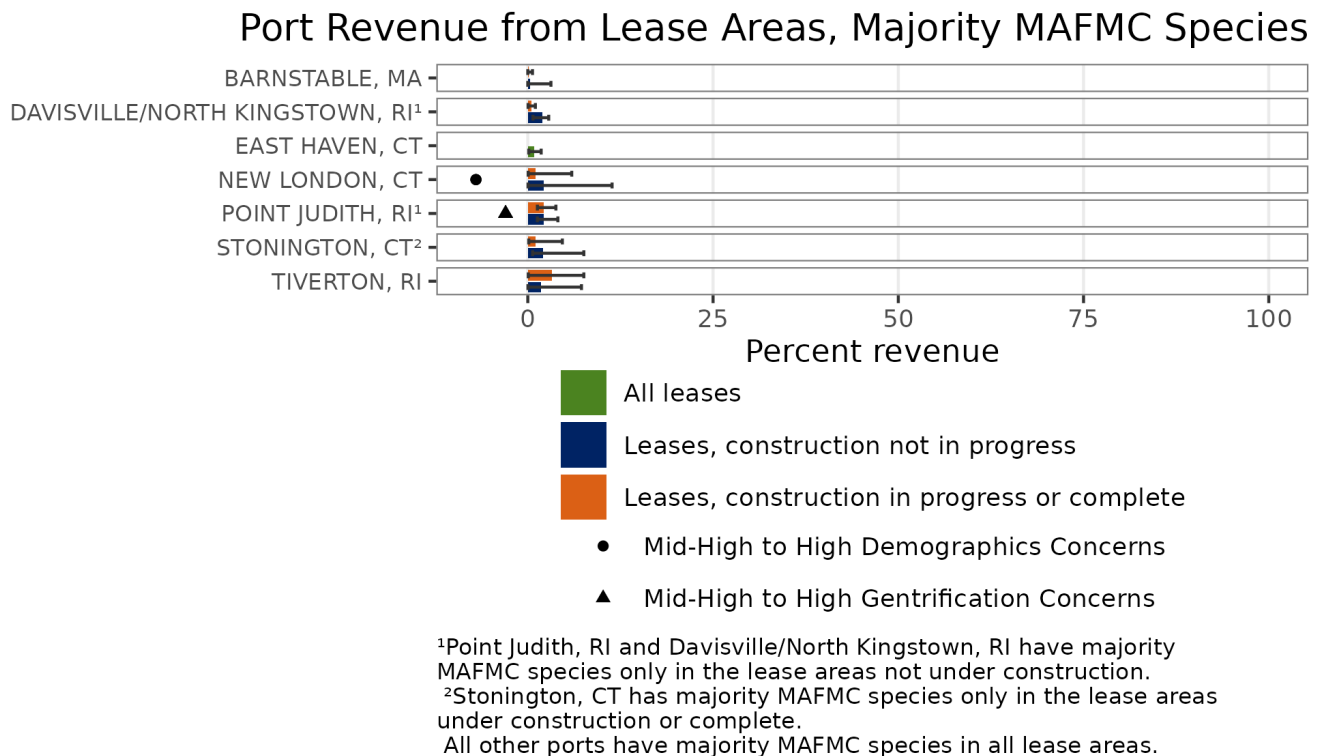


Figure 60: Percent of Mid-Atlantic port revenue with majority NEFMC landings from Wind Energy Areas (WEA) from all leases (green), leases not under construction (blue), and active leases (orange).

If all proposed projects are developed, BOEM reports that cumulative offshore wind development could have moderate impacts on low-income members of communities who work in the commercial fishing and for-hire fishing industry due to disruptions to fish populations, restrictions on navigation and increased vessel traffic, and existing vulnerabilities of low-income workers to economic impacts.

Top fishing communities with both landings from the Active Projects (i.e., Point Lookout, NY and Newport News, VA) and [socio-demographic or gentrification concerns](#) should be recognized as having additional vulnerability from Active Projects. To reduce further social and economic impacts and aid in the resilience and adaptive capacity of these communities, this vulnerability should be considered in decision making. Historically, the introduction of new industries can trigger industrial and socioeconomic gentrification of fishing ports. Competition for port space and potential pivoting of space use for offshore wind development should be monitored closely to ensure



fishing communities are not adversely impacted. Additionally, offshore wind could increase recreational fishing opportunities at the turbines, potentially creating a demand for additional tourism, recreational fishing and boating port space in communities already balancing these uses with commercial fishing infrastructure (e.g., Virginia Beach, VA, Montauk, NY and Barnegate Light, NJ.) Socio-demographic concerns also highlight communities where further resources are needed to reach underserved and underrepresented groups and create opportunities for, and directly involve, these groups in the decision-making process.

## Implications

Current plans for buildout of offshore wind in a patchwork of areas spreads the impacts differentially throughout the region (Fig. 56). Up to 6% of maximum annual fisheries revenue for major Mid-Atlantic commercial species in lease areas could be forgone or reduced and associated effort displaced if all sites are developed. Displaced fishing effort can alter historic fishing area, timing, and method patterns, which can in turn change habitat, species (managed and protected), and fleet interactions. Several factors, including fishery regulations, fishery availability, and user conflicts affect where, when, and how fishing effort may be displaced, along with impacts to and responses of affected fish species.

Proposed wind development areas interact with the region's federal scientific surveys. Scientific surveys are impacted by offshore wind in four ways: 1. Exclusion of NOAA Fisheries' sampling platforms from the wind development area due to operational and safety limitations. 2. Impacts on the random-stratified statistical design that is the basis for scientific assessments, advice, and analyses. 3. Alteration of benthic and pelagic habitats, and airspace in and around the wind energy development, requiring new designs and methods to sample new habitats. 4. Reduced sampling productivity through navigation impacts of wind energy infrastructure on aerial and vessel survey operations.

Increased vessel transit between stations may decrease data collections that are already limited by annual days-at-sea day allocations. In the Northeast region, 14 NEFSC surveys overlap with offshore wind development projects at varying capacities, with each of the 38 existing lease areas overlapping between 4-14 surveys. The Active Projects overlap between 10-12 surveys. Implementation of the region-wide survey mitigation program is underway with requirements to mitigate impacts to surveys included as a condition of most project approvals.

Planned development overlaps NARW mother and calf migration corridors and a significant foraging habitat that is used throughout the year (Fig. 61). Turbine presence and extraction of energy from the system could alter local oceanography and may affect right whale prey availability. For example, persistent foraging hotspots of right whales and seabirds overlap on Nantucket Shoals, where unique hydrography aggregates enhanced prey densities. Wind leases (OCS-A 0521 and OCS-A 0522) currently intersect these hotspots on the southwestern corner of Nantucket Shoals and a prominent tidal front associated with invertebrate prey swarms important to seabirds and possibly right whales. Proposed wind development areas also bring increased vessel strike risk from construction and operation vessels. In addition, there are a number of potential impacts to whales from pile driving and operational noise such as displacement, increased levels of communication masking, and elevated stress hormones.

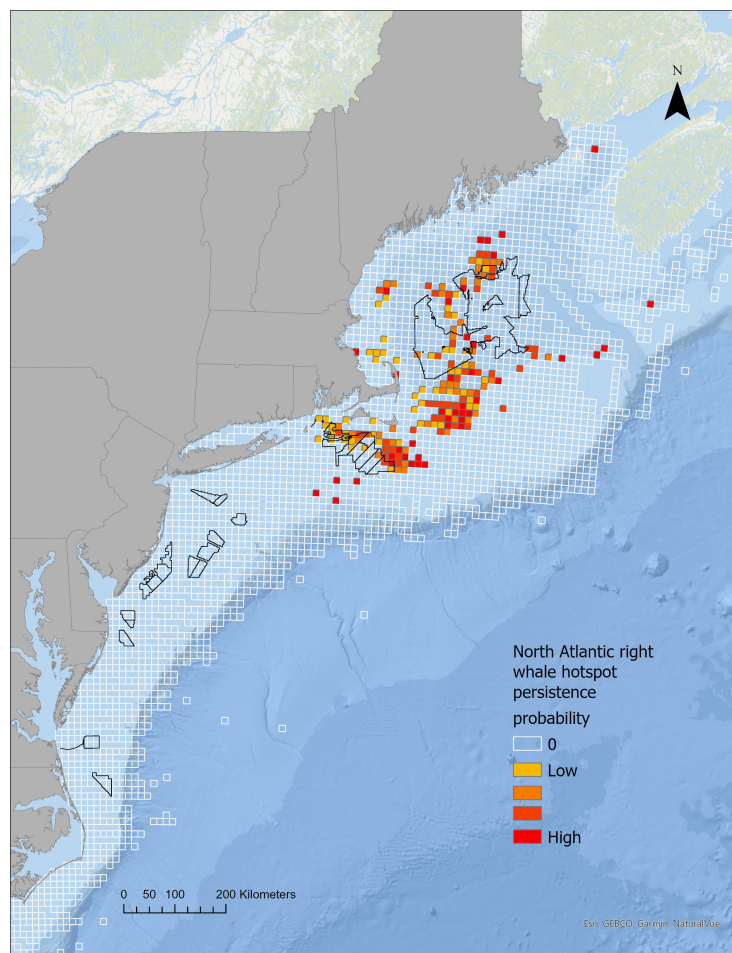


Figure 61: Northern Right Whale persistent hotspots (red shading) and Wind Energy Areas (black outlines).

The increase of offshore wind development can have both positive (e.g., employment opportunities) and negative (e.g., space-use conflicts) effects. Continued increase in coastal development and gentrification pressure has resulted in loss of fishing infrastructure space within ports. Understanding these existing pressures can allow for avoiding and mitigating negative impacts to our shore support industry and communities dependent on fishing. Some of the communities with the highest fisheries revenue overlap with offshore wind development areas that are also vulnerable to gentrification pressure are Beaufort, NC, and Cape May, Barnegat Light, and Long Beach, NJ.

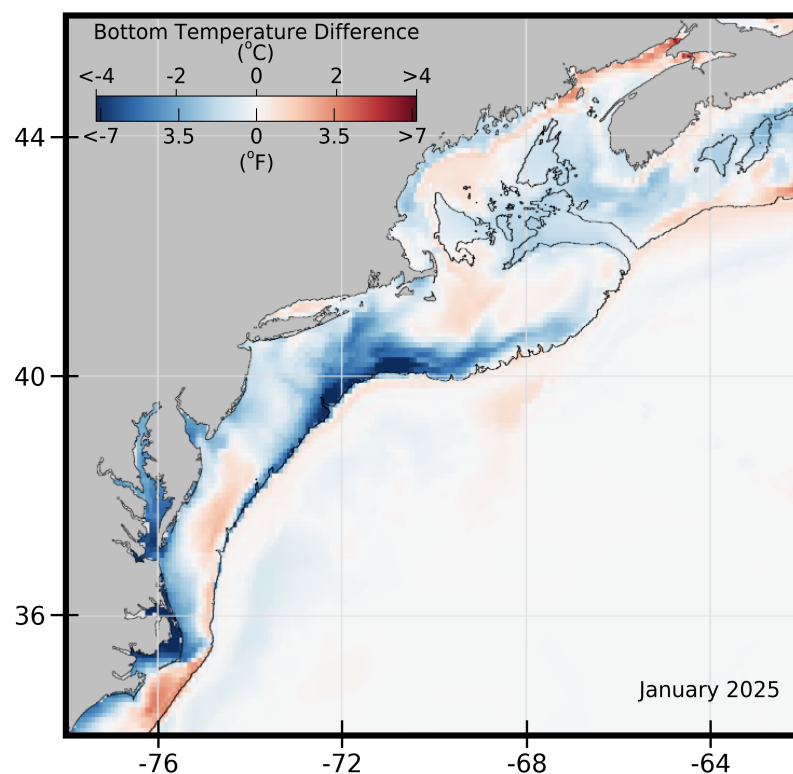
**Marine Aquaculture** Aquaculture fisheries and federally-managed fisheries could both compete or benefit each other with spatial access, shoreside infrastructure, or the supply of seafood. Unlike offshore wind, offshore aquaculture is not regulated by any federal leasing program but is permitted via the U.S. Army Corps of Engineers and the U.S. EPA. Currently, there are no federally-permitted aquaculture projects in the Northeast U.S. The marine aquaculture industry of the Northeast currently occurs in nearshore waters which are regulated by state leasing and permitting processes and federal permitting processes, as applicable. Analyses are needed to quantify the nearshore spatial distribution of aquaculture in the Northeast.

## 2025 Highlights

This section intends to provide a record of [noteworthy observations reported in 2025](#) across the Northeast U.S. region. The full ecosystem and fisheries impacts of many of these observations are still to be determined. They should, however, be noted and considered in future analyses and management decisions.

The Northeast U.S. region experienced colder than average ocean temperatures, despite record warm [global](#) ocean and air temperatures. Similar to 2024, oceanographic and ecological conditions reflected cooler water and changing species abundance, distribution, and timing.

**Northwest Atlantic Phenomena** The below average temperatures observed in 2024 persisted into 2025, although there are seasonal and local exceptions to this pattern. Anomalously cold surface conditions (Fig. ??)(Fig. 62) were recorded throughout the Northeast Shelf and were widespread across the Slope Sea for much of the year, however the waters were not as fresh as recorded in 2024. Winter bottom temperatures were also below average across much of the Northeast Shelf (Fig. 96b). Multiple oceanographic and atmospheric factors can contribute to these cooler conditions including a more southerly Gulf Stream and higher proportions of Labrador Slope and Scotian Shelf water entering the system.



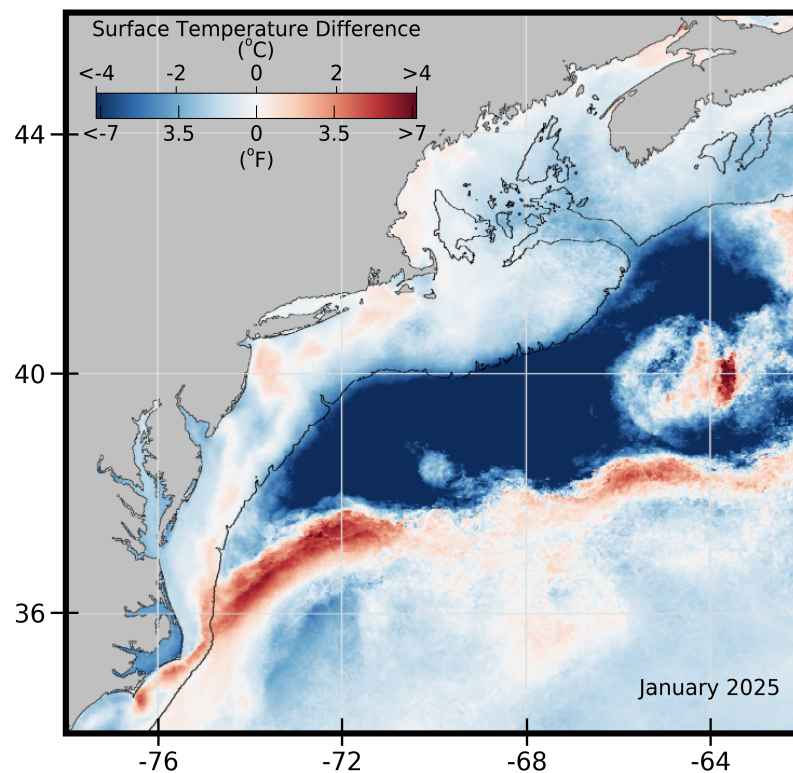


Figure 62: January 2025 sea surface (a) and bottom (b) temperature differences compared to the January climatological means. Sea Surface Temperature (SST) is from the NOAA Advanced Clear-Sky Processor for Ocean (ACSPO) Super-collated SST (climatology range 2000-2020); Bottom temperature is from the GLORYS reanalysis model (climatology range 1990-2020).

In 2024, the Gulf of Maine [source water](#) entering through the Northeast Channel was near equal proportions of Warm Slope Water and cooler Labrador Slope Water (Fig. 97); data are still being processed for 2025. The colder conditions [observed in 2024](#) continued into 2025 and contributed to the increased size and colder temperatures of the Mid-Atlantic [Cold Pool](#).

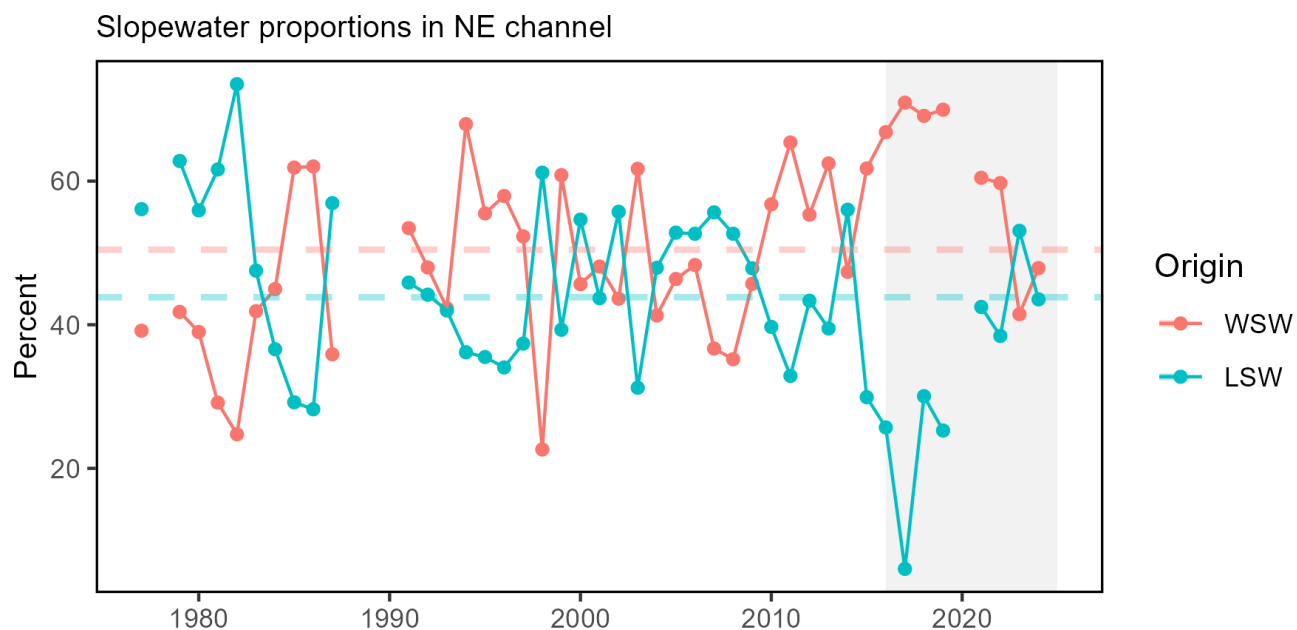
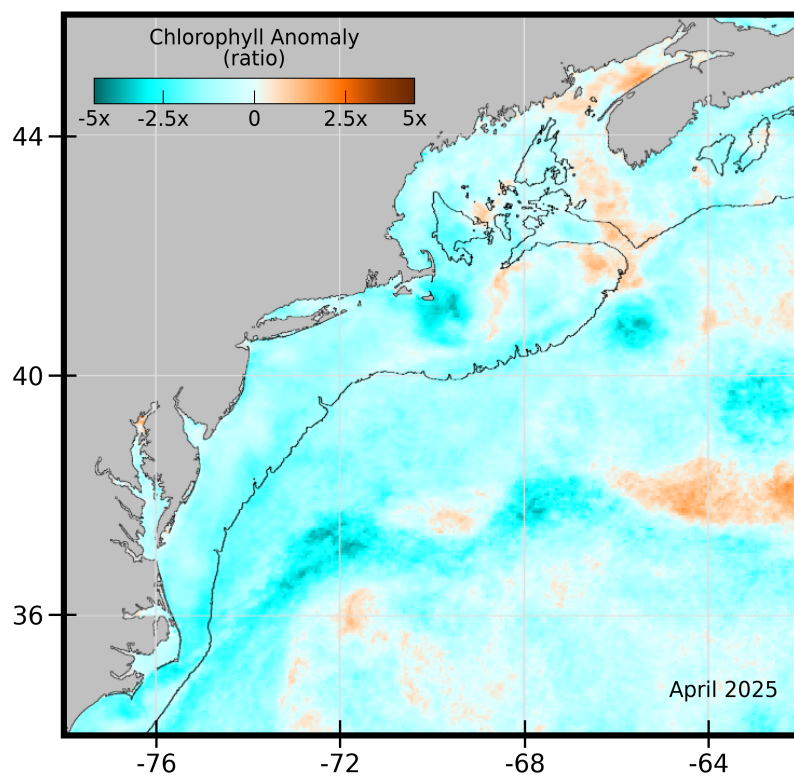


Figure 63: The proportion of Warm Slope Water (WSW) and Labrador Slope Water (LSW) enter the Gulf of Maine through the Northeast Channel from 1977 to 2023. The red and teal dashed lines represent the long-term proportion averages for the WSW and LSW respectively.

2025 total primary production was below average in Georges Bank and the Mid-Atlantic due to lower phytoplankton biomass and cooler sea surface temperatures. [Phytoplankton biomass](#) (shown as chlorophyll a concentration) was also below average for much of 2025 (Fig. ??)(Fig. 64). In particular, the winter-spring bloom period, which typically accounts for a significant proportion to total annual phytoplankton production, was shorter in duration and lower in magnitude across the entire Northeast shelf region. The fall bloom period was above average in the Gulf of Maine and Georges Bank, but near average in the Mid-Atlantic.





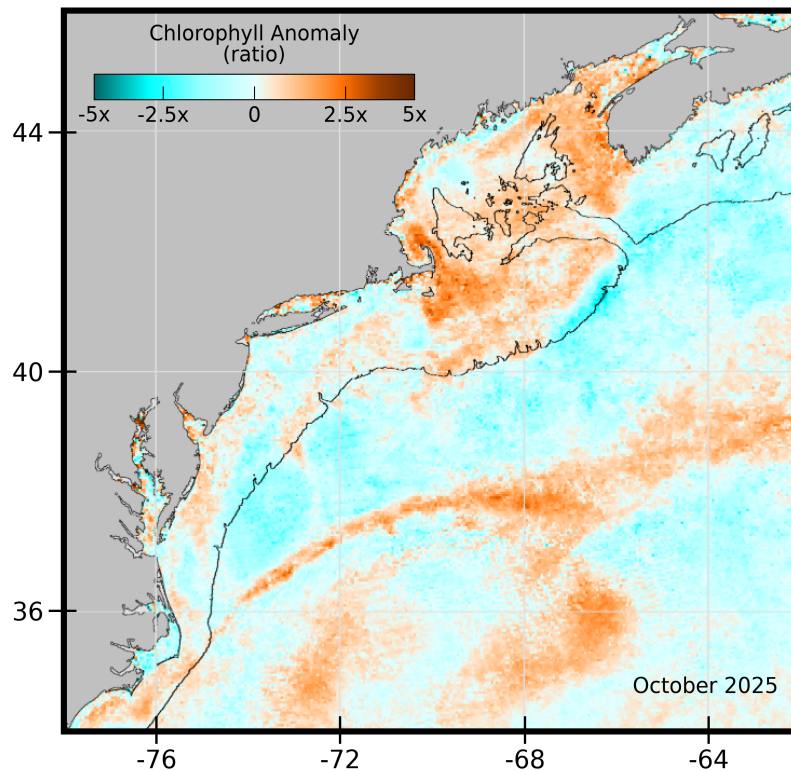


Figure 64: Chlorophyll anomalies shown as the ratio of the monthly concentration compared to the climatological mean (1998 to 2020).

Hurricane Erin was the most notable storm in the region in 2025 and caused significant shoreside and oceanographic disturbances despite not making landfall in the Northeast. Its strong winds and large size caused mixing and weakened stratification throughout the Mid-Atlantic resulting in cooler than average surface waters across the shelf and into the Slope Sea. Along the coast, beaches from Maryland to Maine were closed due to rough surf, large waves, and rip currents, while coastal flooding led to road closures and several water rescues, particularly in New Jersey. Beach erosion was significant in some locations.

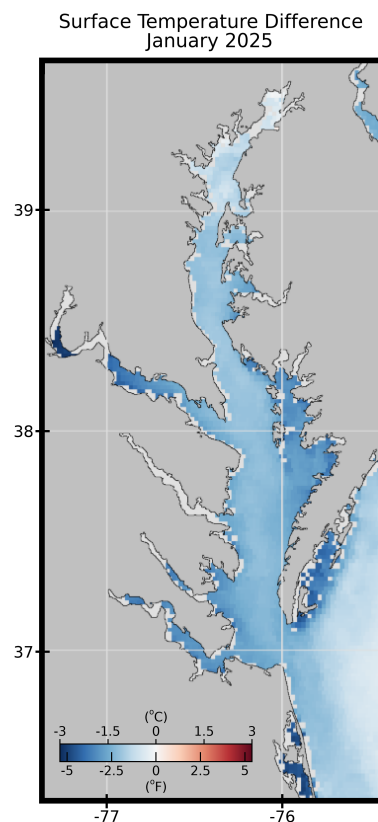
**Northeast Shelf and Local Phenomena** The shift to cooler waters in 2024-2025 is likely linked to multiple observations across the Northeast Shelf including the uncommon presence of Arctic zooplankton species in the Gulf of Maine, delayed migration of many species, and redistribution of some species. These shifts could affect the availability of some species to surveys or fishing, although aggregate species distributions in the cooler 2024-2025 period are tracking on the long-term trend towards northward and deeper waters (Fig. 32).

Mid-Atlantic scallops in the Elephant Trunk region are showing positive signs following the documented die-off. Two-year olds observed in 2024 had good survival into the 2025 survey. The Elephant Trunk region is scheduled to reopen in 2026. There was also good survival of the 2024 recruits in the southeastern Nantucket Lightship Area in 2025. In contrast, large numbers of the scallop predator *Asterias vulgaris* sea stars were linked to an increased sea scallop mortality in 2024 and 2025 on Georges Bank.

Several members of the fishing community noted changes in species composition, distribution, and timing in their typical fishing grounds and attributed it to the cooler temperatures. These observations may not fully represent the entire ecosystem, but provide local context to recent events that may not be represented in other indicators. Some notable examples include: - Several members of the fishing industry reported that it was a “very good year” for billfish. According to the Large Pelagic Survey, it was a record year for white marlin with more than 23,000 fish

caught and released. Billfish effort may have been higher than usual due to the closure of the recreational bluefin tuna fishery in August 2025. - Chesapeake Bay anglers reported good catch of red drum in 2024, followed by low catch and even cold stunned red drum and spotted sea trout in 2025. Scientists working with charter captains in Chesapeake Bay reported low catch rates of striped bass and red drum from June-September, but higher catch rates in the early spring and fall. - Fishers attributed the delayed migration of black sea bass inshore, and scup migrating south for the winter using similar routes as in the early 1970s due to the cooler temperatures. - Members of the bluefish fishery in Rhode Island reported very low landings in 2025 attributed to changes in seasonal migration path or timing. - Some species, such as Atlantic mackerel, Illex squid and sandlance, were observed in higher abundance and wider distributions compared to recent years. - Fishers in the Gulf of Maine and Georges Bank had mixed reports of lobsters and good catch of sea scallops. - Fishers reported fewer warm water species in 2025 along the New Jersey coast. Others, however, noted more new species (e.g., pomapano, spadefish, triggerfish, Spanish and king mackerel) in Delaware Bay. - Anglers also observed low spring and summer catches of gamefish in Mid-Atlantic bays and on the shelf, and high concentrations of shark species near the coast.

In Chesapeake Bay, colder than average winter 2025 temperatures (Fig. 100) were reported by state agencies as a likely cause of higher blue crab mortality rates compared to the previous winter. Colder winters generally indicate good conditions for striped bass spawning, and while the striped bass juvenile index slightly improved, it was still well below the long-term average. Several years of low striped bass recruitment is a growing concern of fisheries managers. Factors that could be influencing striped bass include below average winter-to-spring freshwater flow and above average water temperatures combined with stressful dissolved oxygen values during the summer. The continued presence of invasive blue catfish and the effect they are having on blue crab, alosines, menhaden, and striped bass populations is also a management concern.





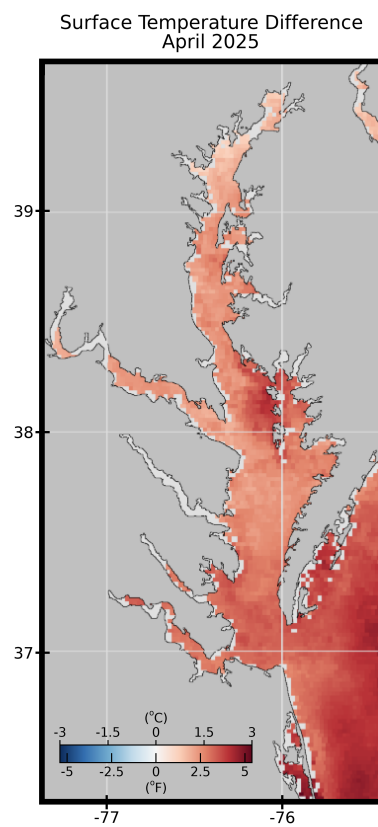


Figure 65: January 2025 (a) and April 2025 (b) sea surface temperature differences compared to the respective monthly climatological means. Sea Surface Temperature (SST) is from the NOAA Advanced Clear-Sky Processor for Ocean (ACSPO) Super-collated SST (climatology range 2000 to 2020).

**Ocean acidification (OA)** risk in 2025 in the Mid-Atlantic was relatively low. Compared to 2023 and 2024, there were only a few locations in 2025 where OA risk was high for Atlantic sea scallops (outer shelf, spring 2025 only), longfin squid (nearshore, summer 2025 only), and pteropods (nearshore and outer shelf, spring and summer). Gulf of Maine surface OA risk in 2025 was below 2024 levels, as indicated by aragonite saturation state ( $\Omega_a$ ) at or near the climatological average (2006-2024).

**Offshore Wind/Social** Active offshore wind projects continue to be developed throughout the region. In Southern New England, South Fork Wind Farm remains the first and only commercial scale project under operation (12 turbines). Vineyard Wind 1 and Revolution Wind continued construction, and Empire Wind 1 in the New York Bight and Sunrise Wind began offshore construction in 2025 (Fig. 56). Coastal Virginia Offshore Wind (CVOW) also continued construction in the Mid-Atlantic. All projects currently under construction are anticipated to be complete by the end of 2026. New London, CT and New Bedford, MA have expanded dedicated space and infrastructure for the offshore wind industry with increased port activity for the first projects under construction in southern New England. There are eight additional projects that have Construction and Operations Plan (COP) approvals (three in Southern New England and five in the Mid-Atlantic/New York Bight) that could begin construction in 2026. However, construction schedules are highly uncertain at this time.

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## Document Orientation

The figure format is illustrated in Fig 66a. Trend lines are shown when slope is significantly different from 0 at the  $p < 0.05$  level. An orange line signifies an overall positive trend, and purple signifies a negative trend. To minimize bias introduced by small sample size, no trend is fit for  $< 30$  year time series. Dashed lines represent mean values of time series unless the indicator is an anomaly, in which case the dashed line is equal to 0. Shaded regions indicate the past ten years. If there are no new data for 2022, the shaded region will still cover this time period. The spatial scale of indicators is either coastwide, Mid-Atlantic states (New York, New Jersey, Delaware, Maryland, Virginia, North Carolina), or at the Mid-Atlantic Bight (MAB) Ecosystem Production Unit (EPU, Fig. 66b) level.

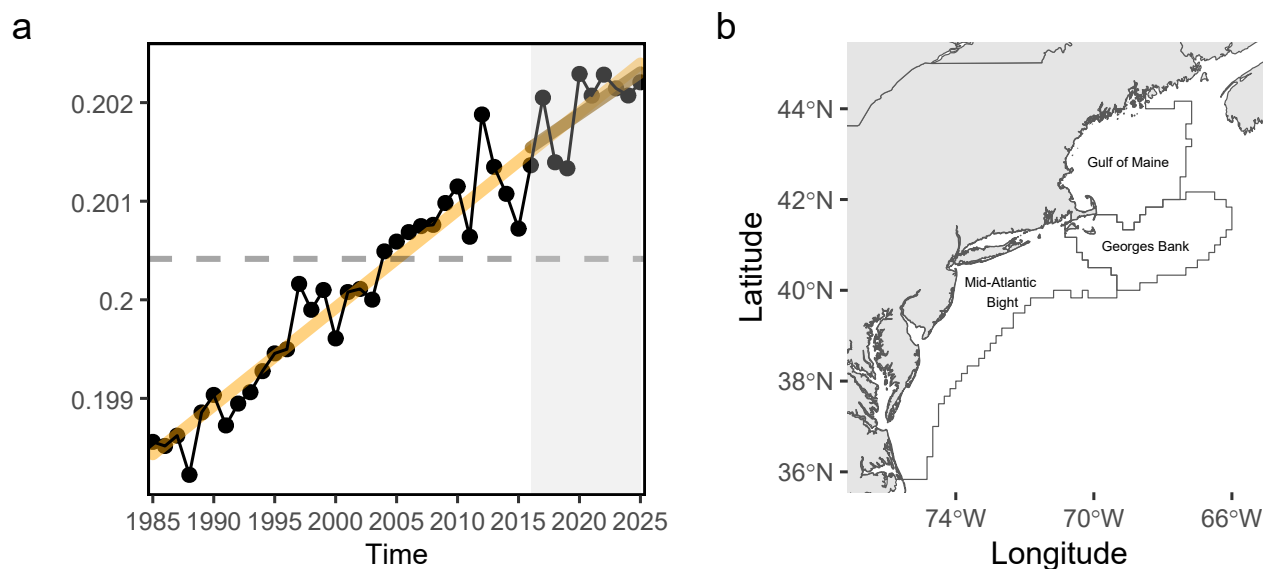


Figure 66: Document orientation. a. Key to figures. b. The Northeast Large Marine Ecosystem.

Fish and invertebrates are aggregated into similar feeding categories (Table 6) to evaluate ecosystem level trends in predators and prey.

Table 6: Feeding guilds and management bodies.

| Guild         | MAFMC                                                                      | Joint                       | NEFMC                                                                                                                                                                                          | State or Other                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|----------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apex Predator |                                                                            |                             |                                                                                                                                                                                                | shark uncl, swordfish, yellowfin tuna, bluefin tuna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Piscivore     | summer flounder,<br>bluefish, northern<br>shortfin squid,<br>longfin squid | spiny dogfish,<br>goosefish | winter skate,<br>clearnose skate,<br>thorny skate,<br>offshore hake,<br>silver hake, atlantic<br>cod, pollock, white<br>hake, red hake,<br>atlantic halibut,<br>windowpane,<br>acadian redfish | sea lamprey, sandbar shark, atlantic angel shark, atlantic<br>torpedo, conger eel, spotted hake, cusk, fourspot flounder,<br>john dory, atlantic cutlassfish, blue runner, striped bass,<br>weakfish, sea raven, northern stargazer, banded<br>rudderfish, atlantic sharpnose shark, inshore lizardfish,<br>atlantic brief squid, northern sennet, king mackerel,<br>spanish mackerel                                                                                                                                                                                                                                                                                                                                                                  |
| Planktivore   | atlantic mackerel,<br>chub mackerel,<br>butterfish                         |                             | atlantic herring                                                                                                                                                                               | harvestfishes, smelts, round herring, alewife, blueback<br>herring, american shad, menhaden, bay anchovy, striped<br>anchovy, rainbow smelt, atlantic argentine, slender snipe<br>eel, atlantic silverside, northern pipefish, atlantic<br>moonfish, lookdown, blackbelly rosefish, lumpfish,<br>northern sand lance, atlantic saury, mackerel scad, bigeye<br>scad, round scad, rough scad, silver rag, weitzmans<br>pearlsides, atlantic soft pout, sevenspine bay shrimp, pink<br>glass shrimp, polar lebbeid, friendly blade shrimp, bristled<br>longbeak, aesop shrimp, norwegian shrimp, northern<br>shrimp, brown rock shrimp, atlantic thread herring,<br>spanish sardine, atlantic bumper, harvestfish, striated<br>argentine, silver anchovy |

Table 6: Feeding guilds and management bodies.

| Guild      | MAFMC                              | Joint | NEFMC                                                                                                                                                                                                                    | State or Other                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benthivore | black sea bass,<br>scup, tilefish  |       | barndoor skate,<br>rosette skate, little<br>skate, smooth<br>skate, haddock,<br>american plaice,<br>yellowtail flounder,<br>winter flounder,<br>witch flounder,<br>atlantic wolffish,<br>ocean pout,<br>crab,red deepsea | crab,unc, hagfish, porgy,red, sea bass,nk, atlantic hagfish,<br>rougtail stingray, smooth dogfish, chain dogfish,<br>bluntnose stingray, bullnose ray, southern stingray, longfin<br>hake, fourbeard rockling, marlin-spike, gulf stream<br>flounder, longspine snipefish, blackmouth bass, threespine<br>stickleback, smallmouth flounder, hogchoker, bigeye,<br>atlantic croaker, pigfish, northern kingfish, silver perch,<br>spot, deepbody boarfish, sculpin uncl, moustache sculpin,<br>longhorn sculpin, alligatorfish, grubby, atlantic seasnail,<br>northern searobin, striped searobin, armored searobin,<br>cunner, tautog, snakeblenny, daubed shanny, radiated<br>shanny, red goatfish, striped cusk-eel, wolf eelpout,<br>wrymouth, fawn cusk-eel, northern puffer, striped<br>burrfish, planehead filefish, gray triggerfish, shortnose<br>greeneye, beardfish, cownose ray, american lobster, cancer<br>crab uncl, jonah crab, atlantic rock crab, blue crab, spider<br>crab uncl, horseshoe crab, coarsehand lady crab, lady<br>crab, northern stone crab, snow crab, spiny butterfly ray,<br>smooth butterfly ray, snakefish, atlantic midshipman,<br>bank cusk-eel, red cornetfish, squid cuttlefish and octopod<br>uncl, spoonarm octopus, bank sea bass, rock sea bass,<br>sand perch, cobia, crevalle jack, vermilion snapper,<br>tomtate, jolthead porgy, saucereye porgy, whitebone<br>porgy, knobbed porgy, sheepshead porgy, littlehead porgy,<br>silver porgy, pinfish, red porgy, porgy and pinfish uncl,<br>banded drum, southern kingfish, atlantic spadefish,<br>leopard searobin, dusky flounder, triggerfish filefish uncl,<br>blackcheek tonguefish, orange filefish, queen triggerfish,<br>ocean triggerfish |
| Benthos    | atlantic surfclam,<br>ocean quahog |       | sea scallop                                                                                                                                                                                                              | sea cucumber, sea urchins, snails(conchs), sea urchin and<br>sand dollar uncl, channeled whelk, blue mussel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |